

Perimeter detection for a 350 mm mobile robot

Time-of-flight, mmWave radar and thermal infrared: a catalogue of every candidate part, the coverage geometry that decides how many of them you need, and what each one can actually tell you about a human, a pet or a box at 1, 3, 5, 8 and 10 feet.

Revision A | research date 2026-09-12 | vendors surveyed: Adafruit, DFRobot, STMicroelectronics, and the wider ToF and lidar market

The answer, first

A 350 mm robot needs **twelve multizone time-of-flight sensors in the perimeter ring, six downward cliff sensors, and eight thermal arrays** to do the whole job the brief describes: avoid every obstacle through 360 degrees, never fall down a stair, know a person is there, and know the difference between a person and a pet. That is nineteen ToF modules and eight people sensors, about USD 909 and about 1.0 A at 5 V.

If the budget only stretches to collision avoidance, the number is **eighteen ToF modules and zero people sensors**, about USD 455. That build is honest about what it is: a bumper made of light. It cannot tell you anything is alive.

	Minimum: collision only	Recommended: collision + animate	Maximum: full classification
ToF modules	18 (12 ring, 6 cliff)	19 (12 ring, 6 cliff, 1 scanner)	24 (16 ring, 6 cliff, 1 head, 1 scanner)
People sensors	0	8 thermal arrays	6 thermal + 4 cameras
Hardware cost	USD 455	USD 909	USD 1,440
Average current at 5 V	724 mA	999 mA	3.8 A
"Will I hit it?"	Yes, 360 deg	Yes, plus dark targets and sunbeams	Yes
"Is it alive?"	No	Yes, to about 2.4 m for a pet, 5 m for a human	Yes, in darkness too
"Human or pet?"	No	By fusion, to about 5 ft	Labelled, to 10 ft

The five things that decide this design

- 1. Use the horizontal field of view, never the diagonal.** Every ST VL53 datasheet leads with a diagonal figure. It is a separately measured number, not the geometric diagonal of the square detection volume, so it cannot be converted by dividing by the square root of two. The VL53L5CX is 63 deg diagonal and **45 deg horizontal**. The VL53L7CX is 90 deg diagonal and **60 deg horizontal**. The VL53L1X is 27 deg diagonal and about **19 deg horizontal**. Sizing a ring from the diagonal understates the count by three to six sensors, and this document's own first-pass research made exactly that error before the verification pass caught it.
- 2. A ring closes only when $n \times \text{FoV} > 360$, and "exactly 360" is the one value to avoid.** At exactly 360 the edge rays of adjacent sensors are parallel, so they never meet at any finite range and the wedge between them is blind forever. Eight VL53L5CX at 45 deg is exactly 360 deg and it does not work.
- 3. The sensors sit on the rim, not at the centre, so even a closed ring is blind near the robot.** On a 175 mm radius the wedges between adjacent sensors close at a range

```
u = r sin(a) / sin(a - s)
a = half the horizontal field of view
s = pi / n
r = the mounting radius, 175 mm
```

measured from the platform centre. Twelve VL53L5CX close at 513 mm from the centre, 338 mm past the edge. Sixteen close at 343 mm from the centre, inside the robot's own 350 mm footprint. Both figures appear in this document and they do not disagree: **twelve** is the recommended build, whose near-field wedge is covered by a compliant bumper, and **sixteen** is what it costs to close the wedge inside the footprint and drop the bumper from the safety argument.

These closure figures were derived twice, independently and by different algebra - once by a research agent working from the platform geometry, once by radar-compare/scripts/geometry.py solving the bisector intersection directly. They agree to the tenth of a millimetre at every count checked: 513.1 mm for twelve at 45 deg, 343.3 mm for sixteen at 45 deg, 670.4 mm for eight at 60 deg, 1170.9 mm for seven at 60 deg, and 343.0 mm for the thirty-eight single-zone VL53L1X sensors it would take to close a ring at 19.27 deg.

4. Mounting height decides what the robot runs over, and it is not a detail. A ring at 800 mm with a 45 deg vertical field of view cannot see a standing cat at 1 ft or at 3 ft at all - the animal is entirely beneath the beam. A ring at 100 to 200 mm sees everything from a sleeping cat upward, at the price of a floor return in every frame. Put the collision ring low.

5. Classification must never gate the stop. There are 167 to 323 ms between the warning-field edge and the protective-field edge for a target walking toward the robot. Micro-Doppler needs about 250 ms of dwell to reach 75 percent;

ST's own people-counting reference uses a 2.1 s window; a thermal frame at 4 Hz is 250 ms by itself. Braking runs on raw geometry at ring frame rate. Classification runs in parallel and changes only what the robot does *after* it has stopped.

What this costs you to ignore

- **24 GHz presence radar is rejected for this robot**, despite being the obvious answer for people sensing. Those modules assume a fixed installation. Their static-clutter rejection is the only thing that makes them useful, and a driving robot gives every static object an apparent velocity, which leaves the filter nothing to reject. They also see through walls, so they report the person in the next room as a perimeter contact. Radar earns its place on this robot only where it can hide behind a printed shell that a ToF sensor cannot see through.
- **Nothing here carries a safety rating.** A certified IEC 61496 Type 3 scanner must prove detection against a 1.8 percent reflectance target inside its protective field. A VL53L8CX does not meet that, which is exactly why a certified scanner sees a black sock and this stack does not. Use the industrial arithmetic; do not claim the rating.
- **No combination in this document tells a cat from a dog without a camera.** Nor does any of it work in direct sunlight above about 5 kLux, which is where ST's published tables stop and direct sun is 100 to 120 kLux. Mirrors defeat all of it.

Two standalone product guides

The operator asked for two parts to be assessed on their own, independent of the ring recommendation. Both chapters read without the rest of the document.

- **ST multizone grid time-of-flight.** The latest grid part as of 2026-09-12 is the **VL53L9CX**, a 54 x 42 zone, 2268-zone stacked imager with MIPI output, announced with mass production from early July 2026. It is a camera module, not a microcontroller peripheral, and it is not a drop-in for anything. The part to actually build with is the **VL53L8CX**.
- **Adafruit 4010** is the **Slamtec RPLIDAR A1M8**, a 360 deg spinning laser scanner at USD 99.95, in stock. It is not a proximity sensor and it is easily confused with the unrelated Vishay VCNL4010. It is the best single "see everything around me" part in its price class and it classifies nothing at all.

How to read the rest

The catalogue chapters list every candidate part with its measurement conditions. The master grid rates each one against an adult, a cat and a 30 mm chair leg at 1, 3, 5, 8 and 10 ft, by a published rule rather than by opinion. The computed appendix carries every geometry table, all of it generated by `radar-compare/scripts/geometry.py` from first principles rather than copied from a vendor page. The last chapter gives a different answer for each of the three robot projects in this repository, because their pin, bus and shell budgets are genuinely different.

What perimeter detection has to do

Perimeter detection on a small indoor robot is four jobs, not one. They have different targets, different update rates and different failure directions, and the commonest error in this field is buying one sensor and assuming it covers all four. It does not.

The robot this document is written against

A mobile robot with a **350 mm square or 350 mm round footprint**, **600-1200 mm tall**, **5-15 kg**, moving at **0.2-0.5 m/s** on hard floor and carpet indoors, among adults, children and pets. The 0.5 m/s upper bound is not arbitrary: it is the household speed ceiling, against which 1.0 m/s on a 10-15 kg robot in a home is called indefensible (safety-and-failure-modes.md).

Three consequences of that shape drive everything downstream.

- **The centre-to-skin radius is 175 mm.** Sensors sit on the perimeter, not at the centre, so a ring of them is blind in a wedge close to the body (computed, coverage-geometry.md).
- **A 350 mm square is 494.975 mm across the corners**, the corner protruding **72.487 mm** past the inscribed circle. A scheme that forgets the 495 mm diagonal clips door frames (computed).
- **The centre of gravity is high.** A robot with its CG at 600 mm cannot brake harder than **2.86 m/s²** without pitching onto its face (inferred, safety-and-failure-modes.md). That, not tyre grip, is the deceleration ceiling, and it sets the stopping distance every detection range must beat.

The speed band is a safety choice. ISO 3691-4 Annex A Table A.1 permits **0.3 m/s** for a driverless industrial truck whose personnel detection is muted (standard-verified against the 2020 first edition, via the TUV Rheinland whitepaper). A 15 kg robot at 1.0 m/s with a rigid shell delivers a mean **1500 N** into a shin, about 11x the 130 N ISO/TS 15066 transient limit; at 0.5 m/s behind a 50 mm compliant bumper it delivers **38 N** (computed, vendor-page-verified limits).

The four jobs

Collision avoidance. Do not hit anything, of any material, at any height the body occupies, in any light. The hard targets are not people: they are a **30 mm chair leg** and a **6 mm cable**. A chair leg at 5 ft subtends **1.32 deg** (computed, coverage-geometry.md Table 16, at that document's 35 mm chair-leg width), while an 8x8 zoned array at 45 deg horizontal gives **5.625 deg** cells, or **149.7 mm** at 5 ft (computed). The array detects a leg; it cannot classify it. This job needs geometry and coverage, not semantics.

Cliff and drop detection. Do not fall down a stair. It is a separate mandatory function: a horizontal scan plane at 150-200 mm sees nothing at a stair nosing, because the floor simply stops. At 0.5 m/s with a 100 ms loop the robot travels **92 mm** after the edge is seen; at 1.0 m/s it overruns the 175 mm half-footprint at every plausible latency and goes over (inferred). Downward time-of-flight at 50 Hz or better, wired so that a lost reading counts as a cliff.

Human awareness. Know a person is there, where, and how far, for the stop and for the interaction. The governing number is not the robot's speed: ISO 13855 uses **K = 1600 mm/s** for a walking approach, so a robot at 0.5 m/s meeting a walking person closes at **2.1 m/s**, not 0.5.

Pet awareness. Know a cat or small dog is there. This is the hardest of the four, because a cat is small, low, fast, silent and usually **lying still**. A sleeping cat is 300 x 130 mm: below every turret-height scan plane, and below every warm-body threshold a radiator also trips. The empirical ceiling is published. In the Oxford robotic-mower study, 19 models were run against hedgehog cadavers and, apart from one incidence not reproducible on retest, **all had to physically touch the carcass to detect it** (peer-reviewed, prior-art-commercial.md).

The subjects, and where their numbers come from

These are **design assumptions, not measurements**, carrying no percentile and no anthropometric source. They are what the geometry appendix and every capability table here are computed against.

Subject	Width (mm)	Height (mm)	Note
Adult torso	450	1700	shoulders 400-500 mm

Subject	Width (mm)	Height (mm)	Note
Adult leg	140	800	what a low ring actually sees
Toddler	250	850	2 to 3 years old
Large dog	250	650	labrador at the shoulder
Cat, standing	140	250	at the shoulder
Cat, lying	300	130	the worst case
Chair leg	30	430	thin vertical obstacle
Power cable	6	6	the classic robot killer
Table top	900	40	overhang, floor gap 700 mm

The overhang matters as much as the cable: a 600-1200 mm robot passes its body under a 700 mm table edge that no perimeter ring mounted below that height can see.

Why one sensor cannot do all four

Start from the honest framing: **none of these parts is safety-rated**. Not one sensor a hobbyist can buy is electro-sensitive protective equipment under IEC 61496, with a declared response time, a declared minimum detectable object and dual outputs that go safe on internal fault (safety-and-failure-modes.md). Every protective field in this reference is industrial arithmetic applied to uncertified parts.

Each modality has a **fail-to-danger** mode, and each has a different physical cause.

- **Optical time-of-flight** is defeated by black. Against black vinyl the VL53L5CX returned only **18.8 % valid measurements at 25 cm** (peer-reviewed-measured). A person in black jeans reads as an empty corridor.
- **2D lidar** is one horizontal slice. The 130 mm lying cat is under it; the 700 mm table edge is over it.
- **mmWave radar** cannot see a motionless person without micro-motion processing, which needs a **1-2 s dwell** (vendor-page-verified), and every static return acquires a Doppler shift once the robot itself moves.
- **Thermal arrays** lose a pet with range: of six thermal-IR arrays in the catalogue, four reach a standing cat at 1 and 3 ft, two at 5 ft, and **one at 8 and 10 ft** (computed from capability.json).
- **PIR** cannot see a still person at all, by design, and fires continuously on a moving platform.
- **Ultrasonic** is the one modality that sees glass and mirrors, and the one that loses people. Most HC-SR04 units are unreliable on soft targets such as a human beyond about **1 m**, where the successful-measurement rate falls **below 50 %** (vendor-page-verified), and eight of them fired round-robin take **264 ms** for one sweep. It is a warning-field part, never a stop part.

Two modalities with *different* physics remove the shared blind spot. That, not accuracy, is the argument for fusion, and the prior art agrees: every top scorer in the one consistent public obstacle-avoidance benchmark pairs an active depth sensor with a camera classifier.

Timing forces the same split. Between the warning-field edge and the protective-field edge the robot has **167-323 ms** to classify a moving target (computed). No thermal frame at 4 Hz and no micro-Doppler classifier fits in that. **Classification must never gate the stop**. Geometry stops the robot; classification decides only what it does afterwards.

The five survey distances

Every capability table here is evaluated at 1, 3, 5, 8 and 10 ft. Each earns its place.

Distance	mm	What it is for
1 ft	305	Inside the stop. Already too late.
3 ft	914	Decision distance: stop, slow or steer.
5 ft	1524	Planning distance: re-route without stopping.
8 ft	2438	Past most ToF: 5 of 9 multizone parts still register a cat.
10 ft	3048	Room-scale awareness, not avoidance.

1 ft is inside the stopping distance. The computed protective field for a VL53L5CX at 15 Hz and 0.5 m/s is **386 mm** with hobby-grade tolerances, **586 mm** if retroreflectors cannot be excluded, and the sensor must see reliably to **561 mm** once the 175 mm half-footprint is added (inferred). Anything first detected at 305 mm is a contact, not a detection.

3 ft is the decision distance, the first survey point outside the protective field at every speed in the band. **5 ft is the planning distance.** The conventional warning field is twice the protective field, which at 0.5 m/s is **772 mm** (computed); 5 ft sits well outside it, so it is where a re-route is chosen rather than where a brake is applied.

8 and 10 ft are where the field thins out. At 10 ft all 14 scanning lidars and 7 of 8 24 GHz radar modules in the catalogue still register a cat; 3 of 9 multizone ToF parts do, 1 of 9 single-point ToF parts, and one thermal array (computed from capability.json).

One rule applies to every field-of-view figure in this reference: **say which one you mean.** ST puts a diagonal on the front page of every VL53 datasheet, and a diagonal is not a horizontal. Where ST publishes the horizontal directly, use it: VL53L5CX and VL53L8CX are **45 x 45 deg** (63 and 65 deg diagonal), VL53L7CX and VL53L7CH are **60 x 60 deg** (90 deg diagonal), VL53L9CX is **55 x 42 deg** (71 deg diagonal). Where ST publishes only a diagonal, convert it: the VL53L1X's 27 deg is labelled diagonal, and the horizontal is **19.27 deg**. ST's 45/45/63 triple is not self-consistent under a pinhole model, so treat the 63 deg as a separate measurement and the 45 deg as the number that sizes a ring.

Every one of those figures is a conditioned measurement, not a hard cone. The VL53L5CX triple is taken at **88 % white reflectance, 1 m, dark, 8x8 resolution, 14 % sharpener, 15 Hz**; against a 17 % grey target in 5 klux the usable angular extent is smaller. Sizing a ring from a diagonal, or from a dark-condition figure, overstates coverage badly.

Where the sources disagree

Cat height. The prose in coverage-geometry.md models a **200 mm** cat and a **35 mm** chair leg; geometry.json, and this chapter, use **250 mm** and **30 mm**. Prefer the data file, because the computed appendix is built from it. The smaller cat is also the conservative case, so the vertical-coverage conclusions carry over with margin.

Stopping distance. geometry.json gives **73.8 mm** at 0.5 m/s for a VL53L5CX 8x8 at 15 Hz; safety-and-failure-modes.md gives a **386 mm** protective field for the same part and speed. They are not the same quantity: the first is the bare kinematic term, reaction plus braking; the second adds the industrial allowances for measurement tolerance, low reflectance and ground clearance. Size a field from the second. Note also that the geometry.json "firm 0.5 g" column is 4.9 m/s², well above the 2.86 m/s² tip-over ceiling for a CG at 600 mm; use 2.0 m/s².

Coverage geometry: how many sensors, and where

Four numbers decide a perimeter ring, and all four are geometry rather than part choice: the beam width at range, the ring count that actually closes, the depth of the near-field blind wedge, and the mount height that puts a pet inside the beam. This chapter derives each one once. Other chapters cite these results; the computed appendix carries the full tables.

One rule governs everything below. **Size a ring on the horizontal field of view, never on the diagonal headline.** The horizontal figures used throughout are: VL53L5CX and VL53L8CX 45 x 45 deg, VL53L7CX and VL53L7CH 60 x 60 deg, VL53L1X about 19.3 deg horizontal, VL53L9CX 55 x 42 deg. All except the VL53L9CX are datasheet-verified in data/sensors.csv; the VL53L9CX figure is vendor-page-verified only. Two diagonal figures are confirmed against a primary datasheet: the VL53L5CX 63 deg, and the VL53L1X 27 deg, which the datasheet labels diagonal and which converts to 19.27 deg horizontal. A "65 deg" diagonal could not be confirmed against any ST primary source, so it is not used here. Every ST field-of-view figure is a conditioned measurement, not a hard cone. The VL53L5CX 45 deg is measured against an 88 percent white target at 1 m in darkness, 8x8 resolution, 14 percent sharpener, 15 Hz; against 17 percent grey in bright ambient the usable angle is smaller, not equal. The VL53L1X 27 deg diagonal is measured with the target covering the full field of view, long-distance mode, a 100 ms budget, no cover glass, in darkness. The remaining horizontal figures carry no published measurement condition in the sources, so treat them the same way.

Beam footprint: what one sensor covers at range

The illuminated width at range d is

$$w = 2 d \tan(\text{FoV}_h / 2)$$

FoV_h (deg)	1 ft (305 mm)	3 ft (914 mm)	5 ft (1524 mm)	10 ft (3048 mm)
19.3 (VL53L1X)	104 mm	311 mm	518 mm	1037 mm
45 (VL53L5CX, VL53L8CX)	253 mm	757 mm	1263 mm	2525 mm
60 (VL53L7CX)	352 mm	1056 mm	1760 mm	3520 mm

The consequence is the whole single-zone-versus-array argument in one line: **a narrow sensor resolves a thin obstacle, a wide one averages it away.** A single-zone part returns one distance for the entire footprint. At 5 ft a 60 deg single-zone sensor is integrating a 1760 mm swath, so a chair leg, a cat and the wall behind them all contribute to one number, and the strongest reflector wins. A chair leg 35 mm wide subtends 1.32 deg at 5 ft. To report it as an object rather than as a weak average, the sensor needs angular cells near that size. An 8x8 array at 45 deg horizontal has a zone pitch of $45/8 = 5.625$ deg, which is 90 mm at 3 ft and 150 mm at 5 ft: enough to detect a human leg as a distinct return, not enough to classify it. Angular resolution comes from zones inside the field of view, never from more sensors around the ring.

Ring closure: $n \times \text{FoV} \geq 360$, full stop

Take n sensors evenly spaced, each with full horizontal field of view FoV, boresight spacing $\Delta = 360/n$, and angular overlap $\Omega = \text{FoV} - \Delta$. Sensor A's left edge and its neighbour B's right edge diverge by exactly $-\Omega$. When $\Omega = 0$ those two edges are **parallel**. Two parallel rays never meet, so the wedge between them is blind to infinite range.

$N = 360/\text{FoV}$ sets $\Delta = \text{FoV}$, which is $\Omega = 0$ exactly. It is not a conservative answer, it is the one value that must be avoided, and it is the formula every builder reaches for first. The condition for a ring to close at any finite range is $\text{FoV} > 360/n$, that is

$$n * \text{FoV} > 360 \text{ and in practice } n = \text{ceil}(360 / (\text{FoV} - \text{overlap}))$$

where overlap is deliberate margin. Ten to twenty degrees is right for consumer time-of-flight parts, for two reasons outside the geometry. A published field of view is the half-power contour, not a hard edge, so $\Omega = 0$ mates the weakest part of one beam to the weakest part of the next. Boresight error of about ± 2 deg per sensor from board placement and bracket tolerance is normal, so two neighbours can sit 4 deg further apart than the CAD model says.

FoV_h (deg)	overlap 0	overlap 5	overlap 10	overlap 20
19.3	19	26	39	impossible

FoV_h (deg)	overlap 0	overlap 5	overlap 10	overlap 20
45	8 (open)	9	11	15
55	7	8	8	11
60	6	7	8	9

The overlap 0 column is the naive count, and at 45 deg it is a ring that never closes.

The near-field blind wedge: the result most builders miss

Sensors sit on the rim, at $r = 175$ mm on a 350 mm platform, not at the centre. A ring that satisfies $n * \text{FoV} > 360$ is therefore still blind in a wedge near the body, because the two edge rays that bound the gap start apart and need range to converge. With a the half field of view and $s = \pi/n$ the half spacing, the wedge tip sits at

```
u = r * sin(a) / sin(a - s) measured from the platform CENTRE
blind depth past the skin = u - r
```

This is the same result as $u = r [\cos(s) + \sin(s) \cot(a - s)]$; the two collapse into each other by the sine addition rule. It has one behaviour worth memorising: for small overlap, $\sin(a - s)$ is nearly $(a - s)$, so **the blind depth is inversely proportional to the overlap**. Halving the overlap doubles the blind wedge.

ring	FoV_h (deg)	overlap (deg)	tip from centre	tip past skin
8 x VL53L5CX	45	0	never closes	never closes
10 x VL53L5CX	45	9	853.6 mm	678.6 mm
12 x VL53L5CX	45	15	513.1 mm	338.1 mm
16 x VL53L5CX	45	22.5	343.3 mm	168.3 mm
8 x VL53L7CX	60	15	670.4 mm	495.4 mm
12 x VL53L7CX	60	30	338.1 mm	163.1 mm

Read the first row carefully, because it is the layout people build. Eight VL53L5CX at 45 deg horizontal total exactly 360 deg. That looks closed on a coverage diagram drawn from the centre. It is not: it has eight permanently open wedges, and a cat on a wedge bisector 400 mm from the robot is invisible at every range. Ten sensors close the gap 678 mm out from the skin, twelve at 338 mm, sixteen inside 168 mm.

Any ring with $n * \text{FoV} < 360$ never closes at any range. That includes **every** ring of single-zone VL53L1X sensors a builder would realistically fit: at 19.3 deg horizontal the first ring that closes at all is $n = 19$, and even that puts the tip about 10.4 m away; bounding it within 350 mm takes $n = 38$. The research lane first read this part's 27 deg as horizontal, which gave 14 sensors; it is the diagonal, and the corrected answer is 38. Where tables in research/coverage-geometry.md are indexed at 63 deg, that column is the VL53L5CX diagonal: read the 45 deg horizontal row instead. The 65 deg column in that file is unverified against any ST primary source and must not be used at all.

The platform radius changes where the wedge closes, never whether it closes. Closure depends only on $a > s$, which contains no r . The radius is a linear scale factor on u : moving twelve 45 deg sensors from the 175 mm face radius to the 247.5 mm corner radius moves the tip from 513 mm to 726 mm. Shrinking the robot does not fix an open ring.

Vertical coverage: what the robot runs over

Horizontal geometry decides what the robot can see around itself. Vertical geometry decides what it drives straight over. For a level boresight at mount height h with vertical half-angle av :

```
z_lo(d) = h - d * tan(av) lowest height visible at range d
z_hi(d) = h + d * tan(av) highest height visible at range d
d_floor = h / tan(av) nearest range at which the beam strikes the floor
```

A standing cat occupies 0 to about 250 mm. It is visible only where $z_{lo}(d)$ is below its head. The cat columns below are the share of a 250 mm standing cat inside the beam, computed from the 250 mm catalogue height in data/geometry.json; the floor-strike column is the research-lane figure.

mount h	FoV_v 45 deg, floor strike	cat at 1 ft	cat at 3 ft	cat at 5 ft
100 mm	241 mm	about 90 percent	full	full
200 mm	483 mm	about 70 percent	full	full
800 mm	1931 mm	not visible	not visible	about 33 percent

The headline is the bottom row. **A ring at 800 mm with a 45 deg vertical field of view cannot see a standing cat at 1 ft or 3 ft at all** - the beam floor is still 674 mm up at 1 ft and 421 mm up at 3 ft. That is the obvious "mount them high where the wiring is easy" choice, and it is blind at exactly the ranges where a collision happens. A ring at 100 to 200 mm sees everything from a lying cat upward, at the price of a floor return from 241 mm or 483 mm outward: past that range the software must reject a floor plane rather than trust free space. The sources differ on cat height, 200 mm in the research lane and 250 mm in data/geometry.json; prefer the 250 mm catalogue figure, which the computed appendix uses, and note the conclusion holds at either height.

One level ring cannot both see a 250 mm pet at 300 mm and see a standing human's torso at 3 m. That forces **two tiers**: a low tier near 200 mm for obstacles, pets and the floor plane, and a high tier near 900 to 1100 mm for human torso detection. Height above the floor is the single most reliable human-versus-pet discriminator available, and a single ring does not measure it.

Square versus round, and the 495 mm diagonal

A 350 mm square platform has an inscribed radius of 175 mm at the face centres and a circumscribed radius of 247.487 mm at the corners. **The diagonal across the corners is 494.975 mm, 41.4 percent wider than the face**, and the corner protrudes 72.487 mm beyond the inscribed circle. A design that reasons about "a 350 mm robot" and forgets the 495 mm diagonal clips door frames with its corners.

Face-centre mounting keeps the mount radius at 175 mm but points every blind bisector at a corner, the part of the machine that sticks out furthest. Corner mounting pushes the radius to 247.487 mm, scaling the whole wedge depth up by 41 percent, and aims the bisector at a face centre, which is where the robot drives into things. Corner mounting is strictly worse at equal field of view. A flush sensor on a flat face is also self-occluded to a 180 deg half-plane at best, while a flush tangential sensor on a round body gets a clean 180 deg because a circle curves away from its own tangent. **Make the platform round if the choice is still open**. If it must be square, mount on face centres and protect the corners with a bumper, a compliant skirt, or a 45 deg chamfer.

Cliff sensors: a separate, mandatory, cheap function

Down-canted cliff sensors are not part of the perimeter ring and must not be traded against it. A sensor at height h canted ψ below horizontal has look-ahead $L = h / \tan(\psi)$, flat-floor range $R_f = h / \sin(\psi)$, and a drop of depth D adds $D / \sin(\psi)$. The contrast D / h is independent of ψ , so the cant only trades look-ahead against incidence angle. Below about 20 deg the return on tile or polished wood goes specular and walks away from the receiver, which reads as a phantom cliff (inferred from reflection geometry, not vendor-published).

At $h = 200$ mm and $\psi = 30$ deg the look-ahead is 346 mm, the flat-floor range 400 mm, and a 180 mm stair riser moves the return to 760 mm, so a "greater than 500 mm means cliff" threshold has wide margin both ways. Required look-ahead is stopping distance plus front overhang: at 0.5 m/s that is $175 + 175 = 350$ mm, met at ψ of 30 deg or less. **Step-up** detection is not the mirror case: a 20 mm threshold strip shortens the range by only 40 mm, inside sensor noise, so it needs a level sensor at 40 to 60 mm or wheel-current and pitch sensing.

The recess penalty: the field of view you buy is not the one you mount

Every field of view above assumes an unobstructed aperture. A sensor set back z mm behind the skin, looking out through a hole of half-width a, is clipped to

$$\text{FoV}_{\text{eff}} = 2 * \text{atan}(a / z)$$

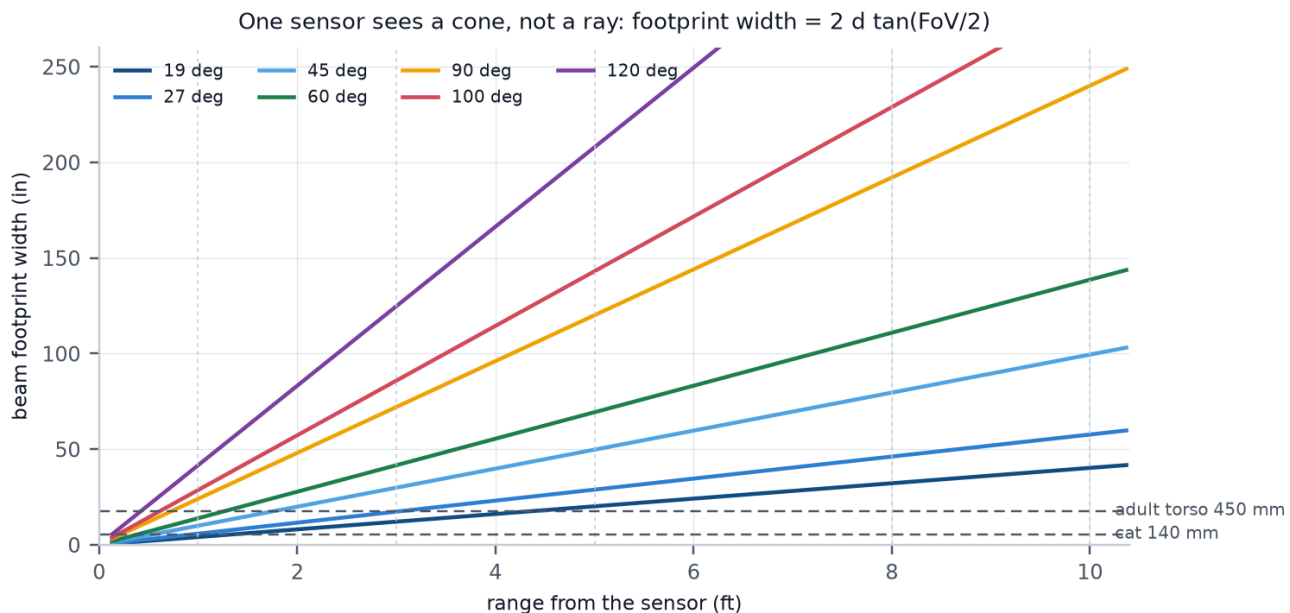
setback z	window half-width 3 mm	5 mm	8 mm	12 mm
3 mm	90.0 deg	118.1 deg	138.9 deg	151.9 deg
5 mm	61.9 deg	90.0 deg	116.0 deg	134.8 deg

setback z	window half-width 3 mm	5 mm	8 mm	12 mm
8 mm	41.1 deg	64.0 deg	90.0 deg	112.6 deg
12 mm	28.1 deg	45.2 deg	67.4 deg	90.0 deg
20 mm	17.1 deg	28.1 deg	43.6 deg	61.9 deg

Read it against the ring counts above. A 60 deg VL53L7CX behind a 5 mm half-width window survives 8 mm of setback, at 64.0 deg, and is destroyed by 12 mm, at 45.2 deg - which moves the part from the 60 deg row of the ring table to the 45 deg row and adds sensors. The same 12 mm of setback clips a 120 deg module to 45.2 deg, turning a four-sensor design into a twelve-sensor problem. **Every millimetre of recess must be budgeted before the ring is sized**, because the clip is silent: the part still ranges correctly, it simply stops seeing the edges of its own stated cone.

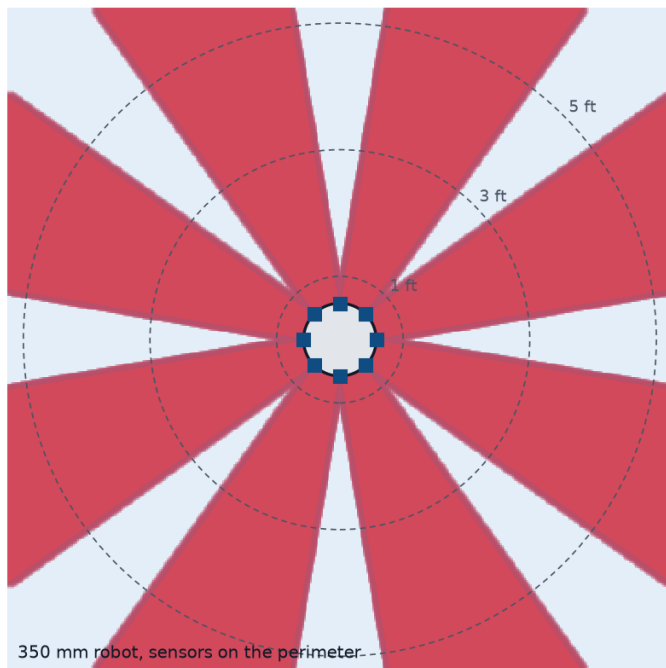
Design rules to carry away

- Size the ring on the **horizontal** field of view. The diagonal is a separately measured number and cannot be converted into coverage.
- $n * \text{FoV} \geq 360$ is necessary and not sufficient. Use $n = \text{ceil}(360 / (\text{FoV} - \text{overlap}))$ with 10 to 20 deg of overlap.
- The blind wedge is the binding constraint, not the ring count: $u = r * \sin(a) / \sin(a - s)$, and blind depth scales as $1/\text{overlap}$.
- Carry these four: eight 45 deg sensors never close; ten close 678 mm out; twelve close 338 mm out; eight 60 deg sensors close 495 mm out.
- Platform radius sets **where** the wedge closes, never **whether**.
- Mount the obstacle and pet ring at about 200 mm with a vertical field of view of 45 deg or more. A ring at 800 mm is blind to a cat inside 5 ft.
- Two tiers are mandatory for human-versus-pet discrimination, because height above the floor is the discriminator.
- A square platform is 495 mm across the diagonal. Mount on face centres, protect the corners, or make the robot round.
- Fit dedicated cliff sensors at about 200 mm high, canted 25 to 35 deg, with a narrow vertical field of view. They are cheap and they are not optional.
- Budget the recess before sizing the ring. $\text{FoV}_{\text{eff}} = 2 * \text{atan}(a / z)$: 12 mm of setback behind a 5 mm half-width window clips any wider part to 45.2 deg.



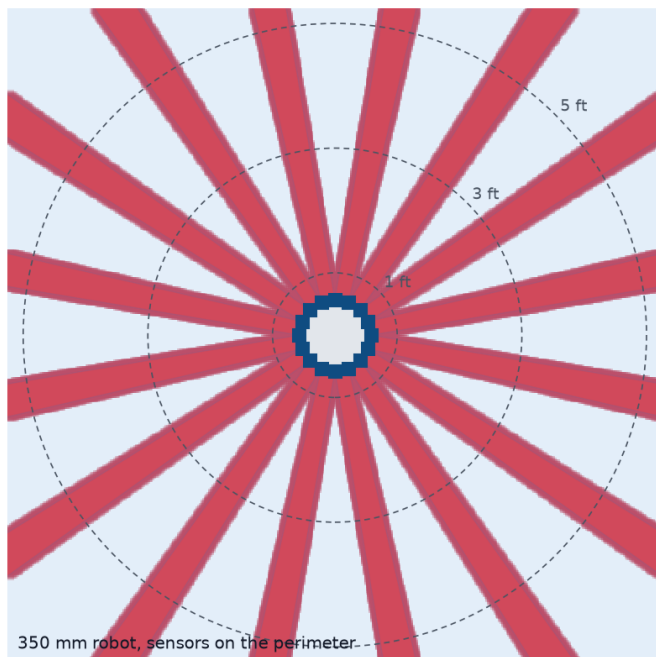
Beam footprint width against range. A sensor does not see a ray; it sees a cone, and at 10 ft a 63 deg cone is 3.7 m wide.

8 x VL53L1X (19 deg horizontal) - the naive ring
blind wedges NEVER close: $8 \times 19 \text{ deg} = 152 \text{ deg} < 360 \text{ deg}$ | 39% covered



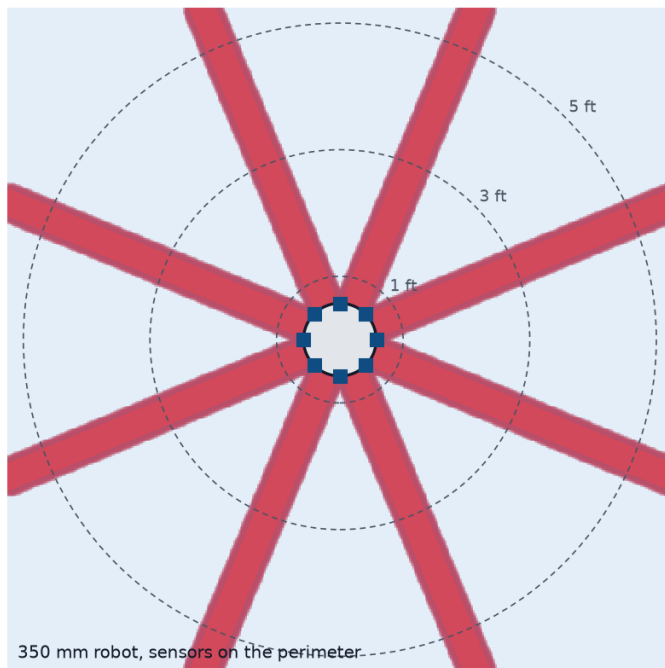
Eight VL53L1X at their true 19 deg horizontal cone cover 152 deg of 360 deg. Most of the perimeter is blind at every range.

16 x VL53L1X (19 deg horizontal) - still 56 deg short
blind wedges NEVER close: $16 \times 19 \text{ deg} = 304 \text{ deg} < 360 \text{ deg}$ | 70% covered



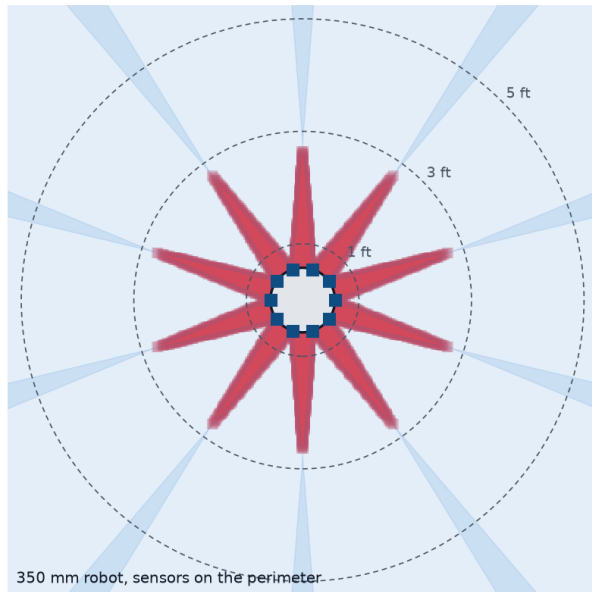
Sixteen of them still total only 304 deg. A ring of single-zone ST parts cannot be closed at any sane count.

8 x VL53L5CX (45 deg H) - exactly 360 deg, no margin
 blind wedges NEVER close: $8 \times 45 \text{ deg} = 360 \text{ deg} < 360 \text{ deg}$ | 84% covered



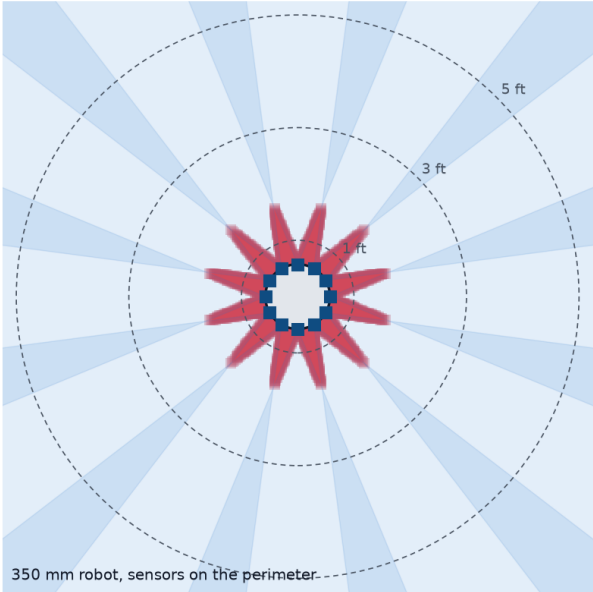
Eight VL53L5CX at 45 deg horizontal total exactly 360 deg. Exactly is not enough: the wedges never close.

10 x VL53L5CX (45 deg H) - 90 deg of overlap
 blind wedges close 679 mm (2.2 ft) past the edge | 96% of the 10.5 ft square covered



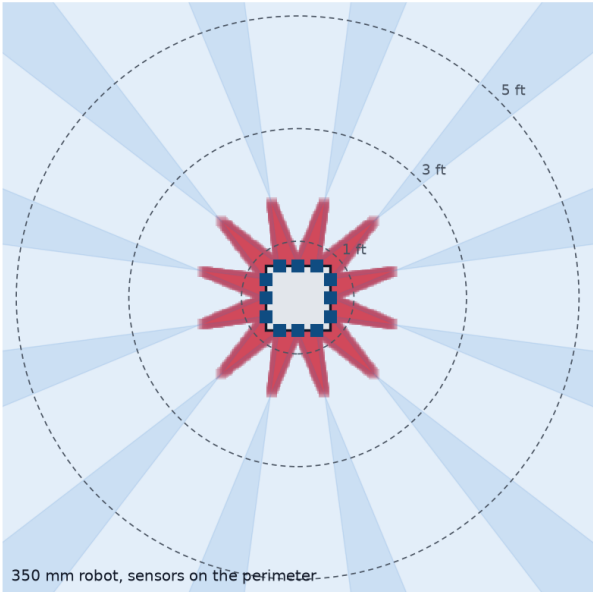
Ten gives 90 deg of overlap and closes the wedges 678 mm out - still beyond the robot at walking pace.

12 x VL53L5CX (45 deg H) - the recommended ToF ring
 blind wedges close 338 mm (1.1 ft) past the edge | 98% of the 10.5 ft square covered



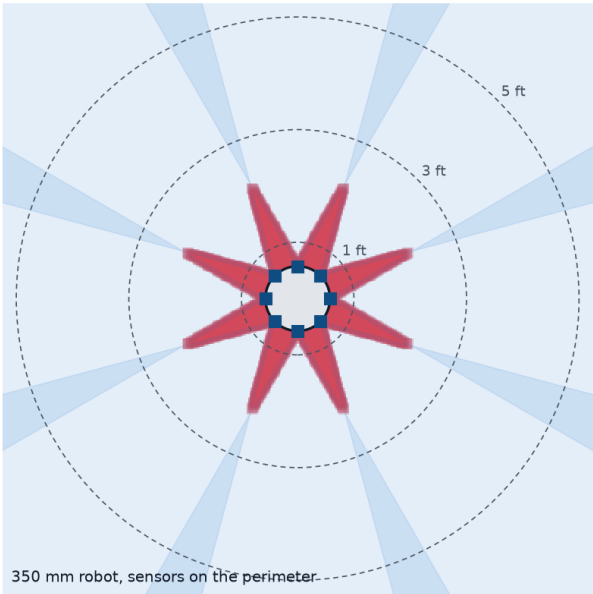
Twelve closes them 338 mm past the edge. This is the ring the recommendation is built on.

12 x 45 deg on a 350 mm SQUARE platform
 blind wedges close 338 mm (1.1 ft) past the edge | 98% of the 10.5 ft square covered

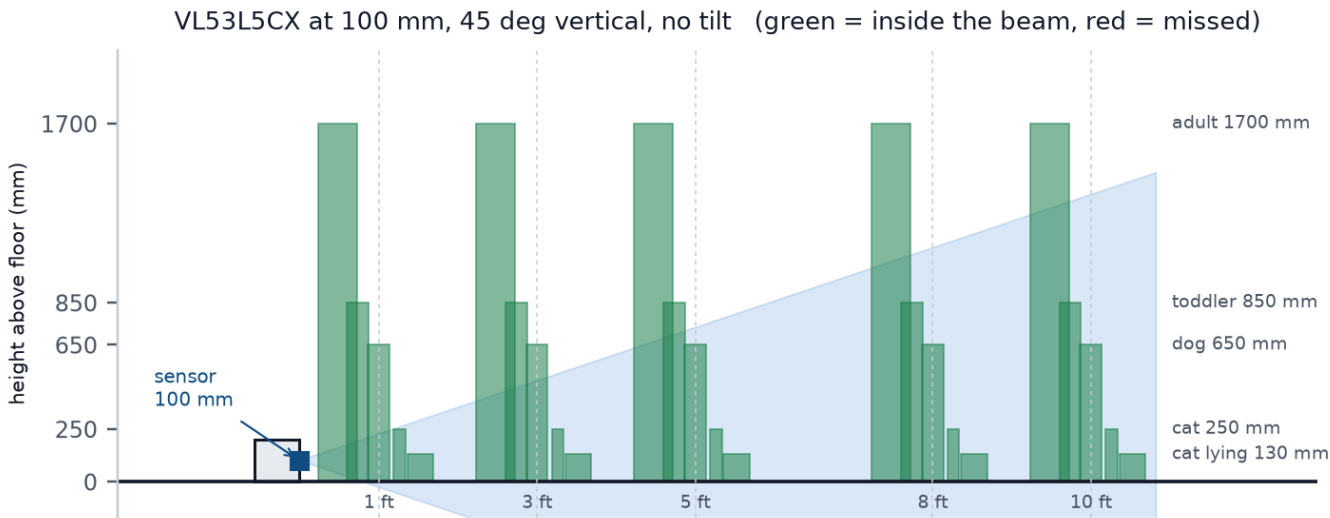


The same twelve on a 350 mm SQUARE platform. The corners push four sensors outboard and skew the wedges.

8 x VL53L7CX (60 deg H) - the cheaper closed ring
 blind wedges close 495 mm (1.6 ft) past the edge | 97% of the 10.5 ft square covered

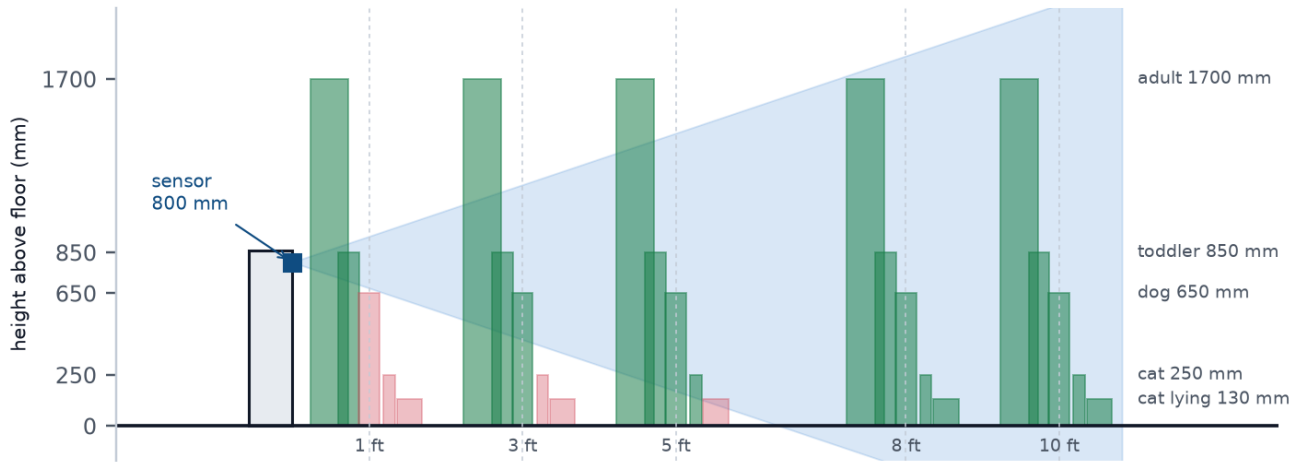


Eight VL53L7CX at 60 deg horizontal close 495 mm out with four fewer parts, four fewer apertures and four fewer addresses.



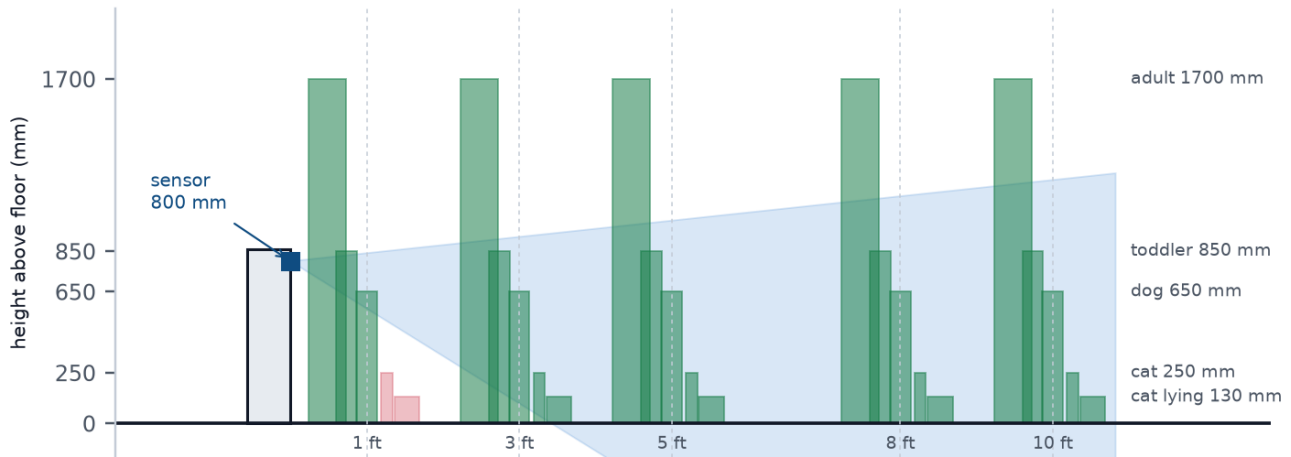
Ring at 100 mm, 45 deg vertical. Everything from a lying cat upward is in the beam, at the price of constant floor return.

sensor at 800 mm, 45 deg vertical field of view, 0 deg down-tilt (green = inside the beam, red = missed)



The same sensor at 800 mm. A standing cat at 1 ft is under the beam entirely - this is what a robot runs over.

sensor at 800 mm, 45 deg vertical field of view, 15 deg down-tilt (green = inside the beam, red = missed)



Adding 15 deg of down-tilt at 800 mm buys back the near field and costs the top of the beam.

How each technology actually works, and what that costs you

Each candidate modality fails in a different place for a different physical reason. Each section below gives the mechanism, separates what is measured from what is inferred, and names the one advantage and the one defeat. Prices are USD list, one unit, read 2026-09-12. Geometry results are cited from `coverage-geometry.md`, not re-derived here.

SPAD direct time of flight, single zone and multizone

A vertical-cavity laser fires a short 940 nm pulse and an array of single-photon avalanche diodes histograms the arrival times of the returning photons; the histogram peak is the range. Three photon rates decide everything: signal rate, proportional to $\rho * f_{\text{fill}} / d^2$, where ρ is reflectance at 940 nm and f_{fill} is the fraction of the cone the target covers; ambient rate, independent of distance, quoted by ST in kcps/SPAD; and dark count rate, negligible indoors. The part measures range per zone and nothing else - height, width, rigidity and class are all inferred from the pattern of zone returns. In the signal-limited case SNR goes as the square root of signal, so range scales as $\sqrt{\rho}$.

ST's own dark table confirms that law: the VL53L1X reaches 360 cm on an 88 percent white target and 170 cm on a 17 percent grey one, long distance mode, 100 ms timing budget, target filling the 27 deg diagonal field ([DocID031281 Rev 3, Table 6](#), datasheet-verified). Grey costs 53 percent of the rated range, and ST's VL53L5CX footnote makes it worse: that 17 percent Munsell chart "measured 13 % in IR at 940 nm". Fill factor compounds it. Every ST range table except VL53L1X Table 9 specifies a target that fills the field of view, and once a target is smaller than the cone its signal falls as $1/d^4$ - the fourth-power law radar obeys. The model $d_{\text{small}} = \sqrt{d_{\text{cross}} * d_{\text{full}}}$ puts the advertised 4 m VL53L1X at about 1.16 m on a 40 x 400 mm chair leg of 88 percent white in the dark, and 0.52 m on the same leg at 200 kcps/SPAD, which is ST's roughly 100 kLux sunbeam case (computed). Fields of view, horizontal x vertical: VL53L1X about 19 deg horizontal (27 deg diagonal), VL53L5CX and VL53L8CX 45 x 45 deg, VL53L7CX 60 x 60 deg.

- Best at: metric range at the lowest latency here, 52-147 ms in the stop path, against 167-200 ms for scanning lidar and 255-380 ms for a thermal array.
- Defeated by: dark matte surfaces. Against black vinyl the VL53L5CX returned 18.8 percent valid measurements at 25 cm (peer-reviewed measured, Caroleo, Albin and Maiolino, [Sensors 26\(5\):1639](#)), and the failure appears as "no return", which a naive filter reads as clear floor.

Scanning lidar

One rangefinder - direct time of flight on the RPLIDAR C1, triangulation on the A1 - is spun behind a mirror at 8-12 Hz on the C1 and 1-10 Hz on the A1 (datasheet-verified), so all emitter power goes into one narrow beam at a time instead of flooding a cone. It measures range and bearing in a single horizontal plane; everything above and below it is unmeasured, not merely unclassified. The concentrated beam shows up as reflectance tolerance: the C1 is specified at 12 m on a white object under 70 percent reflection and 6 m on a black object under 10 percent (datasheet-verified, [Slamtec C1 rev 1.1](#)), so a 7x reflectance change costs 2x range where $\sqrt{\rho}$ predicts 4.5 m. Its ambient limit is 40,000 lux, against a VL53L1X collapse near a 25,000 lux equivalent - the only part here that tolerates an indoor sunbeam.

- Best at: dense 360 deg range for the money - 0.72 deg resolution is 38 mm of arc at 10 ft, about 6.5 hits across a cat (computed).
- Defeated by: the planar slice. No mounting height sees a lying cat at 120 mm, a human torso at 1000 mm and a table top at 720 mm at once.

mmWave FMCW radar at 24 and 60 GHz

The transmitter sweeps a linear chirp, the echo mixes with the outgoing sweep, and the beat frequency is proportional to range while phase change across chirps gives radial velocity. Range resolution depends on bandwidth alone, $dR = c / (2B)$: the 24.00-24.25 GHz ISM allocation is 250 MHz wide and gives a 600 mm range cell - 1.7 robot widths - while a 4 GHz sweep at 60 GHz gives 37.5 mm (computed; the HLK-LD2410 manual states 250 MHz and 0.75 m resolution, datasheet-verified). Radar measures range, radial velocity and, with a MIMO array, coarse angle; species is inferred from a micro-Doppler spectrogram and a trained classifier.

Detection range follows $P_r = P_t * G_t * G_r * \lambda^2 * \sigma / ((4\pi)^3 R^4 L)$, so $R_{\max}$ scales as $\sigma^{(1/4)}$ and a 15 dB cross-section deficit costs a factor of 2.37 in range, not 30. A module rated 12 m on an adult still reaches 7.27 m against a 0.05 m² pet and 4.86 m against a pessimistic 0.01 m² pet (computed). The common claim of 3.5-4 m on a cat needs about a 20 dB deficit; the only figure the sources put behind a cat is a geometric estimate of 11.3 dB, and the range lane calls 10-15 dB better supported than the pessimistic 15-20 dB. The files also disagree on human cross-section: the range lane quotes a 79 GHz median of -11.1 dBsm, spread -20.7 to -4.8; the discrimination lane retrieved both cited sources, found neither contains those numbers, and withdrew them for JRC78619's measured -6.86 to -3.51 dBsm, averaged over 76-81 GHz in an anechoic chamber at 3.4 m. Prefer the withdrawal: it is a primary-source re-check, not a second assertion. No measured cat or small-dog cross-section at either band exists in the sources.

- Best at: radial velocity, in the dark, through a chair back.
- Defeated by: its own ego-motion. Every clutter filter is a moving-target indicator that assumes a static sensor, and on a driving robot every wall acquires Doppler. Even stationary, conventional CFAR after that filter detected quasi-static humans only 51-79 percent of the time (paper-verified, [arXiv 2602.14001](#)).

Thermal infrared arrays

A microbolometer or thermopile array images the 8-14 um long-wave band, and each pixel reports the apparent radiant temperature of whatever fills it. "Alive" and "human" are inferred from contrast, shape and aspect ratio. The governing limit is pixel subtense: nothing smaller than one pixel footprint can be resolved. An AMG8833 is 8x8 over 60 x 60 deg, so each pixel is 7.5 deg and covers 440 mm at 10 ft, making a 450 mm human torso 1.02 pixels wide; an MLX90640-BAB is 32x24 over 55 x 35 deg, 1.72 deg and 99 mm at the same range (computed from datasheet fields of view). Below one pixel of fill the reading dilutes as $T_{\text{apparent}} \approx f * T_{\text{target}} + (1 - f) * T_{\text{background}}$, so a cat that has gone sub-pixel on the AMG8833 at 10 ft reads about 2 K above room against the 8 K true contrast of a clothed body at 30 deg C in a 22 deg C room (computed). Fur compounds that: a long-coated dog's femoral region measures 28.14 +/- 0.31 deg C in a 21 deg C room (paper-verified, [Animals 12\(6\):789](#)), a 6 K contrast against a 22 deg C room that falls to 2 K against a 26 deg C summer room.

- Best at: seeing a motionless warm body. A sleeping cat is invisible to radar clutter filters and invisible to PIR, and plain to thermal.
- Defeated by: a warm room and a window. At 35 deg C ambient the contrast falls inside the MLX90640's own +/-2 deg C accuracy band, and long-wave IR does not pass ordinary glass, so the part needs a germanium aperture or an open hole in the shell.

Single-point thermopiles

A single thermopile junction integrates long-wave IR arriving through a wide aperture and outputs one temperature, or a presence flag derived from it. Unlike PIR it is DC-coupled, so it holds its reading on a motionless body. It measures one scalar for the whole field: no position, no range, no shape. ST's STHS34PF80 states its range with the condition attached, which is unusually honest - "Reach up to 4 meters without lens for objects measuring 70 x 25 cm²", 80 deg field of view (datasheet-verified, DS13916 Rev 2). Scaled by the square root of area, a 25 x 20 cm cat comes to about 2.14 m before any allowance for coat temperature (computed, then inferred).

- Best at: static presence for \$14.95 and microamps.
- Defeated by: having one number. It cannot separate a person from a radiator, or say where either one is.

PIR

A pyroelectric element behind a multi-facet Fresnel lens responds to the rate of change of incident long-wave flux, and fires when a thermal contrast crosses a facet boundary. The output is one scalar voltage: no image, no range, no direction. It is least sensitive to radial motion, so a robot driving straight at a standing person is its worst geometry. Alarm-industry "pet immunity" is a lens cut that removes the lowest floor-looking beams, not classification, and it works only because the unit sits 2.2-2.75 m up (vendor-page-verified, Bosch ISC-BPR2-WP12). At 1.0 m on a robot the geometry inverts: the pet stays in the beam and the person's torso rises above it.

- Best at: cost and power as a wake-up trigger, from \$1.95.
- Defeated by: structural blindness to a still target, and by ego-motion - on a moving platform every scene changes, so the differential principle fires continuously.

Ultrasonic

A piezo transducer emits a roughly 40 kHz burst and times the echo; range is half the round trip times the speed of sound. It measures range along a wide cone with no bearing at all, and being acoustic it does not care whether a surface is optically black, clear or mirrored. That is its whole reason to be aboard: a ToF ranging through a glass door returns the distance to the pane.

- Best at: transparent and specular targets, for \$3.95.
- Defeated by: soft targets and interleave latency. Most HC-SR04 units are unreliable on humans beyond about 1 m, below a 50 percent success rate, and crosstalk forces 33 ms between units, so eight sensors round-robin take 264 ms per sweep - about 3.8 Hz, far too slow for a stop path (vendor-page-verified).

Camera plus on-device model

A lens forms a visible image on a CMOS sensor and a convolutional network run on-device emits class labels and boxes. Nothing metric is measured: range, height and even presence are inferred from pixels. The decisive fact is a dataset one - COCO carries person, cat and dog as separate classes, so a stock detector discriminates human from pet from object with no custom labelling. Compute sets the tier: the TFLite-Micro person detector is a 325 KB binary person/not-person model at about 86 percent on Visual Wake Words (paper-verified), while a Pi 5 runs a YOLO-nano class model at about 7-8 FPS on CPU and, with a \$70 Hailo-8L, YOLOv8n at 136.7 FPS at batch 8 (vendor-page-verified).

- Best at: emitting a class label at all. Nothing else here answers "cat or dog".
- Defeated by: darkness. A camera in an unlit room returns nothing, and near-IR illumination is a different product with a different privacy argument.

Summary

Modality	Measures directly	Can never tell you	Cost per unit	The failure that ends the argument
SPAD dToF single zone	Range in one cone	Shape, class, anything outside the cone	\$14.95	Black matte target reads as empty space
SPAD dToF multizone	Range per zone, 8x8	Temperature, velocity, class	\$19.95-24.95	Same, plus fill-factor range collapse on small targets
Scanning lidar	Range and bearing in one plane	Anything above or below the plane	\$69-160	The lying cat below the slice, the table top above it
24 GHz FMCW radar	Presence and coarse range	Position inside a 600 mm range cell	\$8.90-13.90	600 mm cell, and it detects the neighbour through plasterboard
60 GHz FMCW radar	Range, radial velocity, coarse angle	Species without a trained classifier	\$29.00	Ego-motion Doppler destroys clutter rejection on a moving robot
Thermal IR array	Apparent temperature per pixel	Range, and shape below one pixel of fill	\$44.95-74.95	Warm room collapses contrast into the +/-2 deg C error bar
Single-point thermopile	One apparent temperature	Position, count, class	\$14.95	One scalar cannot separate a person from a radiator
PIR	Rate of change of IR flux	Anything about a still target	\$1.95-9.95	Structural blindness to a motionless person

Modality	Measures directly	Can never tell you	Cost per unit	The failure that ends the argument
Ultrasonic	Range along a wide cone	Bearing, class, soft targets past 1 m	\$3.95-29.95	264 ms for an eight-sensor sweep, too slow to stop on
Camera plus model	Class label and bounding box	Metric range, anything in the dark	\$14.90-150	No light, no detection

Read the last column as the design brief: each entry is a fail-to-danger mode with one physical cause, so every one of them must be covered by a second modality whose physics is different.

Time-of-flight catalogue: every Adafruit part

Adafruit sells no multizone time-of-flight sensor of any kind. Every ToF part in the store is a single-zone rangefinder that returns one number, and the one part that returns a map - the Slamtec RPLIDAR A1 - is a spinning triangulation scanner, not a ToF imager. That single fact governs the whole chapter: a 350 mm robot built from the Adafruit ToF shelf is a tiling problem, and the parts that would solve it in one purchase have to be bought somewhere else.

Nothing in this catalogue classifies. Every part here returns a scalar distance, a scalar proximity count, or a planar point cloud, so a human at 1.5 m and a wall at 1.5 m are the same reading. Human-versus-pet-versus-object has to come from a different modality or from software.

All prices, stock counts and status strings in this chapter were read from adafruit.com on 2026-09-12. Chip-level numbers are datasheet-verified unless the line says otherwise; the ultrasonic figures, the LIDAR-Lite v4 figures and the VL6180X ranging cone are vendor-page figures.

ST single-zone parts

Five ST FlightSense breakouts, all in stock, none discontinued, all with STEMMA QT. Product 3317 carries the STEMMA QT connectors even though the product title does not say so.

PID	Part	Price	Emitt er	Field of view (horizontal)	Condition
3316	VL6180X	\$13.95	850 nm	25 deg ranging cone	vendor-stated (Pololu); ST gives no numeric ranging cone
3317	VL53L0X	\$14.95	940 nm	25 deg system FoV, fixed	datasheet-verified, DocID029104 Rev 1 sec 5.1
3967	VL53L1X	\$14.95	940 nm	about 19 deg (27 deg is the diagonal)	27 deg diagonal is datasheet-verified; 19 deg horizontal is inferred from the square array
5396	VL53L4CD	\$14.95	940 nm	18 deg detection volume	datasheet-verified at a 1000 mm white-88 % target; 22 deg at 100 mm
5425	VL53L4CX	\$14.95	940 nm	18 deg detection volume	datasheet-verified at a 1000 mm white-88 % target; 22 deg at 100 mm

Volume pricing on the four VL53 parts is \$13.46 at 10-99 units and \$11.96 at 100+; the VL6180X is \$12.56 and \$11.16 at the same breaks.

Maximum range, with the reflectance and the ambient-light condition that make each number mean something. Every VL53L0X, VL53L4CD and VL53L4CX row is a 33 ms timing budget; the VL53L1X rows are a 100 ms budget; the VL6180X rows are its worst-case table, measured in an integrating sphere against 80 x 80 mm targets at an SNR limit of 0.1.

Chip and mode	Target	Dark or indoor, no IR	Bright ambient	Ambient condition
VL53L0X	white 88 %	200 cm typ (long-range profile)	80 cm	outdoor overcast, 10 kcps/SPAD = 5 kLux
VL53L0X	grey 17 %	80 cm typ	50 cm	as above
VL53L1X long	white 88 %	360 cm	73 cm	200 kcps/SPAD, direct sun on the sensor
VL53L1X long	grey 17 %	170 cm	68 cm	as above
VL53L1X short	grey 17 %	130 cm	120 cm	as above
VL53L4CD	white 88 %	1200 mm at 90 % detection	550 mm at 90 %	outdoor overcast, 5 kLux
VL53L4CD	grey 17 %	450 mm at 90 % detection	400 mm at 90 %	as above
VL53L4CX	white 88 %	5000 mm at 90 % detection	1600 mm at 90 %	as above
VL53L4CX	grey 17 %	2100 mm at 90 % detection	1100 mm at 90 %	as above
VL6180X	grey 17 %	> 100 mm	> 60 mm	5 kLux diffuse halogen, about 10-15 kLux sun
VL6180X	3 % black	> 100 mm	> 40 mm	as above

VL53L0X (3317). Do not specify it for new work. The VL53L4CD costs the same, has better accuracy, 100 Hz instead of about 30 Hz, and a wider temperature range; the VL53L0X keeps only the wider 25 deg cone and a smaller code footprint. Its advertised 2 m is a white-88 % figure the datasheet itself restricts to "dark conditions (no IR)".

VL53L1X (3967). The most interesting ST part for a robot, because the region of interest can be moved across the 16x16 SPAD array: a 4x4 window swept across it builds a coarse depth image from one sensor, at the cost of range (grey-17 %

drops to 45 cm) and of rate (every window position is a separate cycle). Its short distance mode is the only setting in the ST lane that is nearly immune to sunlight, 130 cm dark and 120 cm on grey-17 % under direct sun through a window. It is not for long range in daylight: long mode collapses about 5x under that condition.

VL53L4CD (5396). A 100 Hz short-range beam, best used low on the chassis for pets and low furniture. The store headline of 1300 mm is a 50 %-detection-rate figure against a white chart indoors; against a dark-grey target at 90 % detection it is 450 mm. It is not a room sensor.

VL53L4CX (5425). The best single-zone choice for forward obstacle beams. 2100 mm against a grey-17 % target indoors at 90 % detection is the most useful "will I see the black cat" number in the ST lane, and the histogram algorithm reports more than one target inside the cone, so one beam returns both a chair back at 1.2 m and the wall behind it. It is not for an Arduino Uno: the driver needs about 50 KB of flash, and Adafruit publishes no Arduino library for it.

VL6180X (3316). A proximity sensor, not a rangefinder: 850 nm instead of 940 nm, 1.7 mA average instead of 16-22 mA, a calibrated lux output, and a guaranteed range of only 0 to 100 mm. It is useless on a perimeter - 100 mm is inside the robot's own bumper line - and excellent pointed down as a cliff and stair-edge detector. Sitting at 850 nm behind its own bandpass filter, it does not interfere with a 940 nm ring (inferred from the filter, not a stated spec).

ST multizone parts, none of which Adafruit sells

A store search for VL53L5CX, VL53L7CX or VL53L8 returns no exact match and falls back to the five single-zone parts. Adafruit designed a VL53LCX STEMMA QT breakout in September 2021 and ran an EYE ON NPI segment on the VL53L5CX, but no product reached the store.

Chip	Zones	FoV H x V	Diagonal	Where to buy	Price read 2026-09-12
VL53L5CX	4x4 or 8x8	45 x 45 deg	63 deg	SparkFun SEN-18642; Pimoroni	\$32.50 (backorder); GBP 16.25 in stock
VL53L7CX	4x4 or 8x8	60 x 60 deg	90 deg	not Adafruit	not read
VL53L8CX	4x4 or 8x8	45 x 45 deg	65 deg	not Adafruit	not read

Quote the horizontal figure and say which one it is. ST prints the diagonal on the front page of every datasheet, and the vendor pages repeat it, but Table 2 of each datasheet publishes the horizontal and vertical figures separately - 45 x 45 deg behind the VL53L5CX "63 deg", 60 x 60 deg behind the VL53L7CX "90 deg". For the VL53L1X, ST publishes only the diagonal, so its 19 deg horizontal is inferred as 27 deg divided by the square root of two. A ring is closed with the horizontal number: at zero overlap that is six VL53L7CX, eight VL53L5CX or VL53L8CX, against roughly nineteen VL53L1X beams. Those are bare tiling counts; coverage-geometry.md computes the overlap a ring needs to actually close its wedges. Sizing from the diagonal overstates the multizone advantage.

The VL53L5CX "up to 400 cm" headline needs its conditions. DS13754 Rev 5 specifies maximum range at a 90 % detection rate, target filling the field of view, 23 deg C, no cover glass, with 5 klux defined as 2 W/m2 of 940 nm target irradiance.

Mode and rate	Target and zone	Dark	5 klux ambient
4x4, 30 Hz	white 88 %, inner	4000 mm typ	1700 mm typ
4x4, 30 Hz	grey 17 %, inner	2400 mm typ	1000 mm typ
8x8, 15 Hz	white 88 %, inner	3500 mm typ	1100 mm typ
8x8, 15 Hz	grey 17 %, corner	1100 mm typ / 600 mm min	650 mm typ / 400 mm min

That last cell - a dark target in a corner zone of a sunlit room at 650 mm - is the number to design to. Note also that the performance tables are specified at 30 Hz for 4x4 and 15 Hz for 8x8; the vendor pages advertise "60 Hz" without saying which map it applies to.

Non-ST direct time of flight: the ams-OSRAM TMF family

Adafruit ships two TMF parts, both single-zone. The multizone TMF882x family appeared as a prototype on the Adafruit blog in August 2024 marked "coming soon" and has no product page.

PID	Part	Price	Stock	Receiver FoV	I2C address
6522	TMF8801	\$9.95	75	37 deg FWHM short, 24 deg FWHM long	0x41
6501	TMF8806	\$12.50	33	52 deg short, 30 deg long (default stack)	0x41

The TMF8801 datasheet publishes a full range-versus-reflectivity-versus-ambient matrix: on an 18 % grey card, 2500 mm under 350 lux fluorescent, 1700 mm at a 2.5 klux sunlight equivalent, 1250 mm at 10 klux and 550 mm at 100 klux. All rows use a 1.5 m x 1.5 m target with light on the target only. Minimum distance 20 mm on an 18 % grey card, about 30 Hz, 27 mA at 30 Hz.

The TMF8806 is the better default of the two. Its conditioned table, from DS001097 v4-00 Table 5:

Mode and iterations	Ambient	Target	Target extent	Max distance
10 m, 1500 k (150 ms)	350 lux fluorescent	90 % white	white wall	10000 mm
5 m, 450 k (33 ms)	350 lux fluorescent	90 % white	1.5 x 1.5 m	5000 mm
5 m, 450 k (33 ms)	350 lux fluorescent	18 % grey	1.5 x 1.5 m	3250 mm
2.5 m, 900 k (33 ms)	830 lux halogen (5 klux sun)	18 % grey	1.5 x 1.5 m	1170 mm
2.5 m, 900 k, tape on glass	170 lux halogen (1 klux sun)	18 % grey	1.5 x 1.5 m	1500 mm

Two things in that table matter more than the headline. The 10 m row is a **white wall** number, not an obstacle number - a wall fills the whole 30 deg cone and a 1.5 m card at 10 m does not, so 10 m must never be quoted as a detection range. And one layer of Scotch Magic Tape on the cover glass costs 340 mm of range on an 18 % grey target at the 1 klux condition, from 1840 mm to 1500 mm. A floor robot's front window collects dust and pet hair continuously.

Adafruit's own store page describes the shipped TMF8806 board as "about 30 degrees short range (under 200 mm) and 21 degrees above" and claims +/-5 % accuracy. That contradicts the datasheet, which gives 52 deg short and 30 deg long for the default optical stack and a tiered accuracy of +/-3 % beyond 200 mm. Prefer the datasheet; Adafruit may be describing their specific cover-glass stack. The same page admits "we weren't able to get results past 5 meter".

The multizone TMF882x, which Adafruit does not sell. TMF8820 is 3x3, TMF8821 adds 4x4 and 3x6, TMF8828 adds 8x8 for 64 zones, with the field of view selected by spad_map_id - 33 x 32 deg (45 deg diagonal) or 41 x 52 deg (63 deg diagonal) among five choices. DS000693 v5-00 conditions the range on zone position, ambient light and iteration count, and the three maps are **not taken at the same iteration count**: the 3x3 and 4x4 tables are 550 k iterations, the 8x8 table only 125 k, so the 8x8 numbers are depressed by a quarter of the integration budget as well as by the smaller zones. On an 18 % grey card, light on the target only:

Map, FoV, iterations, rate	Zone	350 lux LED	700 lux halogen (5 klux sun)	1400 lux halogen (10 klux sun)
3x3, 33 x 32 deg, 550 k, 30 Hz	centre	5000 mm	2000 mm	1500 mm
3x3, 33 x 32 deg, 550 k, 30 Hz	corner	4000 mm	1400 mm	1200 mm
4x4, 41 x 52 deg, 550 k, 15 Hz	centre	4000 mm	1500 mm	1400 mm
4x4, 41 x 52 deg, 550 k, 15 Hz	corner	1400 mm	700 mm	600 mm
8x8, 41 x 52 deg, 125 k, 15 Hz	centre	2000 mm	1000 mm	800 mm
8x8, 41 x 52 deg, 125 k, 15 Hz	corner	500 mm	300 mm	200 mm

The 8x8 corner zones collapse to 200-500 mm on an 18 % grey target, so 64-zone mode is a near-field mode, not a room-scanning mode. Minimum detection distance is 10 mm on an 18 % grey target and 25 mm on a 90 % white one; power is 141 mW at 30 Hz; the default I2C address is 0x41 again, so the family collides with the two TMF parts Adafruit does ship.

Single-point and scanning lidar modules

PID	Part	Price	Stock	Range and condition	Beam
4010	RPLIDAR A1	\$99.95	19	0.15-6 m (R4) or 0.15-12 m (R5), white objects	360 deg
3978	TFmini	\$44.95	No longer stocked	0.3-12 m at 90 % white indoors; 0.3-3 m at 10 % black or 100 klux glare	2.3 deg full
4058	LIDAR-Lite v3	\$129.95	12, max 2 per customer	40 m at a 70 % reflective target	0.46 deg
4441	LIDAR-Lite v4	\$59.95	15, max 2 per customer	5 cm to 10 m, reflectivity condition unverified	4.77 deg

RPLIDAR A1. The only rotating product Adafruit sells and the strongest single perimeter answer in the catalogue: one unit, 1 deg angular resolution, 8000 samples/s, 5.5 Hz typical. It is laser triangulation, not time of flight, so resolution degrades with distance and the range is conditioned on white objects. It sees exactly one plane, so a 200 mm cat is invisible unless the scan plane sits below 200 mm, and a 700 mm table overhang is invisible to a plane at 100 mm. It needs two independent 5 V rails

and adds 170 g. The datasheet's own point-density figures do not reconcile: 8000 Hz at 5.5 Hz implies about 1450 points per revolution, but both the 1 deg resolution and the 5.5 Hz rate are specified at 360 samples per scan. Quote 1 deg and 360 points.

Garmin LIDAR-Lite v3 and v4. Both are the wrong tool here. The v3's 0.46 deg beam lights a 16 mm spot at 2 m and threads straight past a cat's leg to report the wall behind. The v4's 4.77 deg is better but \$59.95 per beam is absurd for a ring, and both are limited to two units per customer, which by itself rules out tiling. Where Adafruit and Garmin disagree, Garmin wins: accuracy is +/-2.5 cm below 5 m and +/-10 cm at and above 5 m, not Adafruit's flat "+/-2.5 cm above 1 m", and 270 Hz is the update rate at a 70 % target while 500 Hz is the repetition-rate maximum.

TFmini, discontinued. Kept here for one table. Its manual publishes the minimum reliable target size as $d = 2 D \tan(1.15 \text{ deg})$: 4 cm at 1 m, 8 cm at 2 m, 20 cm at 5 m. A cat's torso clears that at 2 m; a chair leg or a cat's tail does not. That is the failure mode of every narrow-beam ranger, and a 2.3 deg beam is a 157-sensor problem for 360 deg coverage.

Ultrasonic

Ultrasonic is the only technology here indifferent to optical colour, and the only one that reliably sees clear glass. It is the natural cross-check for a dToF ring, not a replacement.

PID	Part	Price	Stock	Range	Beam
3942	HC-SR04	\$3.95	In stock	2-400 cm, best 10-250 cm	15 deg
4007	RCWL-1601 3V/5V	\$3.95	In stock	2-450 cm, best 10-250 cm	+/-15 to +/-20 deg
4019	US-100	\$6.95	In stock	2-450 cm, best 10-250 cm	under 15 deg
979	MaxBotix LV-EZ0	\$29.95	44	0-254 in (6.45 m), reports from 6 in	widest in the LV-EZ line
982	MaxBotix LV-EZ4	\$28.50	48	0-254 in (6.45 m)	narrowest in the line
4742	RCWL-1601 with I2C	\$3.95	Out of stock	not read	not read
4664	Large ultrasonic with horn, UART	\$28.95	No longer stocked	28-750 cm with horn	40 deg with horn, 75 deg without

The LV-EZ1, LV-EZ2, LV-EZ3, HRLV-EZ0/1/4, MB7092, HR-USB-EZ1 and the panel-mount sonar are all no longer stocked. Of what remains, the LV-EZ0 is the one with published human-detection evidence: MaxBotix states it "can detect people up to approximately 10 feet" and defines the 1-inch dowel beam pattern as the reliable people-detection zone. Against that, a MaxSonar range cycle is 49 ms, so eight sensors round-robin at about 2.5 Hz; 40 kHz transducers on one 350 mm chassis hear each other; fur and wool absorb 40 kHz; and a smooth wall approached at a shallow angle bounces the ping away and reads clear.

The parts people mistake for rangefinders

PID	Part	Price	Stock	Published reach	What it actually is
164	Sharp GP2Y0A21YK0F	\$14.95	65	10-80 cm at a Kodak R-27 90 % white card	analog PSD triangulation
1031	Sharp GP2Y0A02YK0F	\$15.95	17	20-150 cm, same 90 % condition	analog PSD triangulation
1927	Sharp GP2Y0D810Z0F carrier	\$8.95	No longer stocked	20-100 mm	digital high/low, not a distance
3025	Sharp GP2Y0D805Z0F carrier	-	No longer stocked	0.5-5 cm	digital high/low
6064	VCNL4200	\$6.95	Out of stock	0 to 1.5 m, target size not published	reflected-IR count
6491	VCNL4030	\$5.95	In stock	0 to 300 mm	reflected-IR count
4161	VCNL4040	\$5.95	In stock	0 to 200 mm	reflected-IR count
3595	APDS9960	\$7.50	In stock	"a few centimeters"	8-bit proximity plus gesture
5913	TCRT1000	\$4.95	86	millimetres	reflective photo-interrupter

The Sharp analog parts fail for two structural reasons, and one commonly repeated reason that is wrong. The response curve is non-monotonic below the minimum range: output rises to about 3.13 V at 5-6 cm, then falls to 0.4 V at 80 cm, so every far reading has a near twin - about 1.05 V is either 25 cm or 1.6 cm. A robot that drives into a wall gets a reading saying "not close". And the beam width is not published by Sharp, Adafruit or Pololu, so coverage cannot be computed from it. The wrong reason is reflectance sensitivity: these are position-sensitive-detector parts, and Sharp's Figure 2 plots 90 % white and 18 % grey almost coincident across the whole span. Reflectance-blindness is in fact their one advantage over every dToF part

here. Adafruit's "3 V at 10 cm" is also not supportable - Sharp's 1.9 V typical differential puts 10 cm at about 2.3 V.

The VCNL and APDS parts report a reflected-IR count, not a calibrated distance, and their reach depends on how much LED current you burn and how reflective the target is. Adafruit's own VCNL4200 page tempers the 1.5 m datasheet claim to a "practical range 50-100 cm", and Vishay publishes no target size or reflectivity behind the 1.5 m figure. Their legitimate role is cliff detection at the chassis skirt, where every ToF part here fails inside its own minimum range: a VCNL4040 pointed down catches a stair drop-off inside 200 mm, and the TCRT1000 photo-interrupter does the same job at millimetre standoff. The VCNL4010 (PID 466) is no longer stocked. The LTR-329 (5591, \$4.50), LTR-303 (5610, \$4.50) and MAX44009 (6498, \$12.50) have no proximity function at all, but earn a place for another reason: every range figure in this chapter is conditioned on ambient light, and a lux sensor lets the robot detect a sunbeam and derate its own trust accordingly.

What the store page will not tell you

Every VL53 and VL6180X boots at the same address. All five ST parts power up at 0x29 7-bit (0x52 8-bit), and the address is volatile. Adafruit's guide states the trap outright: "You must do this every time you turn on the power, the addresses are not permanent!" The sequence is one microcontroller GPIO per sensor wired to XSHUT; hold all low for 10 ms then high; bring sensor one up alone and assign it an address in 0x30-0x3F; repeat. Every part exposes the call - `Adafruit_VL53L0X::setAddress()`, `VL53L1X_SetI2CAddress()`, `set_address()` in CircuitPython. The practical bus limit is spare GPIOs and the 400 pF bus capacitance a daisy chain of STEMMA QT cables around a 350 mm chassis will approach; 12-16 on one bus is plausible with short cables (inferred - no document read here states a maximum). A TCA9548A multiplexer removes the per-sensor XSHUT GPIO entirely. The same collision hits both TMF parts, which share 0x41.

The timing budget buys range and spends rate. The VL53L1X accepts 20 ms to 1000 ms; 20 ms is short-mode only, 33 ms is the minimum that works in all modes, and 140 ms is required to reach the full 4 m in long mode on a white chart in the dark. The CircuitPython driver accepts only 15, 20, 33, 50, 100, 200 and 500 ms and implements short and long modes only - medium is missing. The VL53L0X exposes four profiles: 20 ms high speed at +/-5 %, 30 ms default, 33 ms long range at 2 m but "only for dark conditions (no IR)", and 200 ms high accuracy at better than +/-3 %. On the TMF8806 the budget is an iteration count: 450 k iterations in 33 ms reaches 3250 mm on an 18 % grey card, 900 k in 66 ms reaches 3750 mm, and the ultra-low-power 10 k setting reaches 665 mm. The budget also sets the ring's update rate, because sensors sharing overlapping space should be time-multiplexed - a 12-sensor ring at a 20-33 ms budget round-robins at roughly 2.5-4 Hz each.

One part needs a firmware blob at every boot. Adafruit's TMF8801 guide states the part "does require a firmware 'patch' on boot" - the host pushes a several-kilobyte image at startup, costing both flash and boot time. Adafruit notes the TMF8806 needs no patch for 5 m mode and that its driver fits on an ATmega328. No ST part here is documented as needing a boot blob in the sources read for this reference; the VL53L4CX constraint is simpler, a 50 KB driver that will not fit an Arduino Uno.

Cover glass is a calibration step, not a cosmetic one. "Crosstalk" in these datasheets means cover-glass optical crosstalk, not sensor-to-sensor interference. The VL53L1X requires RefSPAD, offset and crosstalk calibration whenever a cover glass is added; the VL53L4CX is immune beyond 80 cm and compensates for smudge below it; the TMF882x family needs a window at least 85-90 % transparent at 940 nm.

Glass and mirrors are an unpublished blind spot. No datasheet read for this chapter publishes behaviour against a specular or transparent target - all reflectance data uses diffuse Munsell or grey-card charts. Physically, a mirror at an angle returns nothing to the receiver and clear glass is largely transparent at 850 and 940 nm, so both are usually missed. That is inference from the absence of a spec, not a published result, and it is the strongest argument for keeping one ultrasonic sensor in the design.

mmWave radar catalogue: every DFRobot part

DFRobot sells ten radar-class modules and not one of them is designed to move. Nine are buyable as of 2026-09-12, one is retired, and every human-presence part in the line is an indoor occupancy sensor specified for a fixed mount. The catalogue below is organised by what each module puts on the wire - a bare flag, a range, a range with velocity, or a 60 GHz vital-signs record - because the output decides whether a mobile robot can use it at all.

Five catalogue facts first, to stop wasted searching. There is no DFRobot SKU SEN0611; the storefront resolves it to SEN0609 or SEN0610. Those two are usually quoted the wrong way round: SEN0609 is the 25 m UART part and SEN0610 is the 12 m I2C-and-UART part, confirmed on both product pages. MR24HPC1, MR60BHA2 and MR60FDA2 are Seeed Studio modules DFRobot mentions in blog posts and sells no board based on. DFRobot does not stock the Hi-Link LD2450, the cheap radar that does report target x/y, so nothing here reports a bearing. Two kits, KIT0216-EN (\$89.00) and KIT0234 (\$79.90), contain an unnamed "mmWave Human Presence" module whose SKU is not published.

The two Seeed 60 GHz parts get a paragraph because they are the usual suggestion once the DFRobot line runs out. Both are built on the ADT6101P, which integrates a 57-64 GHz transceiver and a 2T2R microstrip antenna; the module datasheets specify 58-62 GHz, 12 dBm typical, 4 dBi of antenna gain and a -3 dB beam of +/-60 deg in both planes. MR60BHA2 measures breathing and heart rate at a chest distance of 0.4-1.5 m. MR60FDA2 detects a fall out to a maximum radius of 2 m, top-mounted at a hanging height of 2.2-3.0 m, and its datasheet is marked Beta Version. Power is the disqualifier on a battery robot: the ICC table says 600 mA maximum, but the same datasheet's Precautions section requires 3.1-3.5 V with ripple of 50 mV or less and current of 1 A or more. Budget 1 A, not 600 mA.

Class A - bare presence flag

These report one bit. They answer "is something alive-ish in the cone" and nothing else.

SKU	Band and chipset	Moving range	Static range	FoV H x V	Price USD
SEN0395	24 GHz; chipset not published	9 m	9 m (not split)	100 x 40 deg	29.00
SEN0192	10.525 GHz X-band CW Doppler	2-16 m, pot-set	none - cannot see a still target	72 x 36 deg (-3 dB)	8.90
SEN0521	5.8 GHz FMCW and CW	11 m	11 m	120 x 120 deg	retired, listed 0.00

SEN0395 outputs GPIO2 high for present plus a serial \$JYBSS frame. No distance, no bearing, no speed. Factory latencies are 2.5 s to assert and 10 s to clear, reprogrammable in 25 ms steps - 10 s of latched presence on a driving robot is meaningless. Supply 3.6-5 V at 90 mA, 24 x 28 mm, UART 115200. It is the most expensive bare-bit part DFRobot sells, at three times the price of a richer module.

SEN0192 is X-band Doppler, not mmWave, and appears only because the search index returns it. It fires on relative motion, so it cannot see a standing person, and its 48.5 x 63 mm board is large for a 350 mm chassis. SEN0521's page still publishes specifications (11 m, 120 x 120 deg, 22 mA, 20 x 18 mm) but the part is unbuyable. Do not design it in.

The -3 dB note matters. SEN0192 and SEN0306 are the only DFRobot radars publishing beamwidth as an explicit half-power figure; every other angle in this chapter is an unconditioned "beam angle" from a spec table. The two are not comparable, so a ring sized from one is not sized the same way as a ring sized from the other.

Class B - presence plus range

One part sits here: SEN0557, at \$9.90, in stock.

Parameter	Published value	Basis
Band and sweep	24-24.25 GHz, 250 MHz sweep	vendor-page-verified
Detection range	0.75 m to 6 m, moving and static not split	vendor-page-verified
Blind zone	30 cm	vendor-stated as measured
Distance resolution	0.75 m, eight gates	vendor-page-verified
FoV	+/-60 deg; axis not stated , vertical not published	vendor-page-verified
Output	presence bit on GPIO, plus per-gate moving and stationary energy over UART	vendor-page-verified

The per-gate energy vector is the most information-rich cheap output in the catalogue: each 0.75 m gate carries a moving-target energy and a stationary-target energy. That is a crude range profile, and the only place in the DFRobot line where moving and static returns are separated per range cell. Supply 5-12 V at 80 mA average, 7 x 35 mm - the smallest radar footprint DFRobot sells. Its stated 57600 baud default conflicts with the Hi-Link LD2410 datasheet default of 256000; verify on the bench before writing firmware.

The chipset is an inference. The fingerprint - 24-24.25 GHz, 250 MHz sweep, 0.75 m gates, eight gates to 6 m, +/-60 deg, a 256000 baud option - matches the Hi-Link HLK-LD2410 exactly. One source file records that as datasheet-identified, another as inferred. Prefer the cautious reading: DFRobot names no silicon vendor anywhere, so the identification is inferred and the moving-versus-static range split remains not published.

Class C - range plus velocity

This is the useful class, and the cheapest part in it is the best one.

SKU	Band	Moving range	Static range	FoV H x V	Price USD
SEN0691	24-24.25 GHz	11 m	10 m micro-motion	120 x 120 deg	8.90
SEN0610	24 GHz	12 m	8 m	100 x 80 deg	12.90
SEN0609	24 GHz	25 m	16 m	100 x 40 deg	13.90
SEN0306	24 GHz	0.5-20 m, any reflector	n/a - no presence claim	78 x 23 deg (-3 dB)	65.90
SEN0676	77-81 GHz, 4 GHz sweep	n/a - liquid level	0.15-40 m	+/-25 deg, +/-3 deg with lens	59.00

SEN0691 (C4002) reports target state, presence distance and energy 0-99, motion distance, speed, energy and an approach/recede direction, ambient lux, and a per-gate occupancy bitmap. Gates are selectable: 16 of 80 cm, or 26 of 20 cm spanning 5.2 m, with per-gate thresholds 0-99 set independently for motion and presence. Report period is 0.1 s to 25.5 s - DFRobot's example code ships at 1 Hz. Price falls to \$8.30 at 10 or more. The wiki scopes the 11 m and 10 m figures to a named target, "Motion, Micro-Motion/Stationary Human Body", which is more than DFRobot states for most of the line. It still gives no RCS, no target size and no aspect, so 11 m is not a reach against a cat or a chair leg.

SEN0609 and SEN0610 (C4001) report target count, range, signed radial speed and energy, one target maximum, with presence and ranging modes mutually exclusive. Both have a 1.2 m ranging floor - exactly the zone a 350 mm robot cares about - and DFRobot's own C4001 datasheet disclaims the distance verbatim: the sensor is "not specifically designed for distance measurement", the distance is "not calibrated" and "can only be used as a reference". The same datasheet documents a 1 m transition zone, so a configured maximum is soft: set 3 m and targets out to 25 m are still reported with falling probability, while nothing past 25 m is reported at all. The 25 m figure carries the only condition DFRobot puts on it - detection at 25 m needs "a significant movement" and is "depending on the target characteristics" - so it is a best-case, large-motion number. No range accuracy or resolution is published for either.

SEN0306 is the only honest obstacle radar here. It ranges any reflector, static included, at +/-0.1 m accuracy and 0.01 m reporting resolution at 10 Hz, and with the mode pin grounded it emits a 126-line range-FFT magnitude spectrum - the only pre-detection data any DFRobot radar exposes. The costs are a 0.5 m blind zone, over 100 mA, 0-70 degC only, no velocity, and \$65.90.

SEN0676 is a tank-level gauge, listed for completeness. Its 1 mm figure is interpolated sub-bin precision on a specular return from a flat liquid surface that a human body does not give; native resolution from the 4 GHz sweep is 37.5 mm.

Class D - 60 GHz vital signs and fall detection

SEN0623 (C1001), \$29.00, in stock, is the only DFRobot radar that reports anything resembling a class.

Parameter	Published value
Band, power	61-61.5 GHz, 6 dBm
Farthest detection	11 m, condition published : ceiling mount at 2.7 m
Detection angle	100 x 100 deg
Sleep mode, chest	0.4-2.5 m
Breathing and heart rate, chest	0.4-1.5 m
Measurement bands	10-25 breaths/min; 60-100 bpm

Parameter	Published value
Supply	5 V, 100 mA or less; UART 115200

The 11 m figure is the only range in this chapter tied to a published mounting geometry. Two others carry a weaker condition: the C4002's 11 m and 10 m are scoped to a "Motion, Micro-Motion/Stationary Human Body", and the C4001's 25 m is qualified by "a significant movement" and "depending on the target characteristics". Every other range number in Class A, B and C is a bare maximum. No figure anywhere in this chapter, the 11 m included, publishes a target RCS, a reflectivity, a clutter description or an indoor/outdoor qualification, so every range here is an upper bound measured under conditions the vendor declines to state.

The C1001 exposes posture and fall state, static-residency and unmanned timers, install height and tilt configuration, and vital signs. One source file records an x/y track method in the library; another records the API as presence, movement class, moving range and fall bit with no coordinates. Neither reading changes the verdict: the fall algorithm is a posture classifier referenced to a configured downward ceiling geometry, which a horizontally mounted radar on a moving 1 m robot does not have. Its wiki FAQ also says it cannot count multiple individuals, and DFRobot states it is not a certified medical device.

Interface, power and stock

SKU	Interface	Supply V	Current mA	Size mm	Stock
SEN0395	UART 115200 + GPIO	3.6-5	90	24 x 28	in stock
SEN0557	UART 57600 + GPIO	5-12	80	7 x 35	in stock
SEN0609	UART 9600 + IO	3.3-5	not published	26 x 30	in stock
SEN0610	I2C 0x2A/0x2B + UART	3.3-5	not published	22 x 30	in stock
SEN0691	UART 57.6k-1M + OUT	3.6-5.5	not published	22 x 26	in stock
SEN0623	UART 115200 + IO1/IO2	5	100 or less	not published	in stock
SEN0306	UART 57600 TX only	4-8	over 100	34 x 44 x 5	in stock
SEN0192	Gravity digital pin	5 +/-0.25	60 max, 37 typ	48.5 x 63	in stock
SEN0676	UART Modbus	3.5-5	30	35 x 35 x 1.2	in stock

Supply current is not published for any C4001 or C4002 part. That is three of the four parts a robot would actually consider, so a ring's power budget cannot be computed from datasheets and must be measured.

Ring sizing for 360 deg coverage follows from the azimuth beam. With a 120 deg beam, three SEN0691 nominally close the circle and four give overlap: \$26.70 to \$35.60 at the single-unit price of \$8.90, or \$33.20 for four at the 10-or-more price of \$8.30. SEN0557's +/-60 deg gives the same three-or-four count, \$36.80 for four at its 10-or-more price of \$9.20. The 100 deg parts need four with no overlap and five with it: \$64.50 for five SEN0610, \$69.50 for five SEN0609, and \$145.00 for five SEN0395 or five SEN0623, neither of which publishes a volume tier. Power follows the same split. A four-unit SEN0395 ring draws 360 mA at 5 V, or 1.8 W, and a four-unit SEN0557 ring 320 mA, or 1.6 W. A C4001 or C4002 ring cannot be budgeted at all until the current is measured.

Where the sources disagree

- **SEN0691 range.** Store title says 10 m, wiki spec says motion 11 m and static 10 m, the GitHub repository says 11 m motion and 11 m static. Three vendor numbers. Plan against 10 m.
- **C4001 velocity.** The datasheet characteristics list says 0.1 to 3 m/s; the same datasheet's \$DFDMD frame definition says 0 to 10 m/s and both product pages say 0.1 to 10 m/s. Treat 10 m/s as the reported field range and 3 m/s as the specified band. Velocity resolution is not published for any part; the I2C transport carries speed as a signed int16 in cm/s, which is a bus quantum and not an accuracy figure.
- **C4001 configurable maximum.** Sold as 25 m; the library documents a 2000 cm cap and enforces 2500. Anything past 20 m is unverified.
- **SEN0395 transmit power.** Wiki says 13-15 dBm, product page says 9-12 dBm. Both first-party, unresolved.
- **SEN0623 field of view.** One source records 100 x 100 deg only and withdraws an earlier 40 x 40 deg sleep-mode figure as unsourced after a re-fetch on 2026-09-12. Another quotes wiki installation text describing sleep mode as a tilted install covering 40 deg horizontal and 40 deg pitch. The likely reconciliation is a spec-table number and a narrower usable sleep volume in the install guidance. Use 100 x 100 deg as the specification.

Ego-motion clutter

These modules assume a fixed installation, and driving the robot destroys the one capability that justifies choosing mmWave over a passive infrared detector.

An FMCW presence radar separates a person from furniture by Doppler: zero radial velocity is background, non-zero is a candidate. When the radar translates at velocity v , every static scatterer acquires an apparent radial velocity of $-v \cos(\theta)$, θ being the angle between heading and bearing. Three consequences follow.

The whole room becomes a target. Walls, doorframes and table legs enter the Doppler passband, and the C4001's single target slot goes to whichever return has the highest energy - usually a wall, not the person.

A standing person becomes invisible in the worst case. Someone standing still directly ahead of a robot moving at v has apparent radial velocity $-v$. So does the wall behind them. They are Doppler-identical, so clutter rejection tuned at $v = 0$ passes both or rejects both.

Nothing here can compensate. The standard fix is to fit the $v \cos(\theta)$ curve across azimuth and subtract the ego-velocity. No DFRobot radar reports azimuth, elevation or x/y, so with one range-Doppler pair and no bearing there is no closed-form correction.

Several vendor features make it worse. The C4002's `startEnvCalibration()` learns a static background and the wiki requires an empty room while it runs. The C4001's disappearance delay defaults to 15 s, and its library clears a lost target only after about ten empty polls. The C4002's lock time disables detection for 0.2 to 10 s after every occupied-to-unoccupied transition. The SEN0395 FAQ offers "please ensure that the sensor is fixed firmly when using it" as the cure for an output stuck high - a drivetrain supplies exactly that vibration.

DFRobot publishes nothing about moving-platform behaviour for any part. That is not a vendor defect; it is the wrong sensor class for the mount. Only SEN0306 offers a way out, because its 126-bin spectrum lets you run your own ego-motion-aware detector against wheel odometry.

Wall penetration

A radar that works through a plastic shell also works through a plasterboard partition. On a wall-mounted occupancy sensor that is the selling point. On a robot it is a false positive that never clears and cannot be filtered out by anything the module exposes.

DFRobot publishes no penetration figure, no attenuation table and no wall-material guidance for any part, and no through-wall measurement was obtained for any part in this study. The only first-party statement on the subject is SEN0306's, which is described as penetrating non-metallic materials, dust, smoke and fog. What follows from it is a radome specification - ABS, PC or PP, no metal, no carbon fill, no metallic paint - and a radome specification is not a wall-rejection specification. How much of a stud wall these parts see through is unmeasured, so treat the through-wall false positive as an untested risk rather than a quantified one.

The mitigation that works on a fixed installation does not transfer. You can range-gate: SEN0691's `setDetectRange` is capped at 0-1100 cm and its per-gate thresholds run 0-99, so gates beyond the working volume can be suppressed. But a range gate is a radial distance, and a person 2 m away on the far side of a partition sits inside any gate that also contains the room you care about. Separating them needs bearing or a wall model, and there is neither. On the C4001 the gate is not even hard: the 1 m transition zone leaves soft, non-deterministic edges.

The 60 GHz SEN0623 is the one part where band physics should help, but DFRobot publishes no attenuation figure for it either, so better wall rejection at 61 GHz is inferred, not verified.

Micro-motion and the curtain problem

Static-human detection works by finding millimetre-scale chest movement in the zero-Doppler return. The same processing that finds a breathing person sitting still is what makes a curtain in a draught, a ceiling fan or a rolling ball read as a person, and no DFRobot module gives you anything to tell them apart.

The entire feature vector is range, signed radial speed, a scalar energy and - on the C4002 only

- a gate occupancy mask. Energy is dominated by radar cross-section and received power falls as the fourth power of range, so a cat at 1 m and a person at 3 m can return the same number. The computed catalogue makes the size effect explicit: scaled by the cross sections this reference uses (adult standing 0.7 m², cat 0.012 m², chair leg 0.01 m²), SEN0691's 11 m reach on an adult becomes 4.0 m on a cat, SEN0609's 25 m becomes 9.0 m, and SEN0557's 6 m becomes 2.2 m. Speed overlaps completely, and vital signs do not save it either: the C1001 measures 10-25 breaths/min, and resting cats at roughly 20-30/min and resting dogs at roughly 15-30/min overlap that band almost entirely.

The vendor says so itself. The SEN0395 FAQ answers directly: "No, the sensor detects movement of all objects within range by detecting mmWave radar and is very sensitive." Across all six presence parts, DFRobot's pages, wikis and library READMEs contain zero mentions of pets, pet immunity, or false triggers from fans and curtains. The topic is absent in both directions.

The filter also cuts the other way, and this is the part that surprises people. The computed capability grid scores every presence module at level 0 - nothing - against a chair leg at every survey distance: static clutter rejection removes motionless furniture. The processing that admits a breathing curtain rejects a chair. Only SEN0306, which has no presence processing, ranges a static chair leg, at level 2 out to an RCS-scaled 6.9 m.

What radar is genuinely good for on this robot

Three things, and they are specific.

A sealed, windowless enclosure. Radar works through an ABS, PC or PP shell with no aperture, no glass and no dust path. Keep metal, carbon fill and metallic paint out of the radome, and keep the wall near a half-wavelength multiple, about 6.25 mm at 24 GHz and about 2.5 mm at 60 GHz.

A closing-target gate that does not care about light. The signed radial speed from a C4001 or C4002 is a direct approach/recede bit, computed in the module, in darkness or full sun, at \$8.90. Nothing optical gives velocity without differentiating position first.

A stationary-robot watchdog. The strongest case, because it is the duty cycle where every objection above disappears. Parked or docked, there is no ego-motion, the C4002's environment calibration is valid, and micro-motion presence works as specified: a still, breathing person at up to 10 m through the shell, at 10 Hz, for under \$10, with no camera and no privacy problem. Arm it when the wheels stop; disarm it when they turn.

What radar must not be asked to do is identify anything or bound a volume. Species discrimination needs micro-Doppler on a raw range-Doppler map, and no DFRobot module emits raw IQ, an ADC stream or a range-Doppler matrix, so no classifier can be built on any of them. Height, the feature that separates a 300 mm cat from a 1700 mm adult, needs elevation, and no part reports it - a 300 mm pet does not enter a 40 deg vertical beam until 1.65 m with the sensor at 900 mm, per the coverage-geometry chapter. A ring of three or four same-band FMCW modules is an unverified risk too: DFRobot publishes no interference mitigation, no chirp randomisation, no synchronisation pin and no multi-sensor guidance for any part. Mixing the 61 GHz SEN0623 with 24 GHz units is the only way in this catalogue to keep two modules off one band, and DFRobot publishes no guidance for that combination either.

Thermal and infrared people sensing

No thermal or infrared part in this catalogue can serve the obstacle requirement. A chair leg, a door frame and a cardboard box all sit at room temperature, so their radiated contrast against the wall behind them is approximately zero, which is below the noise floor of every device listed here. A thermopile or TMOS array measures radiated flux, not geometry. Thermal is an animate-versus-inanimate channel and nothing else, and it is the only modality here that answers that question about a motionless body. Everything below decides whether that one answer is worth its price, its aperture and its frame rate on a 350 mm robot.

The catalogue

Multi-pixel arrays, the only thermal parts that can classify anything. All prices and stock read from adafruit.com on 2026-09-12.

Part	ID	Price	Array	FoV H x V	Stock
MLX90640 IR Thermal Camera Breakout, 55 Degree	4407	\$74.95	32x24, 768 px	55 x 35 deg	4 in stock
MLX90640 24x32 Breakout, 110 Degree FoV	4469	\$74.95	32x24, 768 px	110 x 75 deg	Out of stock
AMG8833 Grid-EYE Breakout, STEMMA QT	3538	\$44.95	8x8, 64 px	60 x 60 deg	57 in stock
AMG8833 Grid-EYE FeatherWing	3622	\$44.95	8x8, 64 px	60 x 60 deg	In stock
UTI165H handheld imager	4578	\$250.00	160x120	56 x 42 deg	5 in stock
UTI165K handheld imager, USB video	4579	\$499.95	160x120	56 x 42 deg	91 in stock

The MLX90640 field of view is datasheet-verified from Melexis Rev 12 Table 15 at the 50 % sensitivity point, the wider figure being the X axis across the 32-pixel direction. The Grid-EYE 60 deg is the Panasonic "Typical 60 deg" viewing angle, horizontal and vertical alike. Neither is a diagonal.

Single-element presence parts, which detect but cannot locate or classify.

Part	ID	Price	FoV	Stated range	Stock
STHS34PF80 IR Presence / Motion (ST TMOS)	6426	\$14.95	80 deg full, at 50 % IR intensity	4 m for a 700 x 250 mm object, no lens	In stock
MLX90614 contact-less IR, 3 V (ESF-BAA)	1747	\$15.95	90 deg	Not published	No longer stocked
MLX90614 contact-less IR, 5 V (ESF-AAA)	1748	\$15.95	90 deg	Not published	No longer stocked
PIR (motion) sensor	189	\$9.95	120 deg	About 7 m, vendor-stated	Out of stock
Breadboard-friendly Mini PIR, 3-pin	4871	\$3.95	100 deg	2 to 5 m, vendor-stated	In stock
Mini PIR with time and sensitivity, BS612	5578	\$1.95	120 deg	About 5 m, lens focused at 5 m	In stock

Both MLX90614 boards are xAA optics at 90 deg, datasheet-verified from Melexis Rev 021 Table 19, although they carry different order codes. The same datasheet says the reading is "the average temperature of all objects in the Field Of View", which across a 90 deg cone is the room, not the person. Adafruit redirects 1747 to the AMG8833 and 1748 to the STHS34PF80. The three PIR modules report motion only: none holds a static body, and on a driving robot the whole scene moves, so the output saturates.

The two UNI-T handhelds publish no detection range at all, only a 15 cm minimum and a 1 m optimal measuring distance, and their 30 to 45 degC window clips everything at room temperature. They are battery handhelds with no embeddable output, and are excluded from every table below.

Parts the search returns that are not thermal sensors

The search `camera+stema+ir` on adafruit.com returns exactly four items, and one of them is a book. The wider searches return magnetometers with similar part numbers. Named and excluded:

- **Bots! Robotics Engineering (4348, \$14.95)** is a book.
- **MLX90393 (4022, \$9.95)** is a triple-axis magnetometer, +/-5 to 50 mT. The similar Melexis part number is the only connection to the MLX90614 and MLX90640.
- **LIS3MDL (4479, \$9.95)** is also a magnetometer.

- **Raspberry Pi NoIR camera modules (3100, 5659, 5660, 3415, 5658)** have the IR-cut filter removed, which extends a silicon imager to roughly 850 to 1000 nm near-infrared. Body heat radiates at 8 to 14 um, so a NoIR camera cannot see it, and in a dark room it sees nothing without an 850 nm illuminator. It belongs in the camera lane.
- **MLX90641 (16x12)** is not stocked by Adafruit in any form, verified by search. Seeed, Waveshare and Evelta sell it, so it is buyable, but no Adafruit price, stock or breakout exists to quote.
- A **thermopile** search on adafruit.com returns thermoplastic and filament products only.

Pixel subtense: the argument that settles it

Angular pixel pitch is the field of view divided by the pixel count along that axis, and the linear footprint of one pixel at range d is $w = 2 \times d \times \tan(\text{pitch} / 2)$.

- Grid-EYE: $60 / 8 = 7.5$ deg, so $w = 2 \times d \times \tan(3.75 \text{ deg}) = 0.1311 \times d$, or **131 mm per metre**, square.
- MLX90640 55 deg: $55 / 32 = 1.719$ deg H and $35 / 24 = 1.458$ deg V, so **30.0 mm/m H, 25.5 mm/m V**.
- MLX90640 110 deg: $110 / 32 = 3.438$ deg H and $75 / 24 = 3.125$ deg V, so **60.0 mm/m H, 54.6 mm/m V**.

Worked example at the longest survey distance. Grid-EYE at 10 ft (3.048 m): one pixel covers $2 \times 3048 \times \tan(3.75 \text{ deg}) = 400$ mm. A 450 mm adult torso is $450 / 400 = \mathbf{1.1 \text{ pixels}}$ wide. A 250 mm cat is $250 / 400 = \mathbf{0.6 \text{ pixels}}$. Targets in the tables are a 450 mm torso, a 1700 mm standing adult, and a cat 250 mm long by 200 mm tall.

Horizontal pixels across an adult torso:

Range	Grid-EYE 8x8	MLX90640 55 deg	MLX90640 110 deg
1 ft (0.305 m)	11.3 (overflows frame)	49 (overflows)	24.6
3 ft (0.914 m)	3.8	16.4	8.2
5 ft (1.524 m)	2.3	9.8	4.9
8 ft (2.438 m)	1.4	6.2	3.1
10 ft (3.048 m)	1.1	4.9	2.5

Pixels on a cat, horizontal x vertical:

Range	Grid-EYE 8x8	MLX90640 55 deg	MLX90640 110 deg
1 ft (0.305 m)	6.3 x 5.0	27 x 26	13.7 x 12.0
3 ft (0.914 m)	2.1 x 1.7	9.1 x 8.6	4.6 x 4.0
5 ft (1.524 m)	1.3 x 1.0	5.5 x 5.2	2.7 x 2.4
8 ft (2.438 m)	0.8 x 0.6	3.4 x 3.2	1.7 x 1.5
10 ft (3.048 m)	0.6 x 0.5	2.7 x 2.6	1.4 x 1.2

Classification needs shape, and shape needs at least three pixels across the short axis before an aspect ratio means anything. That sets a hard ceiling no software can lift. **Grid-EYE: about 1.2 m for a pet and about 2.5 m for a human. MLX90640 55 deg: about 3 m for a pet, past 7 m for a human. MLX90640 110 deg: about 1.5 m for a pet, about 3 m for a human.** Field of view is bought with range one for one: the 110 deg part has almost exactly the pet range of the 55 deg part at half the distance.

Below one pixel of fill, the number the sensor reports is not a temperature. A pixel returns the flux-weighted average over its whole footprint, so apparent rise = fill fraction x surface delta T. Against a 22 degC room and a cat at 8 degC above ambient, the Grid-EYE reports that cat as 8.00 degC out to 1.5 m, then 3.91 degC at 2.44 m, 2.51 degC at 3.05 m and 0.48 degC at 7 m, while a clothed torso at 6 degC reads 3.77 degC at 5 m. **On an 8x8 array apparent temperature is mostly a range measurement in disguise**, so a cat at 1 m and a human at 5 m can return the same pixel value, and no threshold tuning fixes it. The 55 deg MLX90640 is the exception: a torso fills a pixel out to 15 m and a cat to 7.8 m, so on that part nothing in this section dilutes except a bare face, which at 200 mm across begins to dilute beyond 6.7 m. The 110 deg part is coarser: a bare face dilutes beyond 3.3 m, and a cat beyond 3.7 m, where the vertical axis binds first at 200 / 54.6.

Two source disagreements, both in the reader's favour to know. The computed appendix divides the frame width by half the column count (scripts/geometry.py, zone_width_mm), which makes its centre-pixel footprint 88.0 mm for the Grid-EYE at 0.305 m against 40 mm from the datasheet pixel pitch, a factor of 2.2. Prefer the pitch-derived figures above, because 60 deg

across 8 pixels is a 7.5 deg sampling pitch by construction. Panasonic's per-pixel figure is not a second opinion on that pitch: the datasheet quotes it as a half angle, 7.7 deg horizontal and 8 deg vertical, so one pixel accepts a full cone of about 15.4 deg H, roughly twice the pitch. Adjacent pixels therefore overlap heavily and a point target bleeds into its neighbours, which blurs the image but does not change the 7.5 deg spacing the tables are built on. The computed appendix also models a cat as 140 mm wide at the shoulder rather than 250 mm long, so its cat capability levels are one step more pessimistic than the table above at most distances.

Mounting geometry: the constraint that kills pet detection first

Pixel count is not the binding limit for a pet. Frame placement is. The pet stands 200 to 500 mm tall and the sensor sits on a robot 600 to 1200 mm tall, so take the sensor height H as 0.9 m, the mid-point of that build range. With a vertical half field of view of alpha and a downward tilt t, the floor first enters the frame at $H / \tan(\alpha + t)$, the top of a 200 mm cat at $(H - 0.2) / \tan(\alpha + t)$, and a 1700 mm head stays in frame out to $(1.7 - H) / \tan(\alpha - t)$.

Sensor, vertical FoV	Tilt	Floor first seen	Cat top first seen	Adult head out to
Grid-EYE 60 deg	0 deg	1.56 m	1.21 m	1.39 m
Grid-EYE 60 deg	15 deg	0.90 m	0.70 m	2.99 m
Grid-EYE 60 deg	25 deg	0.63 m	0.49 m	9.14 m
MLX90640 35 deg V	0 deg	2.85 m	2.22 m	2.54 m
MLX90640 35 deg V	15 deg	1.41 m	1.10 m	18.3 m
MLX90640 75 deg V	0 deg	1.17 m	0.91 m	1.04 m
MLX90640 75 deg V	25 deg	0.47 m	0.36 m	3.61 m

Read the level rows. **Mounted level at 0.9 m, the 55 deg MLX90640 is blind to a cat inside 2.2 m**, because the animal is below the frame, and that is worse than the 3 m optical limit computed above. The fix is a 15 to 25 deg downward tilt, which pulls the cat-visible boundary in to 0.5 to 1.1 m and, on the narrow part, extends head-in-frame to effectively unlimited range because the upper edge still clears 1.7 m. The 55 x 35 deg part tilted 20 deg is the best single configuration in this catalogue for a robot of this size. The 110 deg part tilted 25 deg sees a cat from 0.36 m but loses a standing adult's head beyond 3.6 m.

One more geometry term applies to a ring. The Grid-EYE optical axis gap is "within typical +/-5.6 deg", which on a 7.5 deg pitch is three-quarters of a pixel of unit-to-unit boresight error, so fusing six frames into one panorama needs per-unit extrinsic calibration. The MLX90640 is tighter: max 5 deg on the 110 deg part and max 3 deg on the 55 deg part.

Why temperature alone cannot separate a pet from a human

The premise that a 38 degC cat can be told from a 34 degC human fails three times over.

First, these sensors measure surface, not core. A cat's core is 38 to 39.2 degC, but fur insulates: measured fur surface is typically 30 to 36 degC, with ears, nose and paw pads at 35 to 38 degC because they are thinly furred. Human exposed skin is 31 to 35 degC, and a clothed torso in a sweater reads 25 to 28 degC in a 22 degC room. The two distributions overlap almost completely, and the clothed human is often the cooler of the two.

Second, fill fraction swamps the signal, as the dilution figures above show.

Third, the absolute accuracy is the same size as the effect. Grid-EYE accuracy is "Typical +/-2.5 degC". MLX90640 frame accuracy is +/-1 to +/-2 degC, with non-uniformity up to +/-3 degC plus 2 % of abs(To - Ta) in the outer zone. The claimed 4 degC core gap maps to perhaps 1 to 3 degC of surface difference, which is inside that uncertainty.

What does separate them is geometry and kinematics, and every cue needs pixels.

Cue	Human	Cat or small dog	Minimum sensor to see it
Blob centroid above floor	0.9 to 1.2 m	0.1 to 0.25 m	any array with the floor in frame
Aspect ratio, tall to wide	about 3.8:1 standing	about 1.25:1, long and low	3 px in the short axis
Gait or step frequency	1.7 to 2.2 Hz	2.5 to 4 Hz	8 full frames/s

The height-above-floor cue is the cheap decisive one, and it works on a Grid-EYE out to about 2.4 m provided the tilt of the section above puts the floor in frame. The aspect-ratio cue needs a 3 x 3 blob, so about 1.2 m on a Grid-EYE, about 3 m on

the 55 deg MLX90640 and about 1.5 m on the 110 deg part. The gait cue needs 8 full frames per second, which a Grid-EYE has at 10 Hz and a bandwidth-bound MLX90640 ring at about 1.4 Hz each does not.

The four limits that decide whether thermal earns its place

It cannot see through glass or a printed shell. Body heat at 34 degC peaks near 9.4 um. Thermal optics use germanium, which transmits roughly 2.5 to 12 um, which is why a germanium window costs what it does. Soda-lime glass transmits about 0.25 to 3 um and is opaque above roughly 3 um. Polycarbonate, acrylic and PET are all opaque in the long-wave infrared, so a clear cosmetic dome silently zeroes the sensor. A human behind a glass door is thermally invisible, and the door itself is undetectable as an obstacle, because it radiates at room temperature like the wall beside it. The shell needs an **open aperture** at every thermal sensor, or a thin LDPE window with an accepted transmission loss. Polished floors, stainless appliance fronts and glossy cabinets are good long-wave reflectors, so expect ghost blobs beside kitchen appliances that an 8x8 array cannot tell from a second person.

A warm room collapses the contrast. A clothed human against a 22 degC room is about 6 degC of surface contrast. Against a 28 degC room it is about 1 to 2 degC, and diluted across a sub-pixel target at 3 m that falls to roughly 0.4 degC, comparable to the Grid-EYE 10 Hz NETD of 0.16 degC. Thermal detection range is a function of room temperature and drops sharply in summer. Two drift terms push the same way: the MLX90640 needs up to 4 minutes of thermal stabilisation for its accuracy spec and the Grid-EYE 15 s, and a chassis full of motor drivers warms the enclosure air 8 to 10 degC above room over a 20-minute run. Never threshold on absolute temperature; subtract a per-pixel background model with a time constant of tens of seconds.

A sun-warmed floor patch reads as a body. Direct sun through a window commonly drives dark flooring to 35 to 45 degC, squarely inside the human range. A thermal-only classifier chases sunbeams. This is the classic robot-vacuum failure and it has not gone away.

The frame rate is slow against a moving robot. The Grid-EYE offers 10 Hz or 1 Hz and nothing else. At 10 Hz, with a human walking at 1.4 m/s and the robot at 0.5 m/s, two frames to confirm a detection costs 200 ms and 380 mm of closure: acceptable beyond about 1 m, marginal inside it. The MLX90640 is worse than its headline, because the programmed refresh rate applies per subpage, so "16 Hz" is 8 full frames per second and a fast target tears across the two half-frames. One frame is about 1668 bytes, roughly 42 ms of 400 kHz bus time, so a four-sensor ring is bandwidth-bound to about 1.4 Hz each: 714 ms and 1.36 m of closure per confirmed detection.

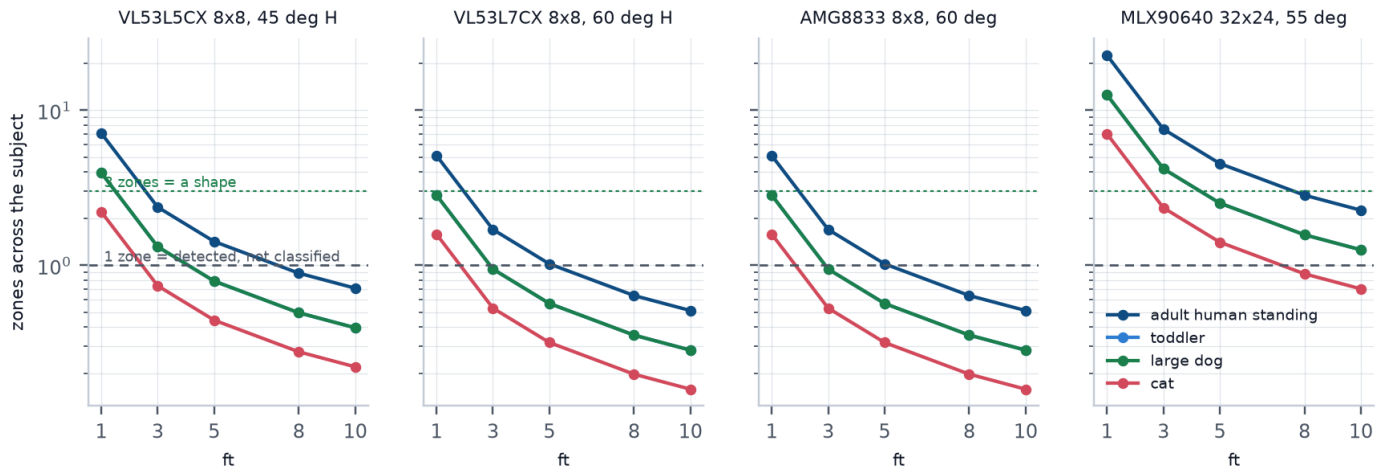
Verdict at the five survey distances

Range	Grid-EYE 3538	MLX90640 55 deg	MLX90640 110 deg	STHS34PF80	PIR
1 ft	Human overfills; cat resolved	Full silhouette both	Full human; cat 13 px	Presence only	Motion only
3 ft	Human blob; cat marginal	Full silhouette; cat 9 px	Human yes; cat 4 px	Presence only	Motion only
5 ft	Human yes; cat = 1 px	Human yes; cat 5 px	Human yes; cat 2.7 px	Presence only	Motion only
8 ft	Human 1x5 strip; cat gone	Human yes; cat 3.4 px	Human 3 px; cat gone	Presence only	Motion only
10 ft	Human 1x4 strip; cat gone	Human yes; cat 2.7 px	Human 2.5 px; cat gone	Presence only	Motion only

On a 350 mm robot, thermal is a short-range channel. The Grid-EYE separates a human from a pet only inside about 1.2 m, and beyond 5 ft a cat is not there at all; six units close a 360 deg ring for \$269.70, but the part answers to only two I2C addresses, 0x68 and 0x69, so the ring also needs a TCA9548A multiplexer. The 55 deg MLX90640 is the only part that holds a real silhouette across the whole survey, and at \$74.95 with 4 units in stock it buys one direction, not a ring: seven units for 360 deg is \$524.65 and a bus it cannot feed. The 110 deg part closes a ring in four units for \$299.80, but it is out of stock and it gives up the pet range to do it. The STHS34PF80 at \$14.95 and 10 uA, measured at 128-sample averaging and a 1 Hz output rate, holds a **stationary** body, which no PIR can, and is the right always-on wake-up gate; five of them cover 360 deg for \$74.75, again behind a multiplexer because the address 0x5A is fixed. The PIR modules and the discontinued MLX90614 contribute nothing a perimeter ring can use. None of them, at any distance, sees the chair leg.

The defensible configuration is therefore not a thermal ring. It is one MLX90640 4407 facing the direction of travel behind an open aperture, tilted 20 deg down so the floor is in frame, taking range and geometry from a time-of-flight lane and contributing the one bit that lane cannot produce: whether the obstacle is alive.

A subject must fill several zones before its SHAPE can be read



How many zones a subject covers. Below one zone the sensor can report presence but never shape, so it can never classify.

Master comparison grid

Every candidate part in one place, rated the same way. The tables in this chapter are generated from `radar-compare/data/sensors.csv` by `radar-compare/scripts/grid.py`. No cell is typed by hand, so no cell can flatter a part the catalogue does not support.

How the capability level is decided

Each cell answers one question: at this distance, against this subject, what can this sensor tell you? The level is computed from the sensor's own published numbers by a fixed rule. The rules are printed here so that a reader who disagrees can argue with the rule rather than with the cell.

Level	What the sensor can tell you
0	nothing
1	presence only
2	obstacle + range
3	living vs not
4	human vs pet
5	full silhouette

The rules, in the order they are applied:

- **Out of range is level 0.** For every optical sensor the rated range is used as published, with its stated target reflectance. Where the catalogue carries a dark-target figure, that one is used instead, because a black sock and a dark-furred cat are the cases that matter.
- **Radar range is scaled by target size, because the physics says so.** The radar range equation makes range proportional to the fourth root of radar cross-section. An adult presents about 0.70 m2 at these frequencies and a cat about 0.012 m2, so a module rated 25 m on a walking adult reaches about 9.0 m on a cat. That scaling is applied to every radar row.
- **A grid imager can only report a shape it can resolve.** The level is set by how many zones or pixels the subject spans across, computed from the centre zone, which is the narrowest. Under half a zone a ToF imager still returns a valid range but nothing else. Under about a third of a pixel a thermal array loses the subject in its own noise.
- **A thermal array and a PIR score 0 against furniture, at every range, forever.** A chair leg is at room temperature. There is no contrast to detect and no amount of processing creates one.
- **A presence radar scores 0 against a motionless inanimate object** unless the catalogue records that it reports static targets. Suppressing motionless returns is what clutter rejection is for; it is the same feature that makes the module immune to a curtain.
- **A 2D scanning lidar sees exactly one horizontal plane.** Anything whose top is below that plane or whose bottom is above it is invisible at every range. The catalogue leaves the scan plane height empty because it is a mounting decision, not a property of the part.
- **Level 4 needs evidence, not optimism.** A vendor radar module reaches level 4 only where its shipped firmware reports an actual class. Telling an adult from a cat on raw range and velocity is a micro-Doppler program someone has to write, and none of these modules exposes the data it would need.

Specifications

Sensor	Vendor / SKU	Technology	USD	FoV H x V	Zones	Rated range	Rate	Interface	Line of sight
VL53L0X ToF Distance Sensor	Adafruit 3317	tof-single	14.95	25 x 25	1	2.0	30	I2C 400 kHz addr 0x29	yes
VL53L1X ToF Distance Sensor	Adafruit 3967	tof-single	14.95	19.3 x 19.3	1	4.0	50	I2C 400 kHz addr 0x29	yes
VL53L4CD ToF Distance Sensor	Adafruit 5396	tof-single	14.95	18 x 18	1	1.3	100	I2C 1 MHz addr 0x29	yes
VL53L4CX ToF Distance Sensor	Adafruit 5425	tof-single	14.95	18 x 18	1 with multi-object	6.0	30	I2C 1 MHz addr 0x29	yes
VL53L3CX Multi-Target ToF	Pololu 3416	tof-single	19.95	25 x 25	1 with up to 4 targets	3.1	30	I2C 1 MHz addr 0x29	yes
VL6180X ToF Proximity and Ambient Light Sensor	Adafruit 3316	tof-single	13.95	25 x 25	1	0.1	150	I2C 400 kHz addr 0x29	yes
VL53L4ED Low-Power ToF	STMicroelectronics VL53L4ED	tof-single		18 x 18	1	1.3		I2C	yes
VL53L5CX 8x8 Multizone ToF	Pololu 3417	tof-multizone	19.95	45 x 45	8x8	4.0	15	I2C 1 MHz addr 0x29	yes
VL53L7CX Wide-FoV 8x8 Multizone ToF	Pololu 3418	tof-multizone	19.95	60 x 60	8x8	3.5	15	I2C 1 MHz addr 0x29	yes
VL53L8CX 8x8 Multizone ToF	Pololu 3419	tof-multizone	24.95	45 x 45	8x8	4.0	15	I2C 1 MHz addr 0x29 or SPI 20 MHz	yes
VL53L8CH Histogram Multizone ToF	STMicroelectronics VL53L8CH	tof-multizone		45 x 45	8x8 plus compact normalised histogram	4.0	15	I2C 1 MHz or SPI 3 MHz	yes
VL53L7CH Wide-FoV Histogram Multizone ToF	STMicroelectronics VL53L7CH	tof-multizone		60 x 60	8x8 plus compact normalised histogram	3.5	15	I2C or SPI	yes
VL53L9CX 2268-Zone dToF Camera	STMicroelectronics STEVAL-VL53L9	tof-multizone	80.39	55 x 42	54x42	8.8	100	MIPI I3C or MIPI CSI-2	yes
TMF8801 ToF Distance Sensor	Adafruit 6522	tof-single	9.95	24 x 24	1	2.5	30	I2C addr 0x41	yes
TMF8806 ToF Distance Sensor	Adafruit 6501	tof-single	12.50	30 x 30	1	5.0	30	I2C addr 0x41	yes
TMF8820 3x3 Multizone dToF	ams-OSRAM TMF8820	tof-multizone		33 x 32	3x3	5.0	30	I2C addr 0x41	yes
TMF8821 4x4 Multizone dToF	ams-OSRAM TMF8821	tof-multizone		41 x 52	4x4	5.0	15	I2C addr 0x41; I3C tolerant	yes
TMF8828 8x8 Multizone dToF	ams-OSRAM TMF8828	tof-multizone		41 x 52	8x8	4.4	15	I2C addr 0x41; I3C tolerant	yes
RPLIDAR A1M8 360 Laser Range Scanner	Slamtec Adafruit 4010	tof-lidar-scanning	99.95	360 x 0	360 per rev	12.0	5.5	UART 115200 8N1 with USB adapter	yes
RPLIDAR C1 360 dTOF Scanner	Slamtec RPLIDAR C1	tof-lidar-scanning	69.00	360 x 0	500 per rev	12.0	10	UART 460800	yes
RPLIDAR A2M12	Slamtec A2M12	tof-lidar-scanning		360 x 0	1600 per rev	12.0	10	UART	yes
RPLIDAR S2	Slamtec S2 / S2E	tof-lidar-scanning		360 x 0	3200 per rev	30.0	10	serial on S2; Ethernet on S2E	yes
RPLIDAR S2L	Slamtec S2L	tof-lidar-scanning		360 x 0	3200 per rev	18.0	10	serial	yes
RPLIDAR S2P	Slamtec S2P	tof-lidar-scanning		360 x 0	3200 per rev	50.0	10	serial	yes
LDROBOT LD06 360 Scanner	LDROBOT LD06	tof-lidar-scanning		360 x 0	450 per rev	12.0	10	UART 230400 one-way stream	yes
LDROBOT LD19 / LD19P 360 dTOF Scanner	LDROBOT LD19	tof-lidar-scanning	99.00	360 x 0	450 per rev	12.0	10	UART 230400 ZH1.5T-4P	yes
Waveshare D500 LiDAR Kit	Waveshare D500	tof-lidar-scanning		360 x 0	500 per rev	12.0	10	UART 230400	yes
YDLIDAR X2	YDLIDAR X2	tof-lidar-scanning		360 x 0	500 per rev	8.0	6	UART	yes
YDLIDAR X4 PRO	YDLIDAR X4 PRO	tof-lidar-scanning		360 x 0	833 per rev	10.0	6	UART	yes
YDLIDAR G4	YDLIDAR G4	tof-lidar-scanning		360 x 0	1800 per rev	16.0	5	UART	yes
YDLIDAR T-mini Plus	YDLIDAR T-mini Plus	tof-lidar-scanning		360 x 0	666 per rev	12.0	6	UART	yes
Livox Mid-360 3D LiDAR	Livox Mid-360	tof-lidar-scanning		360 x 0	200000 points per second	70.0	10	100BASE-TX Ethernet	yes
Benewake TFmini Infrared ToF	Benewake Adafruit 3978	tof-lidar	44.95	2.3 x 2.3	1	12.0	100	UART 115200 8N1	yes
Benewake TF-Luna	Benewake TF-Luna	tof-lidar	29.90	- x -	1	8.0	250	UART or I2C or I/O	yes
Benewake TFmini-S	Benewake TFmini-S	tof-lidar		- x -	1	12.0		UART or I2C or I/O	yes
Garmin LIDAR-Lite v3	Garmin Adafruit 4058	tof-lidar	129.95	0.46 x 0.46	1	40.0	270	I2C addr 0x62 or PWM	yes
Garmin LIDAR-Lite v4 LED	Garmin Adafruit 4441	tof-lidar	59.95	4.77 x 4.77	1	10.0	200	I2C or ANT	yes
Sharp GP2Y0A21YK0F Analog IR Ranger	Sharp Adafruit 164	proximity	14.95	- x -	1	0.8	26	analog voltage	yes
Sharp GP2Y0A02YK0F Analog IR Ranger	Sharp Adafruit 1031	proximity	15.95	- x -	1	1.5		analog voltage	yes
Sharp GP2Y0D810Z0F Digital Distance Switch	Pololu Adafruit 1927	proximity	8.95	- x -	1	0.1	400	digital high/low	yes
Sharp GP2Y0D805Z0F Digital Distance Switch	Pololu Adafruit 3025	proximity		- x -	1	0.05		digital high/low	yes
HC-SR04 Ultrasonic Sonar Distance Sensor	Adafruit 3942	ultrasonic	3.95	15 x 15	1	4.0		trigger/echo GPIO	yes
RCWL-1601 Ultrasonic Distance Sensor 3V/5V	Adafruit 4007	ultrasonic	3.95	30 x 30	1	4.5		trigger/echo GPIO	yes
US-100 Ultrasonic Distance Sensor 3V/5V	Adafruit 4019	ultrasonic	6.95	15 x 15	1	4.5		HC-SR04 trigger/echo mode or UART 9600	yes
MaxBotix LV-MaxSonar-EZ0	MaxBotix MB1000 / Adafruit 979	ultrasonic	29.95	- x -	1	6.45	20	analog Vcc/512 per inch; PWM 147 us/in; serial 9600	yes
MaxBotix LV-MaxSonar-EZ4	MaxBotix MB1040 / Adafruit 982	ultrasonic	28.50	- x -	1	6.45	20	analog or PWM or serial 9600	yes
Large Ultrasonic Sensor with Horn UART	Adafruit 4664	ultrasonic	28.95	40 x 40	1	7.5		UART 9600	yes
VCNL4200 Long Distance IR Proximity and Light Sensor	Adafruit 6064	proximity	6.95	- x -	1	1.5		I2C addr 0x51	yes
VCNL4030 Proximity and Lux Sensor	Adafruit 6491	proximity	5.95	- x -	1	0.3		I2C	yes
VCNL4040 Proximity and Lux Sensor	Adafruit 4161	proximity	5.95	- x -	1	0.2		I2C	yes
APDS9960 Proximity Light RGB and Gesture Sensor	Adafruit 3595	proximity	7.50	- x -	1			I2C	yes
STEMMA Reflective Photo Interrupter TCRT1000	Adafruit 5913	proximity	4.95	- x -	1			digital/analog pin	yes
AMG8833 Grid-EYE IR Thermal Camera Breakout	Adafruit 3538	thermal-ir-array	44.95	60 x 60	8x8	7.0	10	I2C 400 kHz addr 0x69 or 0x68	yes

Sensor	Vendor / SKU	Technology	USD	FoV H x V	Zones	Rated range	Rate	Interface	Line of sight
AMG8833 IR Thermal Camera FeatherWing	Adafruit 3622	thermal-ir-array	44.95	60 x 60	8x8	7.0	10	I2C addr 0x69	yes
MLX90640 32x24 IR Thermal Camera Breakout 55 Degree	Adafruit 4407	thermal-ir-array	74.95	55 x 35	32x24	7.0	8	I2C addr 0x33 programmable	yes
MLX90640 32x24 IR Thermal Camera Breakout 110 Degree	Adafruit 4469	thermal-ir-array	74.95	110 x 75	32x24	7.0	8	I2C addr 0x33 programmable	yes
STHS34PF80 IR Presence and Motion Sensor	Adafruit 6426	thermal-ir-point	14.95	80 x 80	1	4.0	30	I2C addr 0x5A or 3-wire SPI	yes
MLX90614 Contact-less IR Thermometer 3V	Adafruit 1747	thermal-ir-point	15.95	90 x 90	1			I2C addr 0x5A	yes
MLX90614 Contact-less IR Thermometer 5V	Adafruit 1748	thermal-ir-point	15.95	90 x 90	1			I2C addr 0x5A	yes
UT1165H Thermal Camera Imager	UNI-T Adafruit 4578	thermal-ir-array	250.00	56 x 42	160x120		9	2.8 inch 320x240 LCD only	yes
UT1165K Thermal Camera Imager with USB Video	UNI-T Adafruit 4579	thermal-ir-array	499.95	56 x 42	160x120		9	USB-C real-time video out plus LCD	yes
PIR Motion Sensor	Adafruit 189	pir	9.95	120 x 120	1	7.0		digital high/low	yes
Breadboard-friendly Mini PIR Motion Sensor	Adafruit 4871	pir	3.95	100 x 100	1	5.0		digital high/low	yes
Mini PIR Sensor with Time and Sensitivity Control BS612	Adafruit 5578	pir	1.95	120 x 120	1	5.0		digital high/low	yes
mmWave Radar 24GHz Human Presence Detection Sensor 9m	DFRobot SEN0395	mmwave-24ghz	29.00	100 x 40	1	9.0		UART 115200 plus GPIO2 level	no
mmWave C4001 24GHz Presence Sensor 25m UART	DFRobot SEN0609	mmwave-24ghz	13.90	100 x 40	1	25.0		UART 9600 plus configurable IO	no
Gravity mmWave C4001 24GHz Presence Sensor 12m I2C and UART	DFRobot SEN0610	mmwave-24ghz	12.90	100 x 80	1	12.0		I2C addr 0x2A or 0x2B plus UART 9600	no
Fermion C4002 mmWave Motion and Static Presence Module	DFRobot SEN0691	mmwave-24ghz	8.90	120 x 120	26 range gates of 20 cm	11.0	10	UART 57600 to 1 Mbps plus configurable OUT pin	no
C1001 60GHz mmWave Indoor Fall Detection Sensor	DFRobot SEN0623	mmwave-60ghz	29.00	100 x 100	1	11.0		UART 115200 plus IO1 fall level plus IO2 presence level	no
mmWave 24GHz Human Presence Sensing Module 6m	DFRobot SEN0557	mmwave-24ghz	9.90	120 x -	8 gates of 0.75 m	6.0		UART 57600 default plus GPIO OUT 3.3 V	no
mmWave 24GHz Microwave Radar Distance Sensor 20m	DFRobot SEN0306	mmwave-24ghz	65.90	78 x 23	126-bin range spectrum	20.0	10	UART 57600 8N1 TX only	no
Gravity Digital 10.525GHz Microwave Motion Sensor	DFRobot SEN0192	mmwave-24ghz	8.90	72 x 36	1	16.0		Gravity digital pin	no
5.8GHz Microwave Radar Module	DFRobot SEN0521	mmwave-24ghz		120 x 120	1	11.0		UART 115200 or digital I/O	no
mmWave 80GHz Liquid Level Detection Sensor 40m	DFRobot SEN0676	mmwave-60ghz	59.00	50 x 50	1	40.0		UART Modbus	no
MR60BHA2 60GHz Breathing and Heartbeat Module	Seeed Studio MR60BHA2	mmwave-60ghz		120 x 120	1	1.5		UART	no
MR60FDA2 60GHz Fall Detection Module	Seeed Studio MR60FDA2	mmwave-60ghz		120 x 120	1	2.0		UART	no
Person Sensor	Useful Sensors SparkFun SEN-21231	camera	9.95	110 x 110	1 frame; up to 4 faces		7	I2C addr 0x62 up to 400 kbaud	yes
Gravity Offline Edge AI Gesture and Face Detection Sensor	DFRobot SEN0626	camera	14.90	85 x 85	1 frame; up to 10 faces or upper bodies	3.0		I2C plus UART	yes

Beam footprint width at each survey distance

Width is $2 \cdot d \cdot \tan(\text{FoV}/2)$ using the **horizontal** field of view. For a scanning lidar the cone width is meaningless, so the figure given is the arc between adjacent samples, which is what actually limits whether a thin object is hit by a beam at all.

Sensor	FoV H	1 ft	3 ft	5 ft	8 ft	10 ft
VL53L0X ToF Distance Sensor	25	135	405	676	1081	1351
VL53L1X ToF Distance Sensor	19.3	104	311	518	829	1037
VL53L4CD ToF Distance Sensor	18	97	290	483	772	966
VL53L4CX ToF Distance Sensor	18	97	290	483	772	966
VL53L3CX Multi-Target ToF	25	135	405	676	1081	1351
VL6180X ToF Proximity and Ambient Light Sensor	25	135	405	676	1081	1351
VL53L4ED Low-Power ToF	18	97	290	483	772	966
VL53L5CX 8x8 Multizone ToF	45	253	758	1263	2020	2525
VL53L7CX Wide-FoV 8x8 Multizone ToF	60	352	1056	1760	2816	3520
VL53L8CX 8x8 Multizone ToF	45	253	758	1263	2020	2525
VL53L8CH Histogram Multizone ToF	45	253	758	1263	2020	2525
VL53L7CH Wide-FoV Histogram Multizone ToF	60	352	1056	1760	2816	3520
VL53L9CX 2268-Zone dToF Camera	55	317	952	1587	2539	3173
TMF8801 ToF Distance Sensor	24	130	389	648	1037	1296
TMF8806 ToF Distance Sensor	30	163	490	817	1307	1633
TMF8820 3x3 Multizone dToF	33	181	542	903	1445	1806
TMF8821 4x4 Multizone dToF	41	228	684	1140	1823	2279
TMF8828 8x8 Multizone dToF	41	228	684	1140	1823	2279
RPLIDAR A1M8 360 Laser Range Scanner	360	5 (sample pitch)	16 (sample pitch)	27 (sample pitch)	43 (sample pitch)	53 (sample pitch)
RPLIDAR C1 360 dTOF Scanner	360	4 (sample pitch)	11 (sample pitch)	19 (sample pitch)	31 (sample pitch)	38 (sample pitch)
RPLIDAR A2M12	360	1 (sample pitch)	4 (sample pitch)	6 (sample pitch)	10 (sample pitch)	12 (sample pitch)
RPLIDAR S2	360	1 (sample pitch)	2 (sample pitch)	3 (sample pitch)	5 (sample pitch)	6 (sample pitch)
RPLIDAR S2L	360	1 (sample pitch)	2 (sample pitch)	3 (sample pitch)	5 (sample pitch)	6 (sample pitch)
RPLIDAR S2P	360	1 (sample pitch)	2 (sample pitch)	3 (sample pitch)	5 (sample pitch)	6 (sample pitch)
LDROBOT LD06 360 Scanner	360	5 (sample pitch)	16 (sample pitch)	27 (sample pitch)	43 (sample pitch)	53 (sample pitch)

Sensor	FoV H	1 ft	3 ft	5 ft	8 ft	10 ft
LDROBOT LD19 / LD19P 360 dTOF Scanner	360	5 (sample pitch)	16 (sample pitch)	27 (sample pitch)	43 (sample pitch)	53 (sample pitch)
Waveshare D500 LiDAR Kit	360	4 (sample pitch)	11 (sample pitch)	19 (sample pitch)	31 (sample pitch)	38 (sample pitch)
YDLIDAR X2	360	4 (sample pitch)	11 (sample pitch)	19 (sample pitch)	31 (sample pitch)	38 (sample pitch)
YDLIDAR X4 PRO	360	2 (sample pitch)	7 (sample pitch)	11 (sample pitch)	18 (sample pitch)	23 (sample pitch)
YDLIDAR G4	360	1 (sample pitch)	3 (sample pitch)	5 (sample pitch)	9 (sample pitch)	11 (sample pitch)
YDLIDAR T-mini Plus	360	3 (sample pitch)	9 (sample pitch)	14 (sample pitch)	23 (sample pitch)	29 (sample pitch)
Livox Mid-360 3D LiDAR	360	1 (sample pitch)	2 (sample pitch)	4 (sample pitch)	6 (sample pitch)	8 (sample pitch)
Benewake TFmini Infrared ToF	2.3	12	37	61	98	122
Benewake TF-Luna						
Benewake TFmini-S						
Garmin LIDAR-Lite v3	0.46	2	7	12	20	24
Garmin LIDAR-Lite v4 LED	4.77	25	76	127	203	254
Sharp GP2Y0A21YK0F Analog IR Ranger						
Sharp GP2Y0A02YK0F Analog IR Ranger						
Sharp GP2Y0D810Z0F Digital Distance Switch						
Sharp GP2Y0D805Z0F Digital Distance Switch						
HC-SR04 Ultrasonic Sonar Distance Sensor	15	80	241	401	642	803
RCWL-1601 Ultrasonic Distance Sensor 3V/5V	30	163	490	817	1307	1633
US-100 Ultrasonic Distance Sensor 3V/5V	15	80	241	401	642	803
MaxBotix LV-MaxSonar-EZ0						
MaxBotix LV-MaxSonar-EZ4						
Large Ultrasonic Sensor with Horn UART	40	222	666	1109	1775	2219
VCNL4200 Long Distance IR Proximity and Light Sensor						
VCNL4030 Proximity and Lux Sensor						
VCNL4040 Proximity and Lux Sensor						
APDS9960 Proximity Light RGB and Gesture Sensor						
STEMMA Reflective Photo Interrupter TCRT1000						
AMG8833 Grid-EYE IR Thermal Camera Breakout	60	352	1056	1760	2816	3520
AMG8833 IR Thermal Camera FeatherWing	60	352	1056	1760	2816	3520
MLX90640 32x24 IR Thermal Camera Breakout 55 Degree	55	317	952	1587	2539	3173
MLX90640 32x24 IR Thermal Camera Breakout 110 Degree	110	871	2612	4353	6965	8706
STHS34PF80 IR Presence and Motion Sensor	80	512	1535	2558	4092	5115
MLX90614 Contact-less IR Thermometer 3V	90	610	1829	3048	4877	6096
MLX90614 Contact-less IR Thermometer 5V	90	610	1829	3048	4877	6096
UTi165H Thermal Camera Imager	56	324	972	1621	2593	3241
UTi165K Thermal Camera Imager with USB Video	56	324	972	1621	2593	3241
PIR Motion Sensor	120	1056	3168	5279	8447	10559
Breadboard-friendly Mini PIR Motion Sensor	100	726	2179	3632	5812	7265
Mini PIR Sensor with Time and Sensitivity Control BS612	120	1056	3168	5279	8447	10559
mmWave Radar 24GHz Human Presence Detection Sensor 9m	100	726	2179	3632	5812	7265
mmWave C4001 24GHz Presence Sensor 25m UART	100	726	2179	3632	5812	7265
Gravity mmWave C4001 24GHz Presence Sensor 12m I2C and UART	100	726	2179	3632	5812	7265
Fermion C4002 mmWave Motion and Static Presence Module	120	1056	3168	5279	8447	10559
C1001 60GHz mmWave Indoor Fall Detection Sensor	100	726	2179	3632	5812	7265
mmWave 24GHz Human Presence Sensing Module 6m	120	1056	3168	5279	8447	10559
mmWave 24GHz Microwave Radar Distance Sensor 20m	78	494	1481	2468	3949	4936
Gravity Digital 10.525GHz Microwave Motion Sensor	72	443	1329	2215	3543	4429
5.8GHz Microwave Radar Module	120	1056	3168	5279	8447	10559
mmWave 80GHz Liquid Level Detection Sensor 40m	50	284	853	1421	2274	2843
MR60BHA2 60GHz Breathing and Heartbeat Module	120	1056	3168	5279	8447	10559
MR60FDA2 60GHz Fall Detection Module	120	1056	3168	5279	8447	10559
Person Sensor	110	871	2612	4353	6965	8706
Gravity Offline Edge AI Gesture and Face Detection Sensor	85	559	1676	2793	4469	5586

Against an adult human, 450 mm wide and 1700 mm tall

Sensor	Technology	1 ft	3 ft	5 ft	8 ft	10 ft
VL53L0X ToF Distance Sensor	tof-single	2 obstacle + range	0 nothing	0 nothing	0 nothing	0 nothing
VL53L1X ToF Distance Sensor	tof-single	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing
VL53L4CD ToF Distance Sensor	tof-single	2 obstacle + range	0 nothing	0 nothing	0 nothing	0 nothing
VL53L4CX ToF Distance Sensor	tof-single	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing
VL53L3CX Multi-Target ToF	tof-single	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing
VL6180X ToF Proximity and Ambient Light Sensor	tof-single	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
VL53L4ED Low-Power ToF	tof-single	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing	0 nothing
VL53L5CX 8x8 Multizone ToF	tof-multizone	5 full silhouette	3 living vs not	0 nothing	0 nothing	0 nothing
VL53L7CX Wide-FoV 8x8 Multizone ToF	tof-multizone	5 full silhouette	0 nothing	0 nothing	0 nothing	0 nothing
VL53L8CX 8x8 Multizone ToF	tof-multizone	5 full silhouette	3 living vs not	3 living vs not	2 obstacle + range	0 nothing
VL53L8CH Histogram Multizone ToF	tof-multizone	5 full silhouette	3 living vs not	3 living vs not	2 obstacle + range	0 nothing
VL53L7CH Wide-FoV Histogram Multizone ToF	tof-multizone	5 full silhouette	0 nothing	0 nothing	0 nothing	0 nothing
VL53L9CX 2268-Zone dToF Camera	tof-multizone	5 full silhouette	5 full silhouette	5 full silhouette	4 human vs pet	4 human vs pet
TMF8801 ToF Distance Sensor	tof-single	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing
TMF8806 ToF Distance Sensor	tof-single	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
TMF8820 3x3 Multizone dToF	tof-multizone	4 human vs pet	3 living vs not	2 obstacle + range	2 obstacle + range	2 obstacle + range
TMF8821 4x4 Multizone dToF	tof-multizone	4 human vs pet	3 living vs not	2 obstacle + range	2 obstacle + range	2 obstacle + range
TMF8828 8x8 Multizone dToF	tof-multizone	5 full silhouette	4 human vs pet	3 living vs not	0 nothing	0 nothing
RPLIDAR A1M8 360 Laser Range Scanner	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
RPLIDAR C1 360 dTOF Scanner	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
RPLIDAR A2M12	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
RPLIDAR S2	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
RPLIDAR S2L	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
RPLIDAR S2P	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
LDROBOT LD06 360 Scanner	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
LDROBOT LD19 / LD19P 360 dTOF Scanner	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
Waveshare D500 LiDAR Kit	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
YDLIDAR X2	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
YDLIDAR X4 PRO	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
YDLIDAR G4	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
YDLIDAR T-mini Plus	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
Livox Mid-360 3D LiDAR	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
Benewake TFmini Infrared ToF	tof-lidar	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing
Benewake TF-Luna	tof-lidar	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Benewake TFmini-S	tof-lidar	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Garmin LIDAR-Lite v3	tof-lidar	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Garmin LIDAR-Lite v4 LED	tof-lidar	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Sharp GP2Y0A21YK0F Analog IR Ranger	proximity	1 presence only	0 nothing	0 nothing	0 nothing	0 nothing
Sharp GP2Y0A02YK0F Analog IR Ranger	proximity	1 presence only	1 presence only	0 nothing	0 nothing	0 nothing
Sharp GP2Y0D810Z0F Digital Distance Switch	proximity	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
Sharp GP2Y0D805Z0F Digital Distance Switch	proximity	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
HC-SR04 Ultrasonic Sonar Distance Sensor	ultrasonic	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
RCWL-1601 Ultrasonic Distance Sensor 3V/5V	ultrasonic	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
US-100 Ultrasonic Distance Sensor 3V/5V	ultrasonic	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
MaxBotix LV-MaxSonar-EZ0	ultrasonic	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
MaxBotix LV-MaxSonar-EZ4	ultrasonic	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Large Ultrasonic Sensor with Horn UART	ultrasonic	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
VCNL4200 Long Distance IR Proximity and Light Sensor	proximity	1 presence only	1 presence only	0 nothing	0 nothing	0 nothing
VCNL4030 Proximity and Lux Sensor	proximity	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
VCNL4040 Proximity and Lux Sensor	proximity	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
APDS9960 Proximity Light RGB and Gesture Sensor	proximity	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
STEMMA Reflective Photo Interrupter TCRT1000	proximity	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
AMG8833 Grid-EYE IR Thermal Camera Breakout	thermal-ir-array	4 human vs pet	3 living vs not	1 presence only	1 presence only	1 presence only

Sensor	Technology	1 ft	3 ft	5 ft	8 ft	10 ft
AMG8833 IR Thermal Camera FeatherWing	thermal-ir-array	4 human vs pet	3 living vs not	1 presence only	1 presence only	1 presence only
MLX90640 32x24 IR Thermal Camera Breakout 55 Degree	thermal-ir-array	5 full silhouette	5 full silhouette	4 human vs pet	3 living vs not	3 living vs not
MLX90640 32x24 IR Thermal Camera Breakout 110 Degree	thermal-ir-array	5 full silhouette	3 living vs not	3 living vs not	1 presence only	1 presence only
STHS34PF80 IR Presence and Motion Sensor	thermal-ir-point	1 presence only	1 presence only	1 presence only	1 presence only	1 presence only
MLX90614 Contact-less IR Thermometer 3V	thermal-ir-point	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
MLX90614 Contact-less IR Thermometer 5V	thermal-ir-point	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
UTi165H Thermal Camera Imager	thermal-ir-array	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
UTi165K Thermal Camera Imager with USB Video	thermal-ir-array	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
PIR Motion Sensor	pir	1 presence only	1 presence only	1 presence only	1 presence only	1 presence only
Breadboard-friendly Mini PIR Motion Sensor	pir	0 nothing	0 nothing	0 nothing	1 presence only	1 presence only
Mini PIR Sensor with Time and Sensitivity Control BS612	pir	1 presence only	1 presence only	1 presence only	1 presence only	1 presence only
mmWave Radar 24GHz Human Presence Detection Sensor 9m	mmwave-24ghz	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
mmWave C4001 24GHz Presence Sensor 25m UART	mmwave-24ghz	0 nothing	0 nothing	3 living vs not	3 living vs not	3 living vs not
Gravity mmWave C4001 24GHz Presence Sensor 12m I2C and UART	mmwave-24ghz	0 nothing	0 nothing	3 living vs not	3 living vs not	3 living vs not
Fermion C4002 mmWave Motion and Static Presence Module	mmwave-24ghz	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
C1001 60GHz mmWave Indoor Fall Detection Sensor	mmwave-60ghz	0 nothing	3 living vs not	3 living vs not	3 living vs not	3 living vs not
mmWave 24GHz Human Presence Sensing Module 6m	mmwave-24ghz	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
mmWave 24GHz Microwave Radar Distance Sensor 20m	mmwave-24ghz	0 nothing	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Gravity Digital 10.525GHz Microwave Motion Sensor	mmwave-24ghz	1 presence only	1 presence only	1 presence only	1 presence only	1 presence only
5.8GHz Microwave Radar Module	mmwave-24ghz	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
mmWave 80GHz Liquid Level Detection Sensor 40m	mmwave-60ghz	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
MR60BHA2 60GHz Breathing and Heartbeat Module	mmwave-60ghz	0 nothing	3 living vs not	0 nothing	0 nothing	0 nothing
MR60FDA2 60GHz Fall Detection Module	mmwave-60ghz	3 living vs not	3 living vs not	3 living vs not	0 nothing	0 nothing
Person Sensor	camera	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
Gravity Offline Edge AI Gesture and Face Detection Sensor	camera	4 human vs pet	4 human vs pet	4 human vs pet	4 human vs pet	0 nothing

Against a standing cat, 140 mm wide and 250 mm tall

This is the column that separates a specification sheet from a design. Most parts that look capable against an adult collapse here, and they collapse for three different reasons: the cat is too small to fill a zone, too small to return radar energy, or entirely beneath the beam.

Sensor	Technology	1 ft	3 ft	5 ft	8 ft	10 ft
VL53L0X ToF Distance Sensor	tof-single	2 obstacle + range	0 nothing	0 nothing	0 nothing	0 nothing
VL53L1X ToF Distance Sensor	tof-single	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing
VL53L4CD ToF Distance Sensor	tof-single	2 obstacle + range	0 nothing	0 nothing	0 nothing	0 nothing
VL53L4CX ToF Distance Sensor	tof-single	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing
VL53L3CX Multi-Target ToF	tof-single	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing
VL6180X ToF Proximity and Ambient Light Sensor	tof-single	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
VL53L4ED Low-Power ToF	tof-single	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing	0 nothing
VL53L5CX 8x8 Multizone ToF	tof-multizone	3 living vs not	2 obstacle + range	0 nothing	0 nothing	0 nothing
VL53L7CX Wide-FoV 8x8 Multizone ToF	tof-multizone	3 living vs not	0 nothing	0 nothing	0 nothing	0 nothing
VL53L8CX 8x8 Multizone ToF	tof-multizone	3 living vs not	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing
VL53L8CH Histogram Multizone ToF	tof-multizone	3 living vs not	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing
VL53L7CH Wide-FoV Histogram Multizone ToF	tof-multizone	3 living vs not	0 nothing	0 nothing	0 nothing	0 nothing
VL53L9CX 2268-Zone dToF Camera	tof-multizone	5 full silhouette	4 human vs pet	3 living vs not	3 living vs not	2 obstacle + range
TMF8801 ToF Distance Sensor	tof-single	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing
TMF8806 ToF Distance Sensor	tof-single	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
TMF8820 3x3 Multizone dToF	tof-multizone	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
TMF8821 4x4 Multizone dToF	tof-multizone	3 living vs not	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
TMF8828 8x8 Multizone dToF	tof-multizone	3 living vs not	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing
RPLIDAR A1M8 360 Laser Range Scanner	tof-lidar-scanning	3 living vs not	3 living vs not	2 obstacle + range	2 obstacle + range	2 obstacle + range
RPLIDAR C1 360 dTOF Scanner	tof-lidar-scanning	3 living vs not	3 living vs not	2 obstacle + range	2 obstacle + range	2 obstacle + range
RPLIDAR A2M12	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
RPLIDAR S2	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
RPLIDAR S2L	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
RPLIDAR S2P	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
LDROBOT LD06 360 Scanner	tof-lidar-scanning	3 living vs not	3 living vs not	2 obstacle + range	2 obstacle + range	2 obstacle + range

Sensor	Technology	1 ft	3 ft	5 ft	8 ft	10 ft
LDROBOT LD19 / LD19P 360 dTOF Scanner	tof-lidar-scanning	3 living vs not	3 living vs not	2 obstacle + range	2 obstacle + range	2 obstacle + range
Waveshare D500 LiDAR Kit	tof-lidar-scanning	3 living vs not	3 living vs not	2 obstacle + range	2 obstacle + range	2 obstacle + range
YDLIDAR X2	tof-lidar-scanning	3 living vs not	3 living vs not	2 obstacle + range	2 obstacle + range	2 obstacle + range
YDLIDAR X4 PRO	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	2 obstacle + range	2 obstacle + range
YDLIDAR G4	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
YDLIDAR T-mini Plus	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	2 obstacle + range	2 obstacle + range
Livox Mid-360 3D LiDAR	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
Benewake TFmini Infrared ToF	tof-lidar	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing
Benewake TF-Luna	tof-lidar	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Benewake TFmini-S	tof-lidar	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Garmin LIDAR-Lite v3	tof-lidar	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Garmin LIDAR-Lite v4 LED	tof-lidar	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Sharp GP2Y0A21YK0F Analog IR Ranger	proximity	1 presence only	0 nothing	0 nothing	0 nothing	0 nothing
Sharp GP2Y0A02YK0F Analog IR Ranger	proximity	1 presence only	1 presence only	0 nothing	0 nothing	0 nothing
Sharp GP2Y0D810Z0F Digital Distance Switch	proximity	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
Sharp GP2Y0D805Z0F Digital Distance Switch	proximity	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
HC-SR04 Ultrasonic Sonar Distance Sensor	ultrasonic	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
RCWL-1601 Ultrasonic Distance Sensor 3V/5V	ultrasonic	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
US-100 Ultrasonic Distance Sensor 3V/5V	ultrasonic	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
MaxBotix LV-MaxSonar-EZ0	ultrasonic	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
MaxBotix LV-MaxSonar-EZ4	ultrasonic	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Large Ultrasonic Sensor with Horn UART	ultrasonic	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
VCNL4200 Long Distance IR Proximity and Light Sensor	proximity	1 presence only	1 presence only	0 nothing	0 nothing	0 nothing
VCNL4030 Proximity and Lux Sensor	proximity	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
VCNL4040 Proximity and Lux Sensor	proximity	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
APDS9960 Proximity Light RGB and Gesture Sensor	proximity	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
STEMMA Reflective Photo Interrupter TCRT1000	proximity	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
AMG8833 Grid-EYE IR Thermal Camera Breakout	thermal-ir-array	3 living vs not	1 presence only	0 nothing	0 nothing	0 nothing
AMG8833 IR Thermal Camera FeatherWing	thermal-ir-array	3 living vs not	1 presence only	0 nothing	0 nothing	0 nothing
MLX90640 32x24 IR Thermal Camera Breakout 55 Degree	thermal-ir-array	5 full silhouette	3 living vs not	1 presence only	1 presence only	1 presence only
MLX90640 32x24 IR Thermal Camera Breakout 110 Degree	thermal-ir-array	3 living vs not	1 presence only	1 presence only	0 nothing	0 nothing
STHS34PF80 IR Presence and Motion Sensor	thermal-ir-point	1 presence only	1 presence only	1 presence only	1 presence only	1 presence only
MLX90614 Contact-less IR Thermometer 3V	thermal-ir-point	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
MLX90614 Contact-less IR Thermometer 5V	thermal-ir-point	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
UTi165H Thermal Camera Imager	thermal-ir-array	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
UTi165K Thermal Camera Imager with USB Video	thermal-ir-array	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
PIR Motion Sensor	pir	1 presence only	1 presence only	1 presence only	1 presence only	1 presence only
Breadboard-friendly Mini PIR Motion Sensor	pir	0 nothing	0 nothing	0 nothing	1 presence only	1 presence only
Mini PIR Sensor with Time and Sensitivity Control BS612	pir	1 presence only	1 presence only	1 presence only	1 presence only	1 presence only
mmWave Radar 24GHz Human Presence Detection Sensor 9m	mmwave-24ghz	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
mmWave C4001 24GHz Presence Sensor 25m UART	mmwave-24ghz	0 nothing	0 nothing	3 living vs not	3 living vs not	3 living vs not
Gravity mmWave C4001 24GHz Presence Sensor 12m I2C and UART	mmwave-24ghz	0 nothing	0 nothing	3 living vs not	3 living vs not	3 living vs not
Fermion C4002 mmWave Motion and Static Presence Module	mmwave-24ghz	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
C1001 60GHz mmWave Indoor Fall Detection Sensor	mmwave-60ghz	0 nothing	3 living vs not	3 living vs not	3 living vs not	3 living vs not
mmWave 24GHz Human Presence Sensing Module 6m	mmwave-24ghz	3 living vs not	3 living vs not	3 living vs not	0 nothing	0 nothing
mmWave 24GHz Microwave Radar Distance Sensor 20m	mmwave-24ghz	0 nothing	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Gravity Digital 10.525GHz Microwave Motion Sensor	mmwave-24ghz	1 presence only	1 presence only	1 presence only	1 presence only	1 presence only
5.8GHz Microwave Radar Module	mmwave-24ghz	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
mmWave 80GHz Liquid Level Detection Sensor 40m	mmwave-60ghz	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
MR60BHA2 60GHz Breathing and Heartbeat Module	mmwave-60ghz	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
MR60FDA2 60GHz Fall Detection Module	mmwave-60ghz	3 living vs not	0 nothing	0 nothing	0 nothing	0 nothing
Person Sensor	camera	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
Gravity Offline Edge AI Gesture and Face Detection Sensor	camera	4 human vs pet	4 human vs pet	4 human vs pet	4 human vs pet	0 nothing

Against a 30 mm chair leg

The thin, cold, static obstacle. Thermal and PIR score zero by physics. Presence radar scores zero by design. What is left is the honest list of parts that will stop the robot hitting furniture.

Sensor	Technology	1 ft	3 ft	5 ft	8 ft	10 ft
VL53LOX ToF Distance Sensor	tof-single	2 obstacle + range	0 nothing	0 nothing	0 nothing	0 nothing
VL53L1X ToF Distance Sensor	tof-single	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing
VL53L4CD ToF Distance Sensor	tof-single	2 obstacle + range	0 nothing	0 nothing	0 nothing	0 nothing
VL53L4CX ToF Distance Sensor	tof-single	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing
VL53L3CX Multi-Target ToF	tof-single	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing
VL6180X ToF Proximity and Ambient Light Sensor	tof-single	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
VL53L4ED Low-Power ToF	tof-single	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing	0 nothing
VL53L5CX 8x8 Multizone ToF	tof-multizone	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing	0 nothing
VL53L7CX Wide-FoV 8x8 Multizone ToF	tof-multizone	2 obstacle + range	0 nothing	0 nothing	0 nothing	0 nothing
VL53L8CX 8x8 Multizone ToF	tof-multizone	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing
VL53L8CH Histogram Multizone ToF	tof-multizone	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing
VL53L7CH Wide-FoV Histogram Multizone ToF	tof-multizone	2 obstacle + range	0 nothing	0 nothing	0 nothing	0 nothing
VL53L9CX 2268-Zone dToF Camera	tof-multizone	4 human vs pet	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
TMF8801 ToF Distance Sensor	tof-single	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing
TMF8806 ToF Distance Sensor	tof-single	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
TMF8820 3x3 Multizone dToF	tof-multizone	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
TMF8821 4x4 Multizone dToF	tof-multizone	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
TMF8828 8x8 Multizone dToF	tof-multizone	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing
RPLIDAR A1M8 360 Laser Range Scanner	tof-lidar-scanning	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing
RPLIDAR C1 360 dTOF Scanner	tof-lidar-scanning	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing
RPLIDAR A2M12	tof-lidar-scanning	3 living vs not	3 living vs not	2 obstacle + range	2 obstacle + range	2 obstacle + range
RPLIDAR S2	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	2 obstacle + range	2 obstacle + range
RPLIDAR S2L	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	2 obstacle + range	2 obstacle + range
RPLIDAR S2P	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	2 obstacle + range	2 obstacle + range
LDROBOT LD06 360 Scanner	tof-lidar-scanning	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing
LDROBOT LD19 / LD19P 360 dTOF Scanner	tof-lidar-scanning	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing
Waveshare D500 LiDAR Kit	tof-lidar-scanning	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing
YDLIDAR X2	tof-lidar-scanning	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing
YDLIDAR X4 PRO	tof-lidar-scanning	3 living vs not	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
YDLIDAR G4	tof-lidar-scanning	3 living vs not	3 living vs not	2 obstacle + range	2 obstacle + range	2 obstacle + range
YDLIDAR T-mini Plus	tof-lidar-scanning	3 living vs not	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Livox Mid-360 3D LiDAR	tof-lidar-scanning	3 living vs not	3 living vs not	2 obstacle + range	2 obstacle + range	2 obstacle + range
Benewake TFmini Infrared ToF	tof-lidar	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing
Benewake TF-Luna	tof-lidar	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Benewake TFmini-S	tof-lidar	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Garmin LIDAR-Lite v3	tof-lidar	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Garmin LIDAR-Lite v4 LED	tof-lidar	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Sharp GP2Y0A21YK0F Analog IR Ranger	proximity	1 presence only	0 nothing	0 nothing	0 nothing	0 nothing
Sharp GP2Y0A02YK0F Analog IR Ranger	proximity	1 presence only	1 presence only	0 nothing	0 nothing	0 nothing
Sharp GP2Y0D810Z0F Digital Distance Switch	proximity	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
Sharp GP2Y0D805Z0F Digital Distance Switch	proximity	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
HC-SR04 Ultrasonic Sonar Distance Sensor	ultrasonic	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
RCWL-1601 Ultrasonic Distance Sensor 3V/5V	ultrasonic	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
US-100 Ultrasonic Distance Sensor 3V/5V	ultrasonic	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
MaxBotix LV-MaxSonar-EZ0	ultrasonic	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
MaxBotix LV-MaxSonar-EZ4	ultrasonic	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Large Ultrasonic Sensor with Horn UART	ultrasonic	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
VCNL4200 Long Distance IR Proximity and Light Sensor	proximity	1 presence only	1 presence only	0 nothing	0 nothing	0 nothing
VCNL4030 Proximity and Lux Sensor	proximity	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
VCNL4040 Proximity and Lux Sensor	proximity	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
APDS9960 Proximity Light RGB and Gesture Sensor	proximity	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
STEMMA Reflective Photo Interrupter TCRT1000	proximity	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing

Sensor	Technology	1 ft	3 ft	5 ft	8 ft	10 ft
AMG8833 Grid-EYE IR Thermal Camera Breakout	thermal-ir-array	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
AMG8833 IR Thermal Camera FeatherWing	thermal-ir-array	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
MLX90640 32x24 IR Thermal Camera Breakout 55 Degree	thermal-ir-array	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
MLX90640 32x24 IR Thermal Camera Breakout 110 Degree	thermal-ir-array	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
STHS34PF80 IR Presence and Motion Sensor	thermal-ir-point	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
MLX90614 Contact-less IR Thermometer 3V	thermal-ir-point	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
MLX90614 Contact-less IR Thermometer 5V	thermal-ir-point	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
UTi165H Thermal Camera Imager	thermal-ir-array	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
UTi165K Thermal Camera Imager with USB Video	thermal-ir-array	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
PIR Motion Sensor	pir	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
Breadboard-friendly Mini PIR Motion Sensor	pir	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
Mini PIR Sensor with Time and Sensitivity Control BS612	pir	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
mmWave Radar 24GHz Human Presence Detection Sensor 9m	mmwave-24ghz	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
mmWave C4001 24GHz Presence Sensor 25m UART	mmwave-24ghz	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
Gravity mmWave C4001 24GHz Presence Sensor 12m I2C and UART	mmwave-24ghz	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
Fermion C4002 mmWave Motion and Static Presence Module	mmwave-24ghz	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
C1001 60GHz mmWave Indoor Fall Detection Sensor	mmwave-60ghz	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
mmWave 24GHz Human Presence Sensing Module 6m	mmwave-24ghz	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
mmWave 24GHz Microwave Radar Distance Sensor 20m	mmwave-24ghz	0 nothing	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
Gravity Digital 10.525GHz Microwave Motion Sensor	mmwave-24ghz	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
5.8GHz Microwave Radar Module	mmwave-24ghz	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
mmWave 80GHz Liquid Level Detection Sensor 40m	mmwave-60ghz	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
MR60BHA2 60GHz Breathing and Heartbeat Module	mmwave-60ghz	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
MR60FDA2 60GHz Fall Detection Module	mmwave-60ghz	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
Person Sensor	camera	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing
Gravity Offline Edge AI Gesture and Face Detection Sensor	camera	4 human vs pet	4 human vs pet	4 human vs pet	4 human vs pet	0 nothing

What is detectable at 1, 3, 5, 8 and 10 feet

Capability falls off in steps, not smoothly. Each step has a different physical cause, and each removes a whole class of sensor at once. This chapter walks the five survey distances in order: what still reaches, what it resolves, what just dropped out, and what the distance is good for.

Three reference targets are used throughout, from range-capability-modelling.md: a human torso 450 x 700 mm, a cat or small dog 250 x 200 mm in side profile, and a chair leg 40 x 400 mm. Every range figure below is for one of those targets, not for a wall filling the field of view. That distinction is the whole chapter. Fields of view are horizontal x vertical unless "diagonal" appears.

1 ft / 305 mm - every optical modality works, radar does not

Every optical candidate returns something useful at 305 mm, with one standing exception: a thermal array never sees the chair leg, at this or any distance, because furniture sits at room temperature. A VL53L1X at about 19 deg horizontal (27 deg diagonal) covers 103 x 103 mm here, so all three reference targets fill its field of view - fill fraction 1.000 for human, cat and chair leg alike - and the reading is taken under the same condition ST specifies for its 3.60 m dark-room figure (datasheet-verified, [ST DocID031281 Rev 3 Table 6](#): white 88 percent, 100 ms timing budget, dark, target covers full FoV). That full-FoV condition is what 1 ft buys, and it does not survive to 3 ft. The wider multizone fields are already partly unfilled at 305 mm: a cat fills 0.78 of the VL53L5CX field and 0.40 of the VL53L7CX field.

- SPAD 8x8 multizone (VL53L5CX / L8CX, 45 x 45 deg): a human is 7.1 x 26.9 zones, a depth silhouette. A cat is 2.2 x 4.0 zones. A chair leg fills 0.48 of a zone - a valid range, no shape (computed, data/capability.json).
- Thermal AMG8833 (8x8, 60 x 60 deg): the pixel is 44 mm, so a human torso is 10.2 x 15.9 pixels and a cat 5.7 x 4.5 pixels. Both are full silhouettes.
- Scanning lidar (RPLIDAR C1, 0.72 deg at 10 Hz): 3.8 mm arc spacing gives about 65 returns across a cat and 29 across a human shin - but only if the scan plane sits below 250 mm.
- 24 GHz radar: a dead loss. The 250 MHz ISM allocation forces a 600 mm range cell ([Hilink LD2410 manual V1.03](#), "Distance resolution 0.75 m"), so 1 ft sits inside the first bin, and the DFRobot C4001's vendor-stated 1.2 m minimum measurable distance means it reports nothing here at all. A 60 GHz MIMO part (IWR6843 / IWR6432) is the exception: a 4 GHz sweep gives a 37.5 mm range cell and a 51-77 mm cross-range spot at 1 ft, with micro-Doppler on top.

Use 1 ft for the emergency-stop ring, built on short-mode SPAD ToF: VL53L1X short mode holds 136 cm dark and 135 cm at 200 kcps/SPAD, roughly full sun, because its narrow histogram window rejects most ambient photons (datasheet-verified, Table 4).

3 ft / 914 mm - the flood illuminators start to thin

The VL53L0X (0.80 m on a real target in the dark), VL53L4CD (0.45 m) and wide-field VL53L7CX / L7CH (0.80 m) all drop out before 3 ft, and the cause is fill factor, not raw sensitivity. Once a target is smaller than the field of view, the fraction it fills falls as $1/d^2$, so returned signal falls as $1/d^4$ - the fourth-power law radar obeys. The crossover model gives a real reach of $\sqrt{d_{\text{cross}} * d_{\text{full}}}$. Against ST's dark-room numbers (dark, white 88 percent, long mode, 100 ms), a VL53L1X sold as a 4 m sensor reaches 2.45 m on a human torso, 1.55 m on a cat and 1.16 m on a chair leg. Wide fields are worse, not better: the VL53L7CX spreads the same VCSEL power over 60 x 60 deg and manages 1.32 m on a torso under that same dark, white 88 percent condition.

Ambient light compresses all of it. ST Table 7 (long mode, 100 ms, full FoV) measures white 88 percent reach falling 360 cm dark, 166 cm at 50 kcps/SPAD ("a sunny day from behind a window"), 73 cm at 200 kcps/SPAD (the same with direct illumination on the sensor, about 100 klux equivalent). On 17 percent grey it runs 170, 114, 68 cm - bright and dark targets converge on one 70 cm floor once ambient dominates. In a sunbeam, no SPAD part here reaches 3 ft on anything in long mode. Short mode is the single exception: it holds 135 cm at 200 kcps/SPAD, and it is capped there.

Thermal and lidar are untroubled: an AMG8833 puts a human at 3.4 x 5.3 pixels and a cat at 1.9 x 1.5 pixels, so aspect ratio still separates them; an MLX90640 55 x 35 deg gives 15 x 29 pixels on the human; the C1 draws about 10 returns per shin.

Radar does not change across this step. A 60 GHz MIMO part still gives a human range, angle and micro-Doppler, with a 37.5 mm range cell and a 153-230 mm cross-range spot at 3 ft. A 24 GHz ISM module still has one 600 mm bin and, in the LD2410 class, no angle estimate at all.

Use 3 ft as the classification handoff - the last distance at which a cheap suite calls human versus pet versus object with high confidence.

5 ft / 1524 mm - 8x8 thermal stops classifying

Two things end here. The AMG8833 falls to 2.05 x 3.18 pixels on a human and 1.14 x 0.91 pixels on a cat, a one-pixel blob. The VL53L5CX is out on real targets, at 1.30 m reach in the dark.

Sensor	Human	Cat	Chair leg
VL53L1X (19 deg H)	obstacle + range, dark only	marginal, dark only	out
VL53L8CX / L8CH (45 x 45 deg)	1.4 x 5.4 zones, living vs not	0.44 zone, range only	0.10 zone, range only
VL53L9CX (55 x 42 deg, 2268 zones)	7.7 x 39.2 zones, silhouette	2.4 x 5.8 zones, living vs not	0.51 zone, range only
MLX90640 BAB (55 x 35 deg)	9.1 x 17.5 px	5.0 x 5.0 px, pet vs human	not thermal
MLX90640 BAA (110 x 75 deg)	3.3 x 7.2 px, pet vs human	1.8 x 2.1 px, size only	not thermal
RPLIDAR C1	about 6 hits per shin	about 13 hits	2.1 hits

Radar is not range-limited here. Received power scales with radar cross section to the first power but maximum range only to its fourth root, so a 15 dB RCS deficit costs a factor of 2.37 in range, not 30: a module rated 12 m on a human still makes 4.9 m on a 0.01 m^2 pet and 7.3 m on a 0.05 m^2 pet. What stops a hobby radar seeing a pet is the elevation pattern. A C4001 with a vendor-stated 100 x 40 deg beam, mounted 600 mm up and aimed level, puts its lower half-power edge on the floor at 1.65 m; a nearer cat is below the beam.

The sources disagree on RCS. data/capability.json carries 0.7 m^2 for an adult; the research lane uses the EU Joint Research Centre's measured frequency-averaged mean of -4.3 dBsm, about 0.37 m^2, at 23-28 GHz (JRC78619). Prefer the JRC figure: it is measured, where 1.0 m^2 is the top of RFbeam's vendor pedestrian band of 0.1 to 1 m^2, and 0.7 m^2 sits inside that band with no measurement behind it. No measured pet RCS at 24 or 60 GHz was found, so every cat RCS here is inferred.

Use 5 ft for approach planning and for "is a person in the room", not for shape.

8 ft / 2438 mm - optical shape information is essentially gone

The VL53L1X and VL53L3CX drop out at 1.70 m on real targets in the dark. Only a white 88 percent torso in a dark room takes a VL53L1X to 8 ft at all, and its 2.45 m clears 2438 mm by 12 mm, which is not a margin to design to. What remains optically is single-zone range from the TMF8801 and TMF8806, sub-zone fill from the TMF8820 / 8821 and VL53L8CX, and the VL53L9CX, which still shows a human at 4.8 x 24.5 zones and a cat at 1.5 x 3.6 zones. Treat the VL53L8CX row with suspicion: the catalogue credits it with 2.45 m dark-room reach, but the research lane could not obtain ST's VL53L8CX ranging tables at all, so that figure is interpolated between the L5CX and L7CX rather than datasheet-verified. Do not design to it.

Thermal splits cleanly. The MLX90640 55 x 35 deg holds a human at 5.7 x 10.9 pixels and a cat at 3.1 x 3.1 pixels, and still classifies. The AMG8833 does not: a cat covers 0.40 of one pixel, so the pixel averages cat against cooler floor and the apparent rise falls to 3.23 K against an 8 K true contrast. That clears the 0.05 C NETD

easily, so the cat is detected - but it is indistinguishable from a sunlit carpet patch or a laptop charger, and ± 2.5 C absolute accuracy gives no help. Reality is worse: fur insulates, so coat surface temperature in a 22 C room is nearer 26-29 C than 34 C skin, true contrast nearer 4-6 K, and the 8 ft apparent rise nearer 1.6-2.4 K once the same 0.40 fill fraction is applied (inferred; the research lane states this case at 10 ft as 1.0-1.6 K).

Lidar thins the same way: 3.6 returns across a shin and 1.3 across a 40 mm chair leg, so any cluster filter with a two-point minimum deletes the chair leg.

Use 8 ft for room-level awareness and path pre-planning, never for anything the robot must act on inside a second. Thermal arrays are slow - the AMG8833 caps at 10 fps and the MLX90640's 0.1 K NETD is specified at 1 Hz, which is 500 mm of travel per frame at 0.5 m/s.

10 ft / 3048 mm - presence, plus exactly one honest classifier

At 3048 mm an AMG8833 pixel covers 440 mm, so a 450 mm human torso is 1.02 pixels wide and 1.59 tall. No shape survives; the array reports "something warm in that direction", which is what a two-dollar PIR reports. Panasonic's "human detection distance: 7 m or less (reference value)" is not wrong. It answers detection, not classification.

- SPAD ToF: gone, except the VL53L9CX at 3.8 x 19.6 zones on a human and one zone on a cat, and single-zone parts such as the TMF8806 that return a range with no shape.
- MLX90640 BAB: human 4.5 x 8.7 px, cat 2.5 x 2.5 px - the only thermal option in the study that classifies at 3 m. A 360 deg ring needs 7 units at \$74.95, or \$524.65 (vendor-stated, 2026-09-12). The 110 deg BAA variant needs 4 units but puts a cat at 0.92 x 1.03 px, sub-pixel, and was out of stock on the same date.
- Lidar: about 2.9 returns per shin, 6.5 across a cat, 1.0 on a chair leg - and it is a horizontal slice carrying no class information. A 110 mm circular return is a shin, a table leg or a sitting cat.
- Radar: 24 GHz gives presence and a 600 mm range gate, 1.7 robot widths. A 60 GHz MIMO part gives range to about 37.5 mm and a cross-range spot of 509-766 mm here. Classification needs a velocity-resolved micro-Doppler spectrogram and a trained classifier, which TI offers only in its high-performance mode, and which stops working the moment the animal holds still.

The honest statement. At 10 ft a hobby budget can know that something warm and low is over there and roughly in which direction, and a \$525 MLX90640 55 deg ring can add that the blob is pet-sized rather than person-sized - but nothing in this price class can confirm that a stationary, sleeping cat at 10 ft is a cat rather than a sunlit cushion, because the one channel that separates them, gait micro-Doppler, requires the cat to move.

Telling a human from a pet from a box

The discriminant that works on this budget is **height above the floor, gated by warmth**. Thermal answers "is it alive". An absolute height answers "is it a person". Nothing cheap answers "is it a cat or a dog", and the published methods that come closest need raw radar data that no module in this catalogue emits. Design to that and the robot behaves correctly. Design to respiration rate, or to a pet-immune motion sensor, and it will call a calm adult a dog. Five physically independent discriminants exist at this price: surface temperature, sub-centimetre periodic body motion, gross motion kinematics, geometry, and learned appearance.

mmWave micro-Doppler: excellent in the literature, unavailable in the parts

Wavelength sets what a radar can see. Doppler shift for a radial velocity v is $f_d = 2v/\lambda$, and the phase written by a chest displacement dx is $4\pi dx/\lambda$:

Band	Wavelength	Doppler per 1 m/s	Phase per 1 mm of chest motion
24 GHz	12.49 mm	160 Hz	1.01 rad (57.6 deg)
60 GHz	4.997 mm	400 Hz	2.51 rad (144 deg)
77 GHz	3.893 mm	514 Hz	3.23 rad (185 deg)

Exact arithmetic from $c = 299\,792\,458$ m/s. That ratio is why 60 GHz displaced 24 GHz for vital-signs work: an assumed 0.5 mm cardiac displacement - assumed, see the folklore note below - writes a 72 deg phase excursion at 60 GHz against 29 deg at 24 GHz.

Respiration rate does not separate a human from a dog. Measured by IR-UWB radar against a contact reference ([PMC7070589](#)), four resting beagles breathe at 0.219-0.412 Hz, or **13-25 per minute**; five cats at 0.447-0.556 Hz, or **27-33 per minute**. A resting adult sits near 0.35 Hz, about 21 per minute ([PMC6888617](#)). The popular "human 12-20, pet 20-40" heuristic holds for cats and panting dogs only, so if `respiration_rate > 20` then pet mislabels in both directions. The rate is *measured* well - over 95 percent against a contact pressure sensor - but it is not discriminative. Heart rate separates better (human 60-100 bpm, dog 98-126, cat 119-173), yet it needs a sub-millimetre displacement, and the 1.54 percent error against ECG came from **one beagle under isoflurane anaesthesia**; those authors state the method fails on animals in motion. The chest displacements everyone repeats, 5-10 mm respiration and 0.5-1 mm cardiac, are **unverified folklore**: the cited paper contains neither them nor the filter bands.

Two published methods do separate a human from a pet, and both need the radar and the target still. The respiratory-and-heartbeat energy ratio (RHER) exploits the fact that a biped thorax couples cardiac motion to the body surface differently from a quadruped one; humans scored significantly higher than dogs, cats and rabbits on a XeThru/Novelda X4M02 IR-UWB radar, 7.29 GHz centre, 1.4 GHz bandwidth, every subject at **1 m** - 10 humans, 5 dogs, 5 cats, 5 rabbits ([PMC6888617](#)). Those authors state it cannot separate a person and a pet sleeping together. The second is a 19-feature set - energy, frequency, wavelet entropy, correlation coefficient - cut to the top 8 by recursive feature elimination and fed to an SVM ([EURASIP JASP 2021](#)); its headline accuracy could not be retrieved past the publisher's auth redirect, so no accuracy figure is quoted for it here.

Gait classification works and costs time you do not have. For a moving target the discriminant is the limb spectrogram, not a rate. FMCW micro-Doppler reached **97.66 percent** on human versus dog, horse and cow ([UCT MSc](#)) - outdoors, single-target, large animals. In a cluttered operational environment a GMM plus HMM on mel-cepstrum coefficients reached **75 percent at 250 ms of dwell, rising to only about 90 percent at 1.25 s** ([Expert Syst. Appl. 2018](#)). A robot braking before a collision has 200-400 ms, the 75 percent end of that curve. Cadence alone is weaker again - about 2 Hz human step rate against roughly 4 Hz footfall for a trotting small dog, with a jogger, a walking dog and a child all inside the ambiguity band.

Three hard limits sit on top:

- **Static-clutter rejection deletes still people.** Every FMCW pipeline opens with a moving-target indicator that removes the zero-Doppler bin, where a standing person lives. On an Infineon BGT60TR13C, CA-CFAR after MTI detected quasi-static humans at **68.3 / 51.3 / 57.7 percent** across three subjects, OS-CFAR at 78.8 / 68.3 / 69.9 percent ([arXiv 2602.14001](#)). One still person in three is missed, and the same filter deletes a sleeping cat.

- **Small animals fall below the detection floor.** A TI IWR6843ISK against a 3 kg, 400 mm dog in a 4 m room: **46.1 percent sensitivity for one radar without tracking**, 75.2 percent tuned, 97.10 percent only with **four radars fused and tracked** (PMC10975529). For scale, human RCS at 77 GHz is **-6.86 to -3.51 dBsm** (frequency-and-azimuth average, 2 adults, 4 garments, anechoic chamber, 3.4 m, antennas 0.8 m, JRC78619). No published mmWave RCS for a cat or small dog was found by the source lane, which records the result of a search and not a proven absence; either way, do not invent one by scaling for mass.
- **A moving robot invalidates all of it.** MTI assumes a static sensor; on a driving robot every wall has Doppler, and the hobby modules run a fixed, non-compensated MTI in closed firmware.

The blunt point: no DFRobot module exposes the raw data any of this needs. From the output column of data/sensors.csv, the C4001 family (\$12.90-\$13.90) emits target count, range, radial speed and energy; the Fermion C4002 (\$8.90) adds a per-gate energy bitmap; the C1001 60 GHz module (\$29.00) emits a posture, a fall flag and a **scalar** breathing and heart rate. The only spectral output in the catalogue, from the 24 GHz 20 m distance sensor (\$65.90), is a 126-bin range-FFT on a part marked as not detecting static humans at all. **None emits a range-Doppler map, a slow-time IQ stream or a spectrogram**, so every result above is unreachable from these modules at any budget.

Thermal: says alive, and little about which

Thermal has one decisive advantage over radar and PIR: **it sees a motionless target**. A sleeping cat is invisible to MTI radar and to PIR, and plain to a thermal array.

Surface	Temperature	Condition
Human forehead, bare	36.2 +/- 0.16 degC at 20 degC ambient; 36.0-36.5 degC over 14-32 degC ambient	climate chamber, handheld IR imager
Dog femoral, short hair	31.77 +/- 0.19 degC	21 degC room
Dog femoral, long double coat	28.14 +/- 0.31 degC	21 degC room
Dog / cat ocular region	32.3-36.9 / 33.9-37.3 degC	min/max within one image of one animal , not a cohort
Emissivity used in canine IRT	0.95	standard assumption

Human PMC9740153, animals PMC8944468, emissivity PMC10497125; all paper-verified.

Fur is insulation, so a pet is barely warm. A long-coated dog's flank at 28.1 degC is 6 K above a 22 degC room and **2 K** above a 26 degC summer room. Only eyes, nose, inner ear and paw pads run near core temperature, and those are a few square centimetres. A human is easier only because bare skin holds 36.2 degC almost independently of ambient - and a robot mostly sees clothing, not skin. **Fill-factor dilution finishes the job:** a pixel reports about $T_{\text{apparent}} = f T_{\text{target}} + (1 - f) T_{\text{background}}$, so a 30 degC cat filling a quarter of a pixel against a 22 degC room reads 24 degC - inside the AMG8833's +/-2.5 degC accuracy band, and indistinguishable from a laptop or a sunlit floor patch.

Pixel count is therefore the whole specification. Applying the simplified Johnson criteria (about 1 cycle to detect, 3 to recognise, 6 to identify; 1 cycle is about 2 pixels across the critical dimension) to the computed footprint tables:

Task	Grid-EYE 8x8, 60 x 60 deg, \$44.95	MLX90640 32x24, 55 x 35 deg, \$74.95
Detect a human (2 px on a 400 mm torso)	out to about 5 ft	well beyond 10 ft
Recognise a human as human-shaped (6 px)	never, at any range	beyond 3 ft, comfortably at 10 ft
Detect a cat (2 px on 230 mm)	out to about 3 ft	out to about 10 ft
Recognise a cat as a cat (6 px)	never	out to about 5 ft

An 8x8 array is a warm-blob presence detector, not a classifier. A 32x24 array is the cheapest thermal part carrying a height-and-aspect classifier. Both fields of view above are horizontal by vertical, not diagonal.

Multizone ToF: the silhouette, and the zone counts it needs

ToF supplies the metric quantity everything else lacks: **absolute height above the floor**, from the zone elevation angle and the measured range. Three discriminants follow, most robust first.

1. **Full-column occupancy.** Below about 5 ft a human returns depth in every row of an 8x8 grid; a cat returns depth in the bottom one or two rows with free space above. A row-histogram threshold, not machine learning, and the most reliable

- cheap human/pet discriminant available. It needs the sensor low and pitched up, which a 600-1200 mm robot can do.
- Depth-profile rigidity.** A box gives a flat depth plane with near-zero frame-to-frame variance; a living body gives a curved surface with 10-40 mm of per-zone jitter from breathing and micro-sway. Per-zone variance over 1 s separates furniture from creature with no temperature or Doppler information at all.
 - Top-of-return height**, which makes "is the top of this above 1.0 m" directly answerable.

Zone counts from data/capability.json, using the working figure of **45 x 45 deg horizontal by vertical** (63 deg diagonal for the VL53L5CX, 65 deg for the VL53L8CX; the diagonal is separately measured and cannot be converted):

Part	1 ft	3 ft	5 ft	8 ft
VL53L5CX 8x8, \$19.95	human 7.1 x 26.9 zones; cat 2.2 x 4.0	human 2.4 x 9.0; cat 0.74 zone	past 1.30 m dark reach	-
VL53L8CX 8x8, \$24.95	human 7.1 x 26.9; cat 2.2 x 4.0	human 2.4 x 9.0; cat 0.74	human 1.4 x 5.4; cat 0.44	human 0.89 zone
VL53L9CX 2268 zones, \$80.39	human 38 x 196; cat 11.9 x 28.8	cat 4.0 x 9.6 zones	cat 2.4 x 5.8	human 4.8 x 24.5

8x8 gives a height profile, not a silhouette, past about 3 ft; a cat drops below one zone at 3 ft and is unresolvable by 8 ft. The 4x4 mode is a collision sensor only. ST's own people-counting case study on this hardware reports 98.14 percent on a three-class problem - but from **32 successive frames at 15 Hz**, a 2.1 s window, not a single frame.

Every reach in that table is the catalogue's **dark-room figure on a real target, not on a white 88 percent chart**: 1.30 m for the VL53L5CX and 2.45 m for the VL53L8CX, falling to 1.10 m and 1.55 m in sunlight (data/sensors.csv). The 4 m datasheet maximum belongs to a white 88 percent target filling the field of view in the dark, and is not a room figure. A black cat is a low-reflectance target and shortens all of them again.

PIR: a wake-up, and a masking trick that does not transfer

A pyroelectric element outputs one scalar voltage proportional to the rate of change of incident long-wave IR flux: no image, no range, no rate, no direction. It cannot classify, and it is *least* sensitive to radial motion, so a person walking straight at the robot is its worst case. It also does not see through glass, must not be aimed at a window, and responds to warm airflow - on a mobile robot its own motor and driver heat, and the draught past the lens, are false-trigger sources.

Alarm-industry "pet immunity" is a **geometric** trick, not a classification one, and it costs nothing to copy: the Fresnel lens omits the lowest floor-looking beam layers, so an animal below the lowest beam cannot trigger. The Bosch Blue Line Gen2 ISC-BPR2-WP12 rates two animals to 45 lb total on a 77-zone, 7-layer lens - **mounted at 2.2-2.75 m**; the Resideo IS335 rates 80 lb. The mechanism is entirely the mounting height: at 2.2 m, with the lowest retained beam taken to meet the floor at 12 m - an illustrative intercept, not a Bosch specification - that beam is still 1.10 m up at 6 m range - above any dog, while catching a 1700 mm human.

At 1.0 m on a robot the trick inverts. The beam is then 0.50 m up at 3 m range, so a cat and a human knee subtend similar depression angles, and the human's torso and head sit *above* the sensor, outside a downward-fanning lens entirely: a pet-immune PIR on a low robot can see the pet and miss the person. **Buy a plain PIR at \$1.95-\$3.95 as a near-zero-power wake-up, and put the height gate in the ToF domain where a real elevation measurement exists.** The concept transfers; the part does not.

Camera plus a person model: the only cheap true classifier

The advantage nobody states plainly: **COCO contains** person, cat **and** dog **as separate classes**. Any stock YOLO or MobileNet detector trained on COCO does the whole discrimination out of the box, with no custom dataset and no signal processing.

Tier	Model and result	Compute cost
MCU, no accelerator	TFLite-Micro person detection, MobileNetV1 a=0.25, 96x96, 325 KB: about 86 percent on Visual Wake Words (MLPerf Tiny floor 80 percent)	\$5-15
Purpose-built module	DFRobot SEN0626: faces and upper bodies to 3 m, 85 deg diagonal, offline	\$14.90
SBC, CPU only	Pi 5, YOLO-nano class at 640x640 via NCNN: 7-8 FPS	\$80
SBC plus NPU	Pi 5 + Hailo-8L, YOLOv8n: 136.7 FPS at batch 8	+\$70

Three honest costs. **The MCU tier cannot do this**: the 325 KB model is binary person / no-person with no pet class, and the SEN0626 outputs faces, upper bodies and gestures only. **Light**: a camera in a dark room returns nothing, disqualifying at night unless IR illumination is added. **Privacy**: a camera on a home robot is a materially different product, legally and socially. Copy the retired Person Sensor's architecture - run the model on-device, emit only {class, bbox, confidence}, never store or transmit frames - and write that constraint down before any code.

Fusion: the decision table

Assumed stack: an 8x8 ToF pitched up with the sensor at 400 mm, a co-boresighted MLX90640 32x24, and optionally a 60 GHz radar used **only after the robot has been stationary for 3 s**. Every threshold is a calibration starting point, not a published value.

```
TOF r = median cluster range, m
h_top = h_sensor + r * sin(theta_zone_top) # absolute height, m
w = cluster angular width * r # m
jitter = stddev of per-zone range over a 1.0 s window, mm
planar = best-fit plane residual < 15 mm
THERM dT = peak cluster temperature minus 10th-percentile scene temperature, K
RADAR rr = dominant spectral peak, 0.1-0.7 Hz band, breaths/min (stationary only)
```

Class	h_top	w	jitter	dT	Robot action
COLLISION	any	any	any	any	Stop. $r < \text{braking_distance}$. ToF alone, never gated on class
OBJECT	any	any	< 8 mm and planar	< 1.5 K	Plan around it, no announce
HUMAN	> 1.20 m	> 0.25 m	any	>= 2.0 K	Yield, announce, slow to 0.3 m/s
PET	< 0.75 m	< 0.60 m	> 8 mm	>= 1.5 K	Slow, do not announce, widen clearance
UNKNOWN_LIVING	0.75-1.20 m	any	any	>= 1.5 K	Treat as HUMAN. Crouching adult, toddler, large dog
UNKNOWN	any	any	any	< 1.5 K	Treat as obstacle, keep tracking

Radar, when valid, only raises confidence on an existing call: $8 \leq rr \leq 26$ with adequate peak SNR confirms *alive*. The rate never selects the class, for the reason at the top of this chapter.

Three rules are embedded there. **Collision avoidance never waits on classification** - it runs on raw ToF at frame rate, and classification changes behaviour, never braking. **Height is the primary human/pet discriminant**, not temperature and not breathing rate. **The ambiguous band defaults to human**, because the safe failure is over-classifying.

Scenario	What fools it	Mitigation
Person sitting still on a sofa	Radar MTI deletes them; PIR sees nothing	Thermal sees them; never trust radar or PIR for a still human
Long-haired dog, warm room	Flank only 2 K above ambient	Fall back to ToF height plus jitter
Laptop, radiator, sunlit floor	$dT > 2$ K defeats a thermal-only rule	Require jitter > 8 mm before calling anything living
Black cat	Low ToF reflectance, small thermal area	Accept reduced range; claim no untested dark-target distance
Robot's own motor heat	Thermal baseline drift	Use scene-relative dT, never absolute temperature
Robot in motion	All Doppler features void	Gate every radar feature on robot_stationary

The pairing that wins per dollar

Multizone ToF plus a 32x24 thermal array: VL53L5CX at \$19.95 plus MLX90640-D55 at \$74.95, about \$95. ToF gives metric height, width and rigidity but cannot tell warm from cold; thermal gives alive-or-not but has no range and poor geometry. Together they answer will-I-hit-it, is-it-alive, is-it-a-human and a usable is-it-a-pet, in darkness, with no camera and no privacy exposure. The decision table runs entirely on that pair.

One disagreement to flag: the research lane's fusion table prices this pair at "about \$20 plus about \$45", which is the AMG8833 8x8 - yet the same document's Johnson analysis concludes an 8x8 can *never* recognise a human or a cat at any range. **Prefer the 32x24.** The 8x8 stays defensible as a pure alive-or-not gate feeding ToF geometry, but only if described that way.

The runner-up is **ToF plus a camera running a COCO detector**, about \$20 plus \$150 of Pi 5 and Hailo-8L. It wins on absolute capability - the only option here that separates cat from dog - and loses on three counts: \$150 of compute alone, about 1.6 times the price of the whole winning pair; nothing returned in a dark room; and it turns a home robot into a camera

product. Choose it only if night operation is not required and the privacy architecture is designed in from the start. Skip **radar plus ToF**, the weakest pair here: both are geometric, and radar adds only closing velocity that frame differencing already gives. **Thermal plus radar** does show a real gain - thermal-only mAP 0.527, radar-only 0.194, fused 0.644 ([arXiv 2307.03623](#)) - but from over \$1000 of FLIR Boson 640 and Vayyar vTrigB hardware.

What is not achievable on this budget

1. **Cat versus dog without a camera.** No combination of ToF, a 32x24 thermal array and one hobby radar will do it; the published species work uses labelled datasets and per-installation training.
2. **Radar vital signs on a moving robot.** Ego-motion writes a Doppler bias orders of magnitude larger than a chest excursion. Stationary-robot feature only.
3. **Reliable small-animal detection from one mmWave radar.** 46.1 percent against a 3 kg dog, 75.2 percent tuned, 97.10 percent only with four fused radars.
4. **Human detection through a standard static-clutter filter.** CFAR after MTI misses a still person roughly a third of the time.
5. **Any classification at collision-avoidance latency.** 75 percent at 250 ms of micro-Doppler dwell; 2.1 s for ST's ToF people counter; about 200 ms for the Person Sensor before filtering.
6. **A pet-immune PIR that is pet-immune on a robot.** The geometry needs 2.2 m of mounting height a 600-1200 mm robot does not have.
7. **Raw radar signal processing from any module in this catalogue.** Every micro-Doppler result cited here needs a development board such as the IWR6843ISK or BGT60TR13C and the processing to go with it.

Wiring, bus, aperture and power

I2C address collisions: every VL53 boots at 0x29

Two VL53 parts on one bus collide silently: both ACK, and a read returns their wire-AND, so you get wrong distances rather than a fault code. The VL53L1X boots at 7-bit 0x29 (8-bit write 0x52), 400 kbit/s maximum ([VL53L1X DS DocID031281 Rev 3](#) Section 4, datasheet-verified); L0X, L4CD, L4CX, L5CX, L7CX and L8CX share it. The remedy is volatile: the address lives in RAM, not NVM, and is lost on every power cycle and every XSHUT assertion ([Adafruit](#), vendor-page-verified).

Approach A: XSHUT sequencing

All SDA/SCL on one shared bus; each sensor's XSHUT (active-low reset) on its own GPIO, driven, not floating. Breakouts pull XSHUT to VDD, so at power-up every sensor wakes at 0x29 together. N sensors costs N GPIOs, hence an MCP23017 expander (16 I/O at 0x20-0x27) on a 12-node ring.

```
1. Drive ALL XSHUT low. // every sensor in reset, bus empty
2. delay(10 ms). // Adafruit's published settling delay
3. for i in 0..N-1:
  drive XSHUT[i] HIGH // exactly one sensor wakes, at 0x29
  delay(10 ms)
  sensor[i].begin(0x29)
  VERIFY: model-ID register 0x010F // VL53L1X returns 0xEA, 0xCC, 0x10
  sensor[i].setAddress(0x30 + 2*i)
  VERIFY: model-ID at the NEW address, and that 0x29 no longer ACKs
  // if either verify fails: ABORT LOUDLY. Do not continue.
```

Verify every step: a missed write leaves a second device at 0x29, and every later `begin(0x29)` then talks to two chips silently. Space addresses by 2, because libraries are ambiguous about 7-bit versus 8-bit ([ST Community](#), vendor-page-verified). Use 0x30-0x3F; 0x00-0x07 and 0x78-0x7F are reserved, 0x70-0x77 is the mux block, 0x20-0x27 the MCP23017. Probe 0x29 in service - an ACK means a brown-out has reset a node, so re-run the whole sequence.

Approach B: TCA9548A / PCA9548 multiplexer

Part	Channels	USD 1-off	Source
Adafruit TCA9548A, PID 2717	8	6.95	product/2717
Adafruit PCA9546 QT, PID 5664	4	3.95	product/5664
SparkFun Qwiic Mux, BOB-16784	8	6.95	sparkfun

Vendor-page-verified, read 2026-09-12; all occupy 0x70-0x77. Host bus to the trunk, one SDn/SCn branch per sensor, all sensors left at 0x29, XSHUT tied high.

```
void tcselect(uint8_t ch) { // ch = 0..7
  Wire.beginTransaction(0x70);
  Wire.write(1 << ch); // BITMASK, not an index
  Wire.endTransmission();
}
tcselect(i); // before EVERY sensor transaction
sensor.readRangeContinuousMillimeters();
```

Writing 0x03 enables two channels at once and recreates the collision the mux was bought to prevent. Wire mux RESET to a GPIO: a wedged mux takes the ring down, and this is the only recovery short of a power cycle. Interrupts are not muxed - the data-ready line bypasses it, so poll or give each its own pin. A channel select costs 45 us at 400 kHz (computed), negligible against 33 ms.

Which to use

Sensors	Choice	Reason
4	One PCA9546, 3.95 USD	Cheaper than 4 GPIOs plus boot code
8	One 8-channel, 6.95 USD	Branch isolation keeps the bus in spec
12	8-channel + 4-channel, 10.90 USD	Cascades to 64 branches; beats XSHUT for the remainder

XSHUT costs N GPIOs, re-runs every boot, gives no fault isolation, and leaves every branch's cable capacitance permanently on the bus. Recommended (inferred): mux for addressing, XSHUT still wired to GPIOs as a per-node hardware reset.

Bus integrity over 300-800 mm of cable

The 3.3 V ceiling is about 366 pF at 400 kHz, not the 400 pF of the I2C specification. From $t_R = 0.8473 R_p C_b$, V_{OL} 0.4 V at 4 mA and the 3 mA sink limit, $R_{p,min} = 967 \text{ ohm}$, so $C_{b,max} = 300 \text{ ns} / (0.8473 * 967) = 366 \text{ pF}$ at 400 kHz and 1220 pF at 100 kHz (computed from VL53L1X DS Tables 10 and 17, datasheet-verified). Qwiic cable capacitance is **not published**; 28 AWG 4-conductor is typically 40-70 pF/m (inferred - measure yours). Adafruit tops out at 400 mm (1.50 USD, PID 4210), SparkFun's flexible 500 mm is 2.75 USD (PRT-17257), and a base-to-dome run is 600-900 mm. Budget for 8 sensors on 800 mm branches at 60 pF/m, 10 pF per device pin, 5 pF per board (computed):

Topology	Cable	Devices + boards	Host	Total
One shared bus (XSHUT)	384 pF	120 pF	20 pF	524 pF
Muxed, one branch live	66 pF	30 pF	20 pF	116 pF

The shared bus fails at 400 kHz and is marginal at 100 kHz. A mux divides capacitance by the branch count; that, not addressing, is the decisive argument for it. Under 250 pF use 400 kHz with 2.2 kohm pull-ups; at 250-366 pF use 1 kohm pull-ups or drop to 100 kHz.

Above 366 pF, inside the chassis, add one **LTC4311** active pull-up at the far trunk end, 9.95 USD (Adafruit 4756), 1.6-5.5 V, 400 kHz maximum. Sources disagree on its ceiling: Adafruit says 4000 pF, the [datasheet](#) says only "well beyond the 400 pF I2C specification limit" and tabulates one condition, 300 ns rise at 400 pF and 3 V. Prefer the datasheet and design to a measured rise time. A **P82B715** pair is for runs leaving the chassis and is overkill on a 350 mm robot; sources disagree there too, its features claiming 3-12 V against 4.5-12 V guaranteed, so it is out of range on a 3.3 V ring (**NXP**). Neither part fixes an address collision and neither raises the 400 kHz ceiling.

A marginal bus never fails cleanly. Reads of 0xFF; a sensor that works alone and fails with a neighbour plugged in; faults only while driving, as cables flex; NACKs that track temperature; and worst, stable but wrong distances. Scope SDA and measure rise time before believing any software explanation.

Optical crosstalk and time-multiplexed firing

Sensor-to-sensor crosstalk is not cover-glass crosstalk, and no calibration fixes it: A's 940 nm photons reach B's SPAD array, indistinguishable from ambient. ST's engineers say sensors at different angles do not meaningfully interfere, but that with directly overlapping fields "the interference can be very large and the data unusable" ([ST Community](#), vendor-page-verified). The price is in the ambient table (DS Table 7, long-distance mode, 100 ms budget, datasheet-verified):

Target reflectance	Dark	50 kcps/SPAD	200 kcps/SPAD
White 88 %	360 cm	166 cm	73 cm
Grey 54 %	340 cm	154 cm	69 cm
Grey 17 %	170 cm	114 cm	68 cm

Up to 80 % of range is lost on a bright target and 60 % on a dark one. Worse is the phantom close target: an interferer's pulse inside the victim's histogram window reports a range that never existed, and the robot stops intermittently in an empty room.

- Geometric separation.** Aim the cones so they do not overlap inside the first 0.5 m. ST's ROI table (DS Table 9) gives 27, 20 and 15 deg full field of view for 16x16, 8x8 and 4x4 regions of interest, and those are DIAGONAL angles; the horizontal equivalents are **not published**, so do not size a ring from a horizontal figure nobody publishes. A smaller ROI shrinks cone and range together, 170 cm to 45 cm on 17 % grey, dark, 100 ms, partial-field. An 8x8 ROI - 20 deg, 119 cm on 17 % grey in the dark - is the better compromise for a perimeter.
- Group-parallel firing.** Two optical groups of non-adjacent sensors, fired alternately: overlap removed at 2x latency instead of 8x. Recommended (inferred).
- Strict time-multiplexing**, stopping and starting ranging through the driver, never by toggling XSHUT, which loses the assigned address.

4. **A longer timing budget** averages interference down but opposes the latency budget; **decorrelated periods** (40, 41, 43, 47 ms) leave outliers a 3-of-5 median filter removes (inferred).

Ambient loss is worst on low-reflectance targets, so ignoring this lands hardest on a dark-furred cat on a dark carpet.

Apertures: a ToF sensor cannot see through a printed shell

Every ToF sensor needs its own clear aperture. A printed wall is either opaque or translucent and scattering, and the second is worse: it diffuses emitted photons back into the receiver and the sensor reports a fixed short range. ST specifies keep-out cones from a datum at the module cap: emitter 36.50 deg full angle from 0.84 mm diameter, receiver 39.60 deg from 1.08 mm, optical centres 3 +/- 0.02 mm apart (DS Figure 19, drawing DM00319387 Rev 3.0, datasheet-verified). Aperture at standoff H, from $D_{em} = 0.84 + 0.660 \cdot H$ and $D_{rx} = 1.08 + 0.720 \cdot H$ (computed):

Standoff H (mm)	Emitter dia.	Receiver dia.	Single oval
0.5	1.17 mm	1.44 mm	4.31 x 1.44 mm
1.0	1.50 mm	1.80 mm	4.65 x 1.80 mm
2.0	2.16 mm	2.52 mm	5.34 x 2.52 mm
3.0	2.82 mm	3.24 mm	6.03 x 3.24 mm
5.0	4.14 mm	4.68 mm	7.41 x 4.68 mm

A narrower hole clips the cone, which vignettes the field, kills range asymmetrically, and raises crosstalk as the clipped photons scatter off the aperture edge.

Cover rule	Value	Confidence
Air gap, single-zone	As small as possible; gasket or light barrier above ~0.5 mm	ST engineer citing AN5231, vendor-page-verified
Air gap, VL53L5CX	Below 0.4 mm; gasket above 0.7 mm	AN5856 second-hand, unverified
Transmission, 930-950 nm	Above 85 %	AN5231 second-hand, unverified
Cover	About 1 mm thick; smooth and polished	Thickness ST cover-glass article, vendor-page-verified; surface inferred
Crosstalk calibration	Once per unit, real cover fitted, after assembly	Required, vendor-page-verified; target distance not published
Protective liner	Remove immediately before fitting the cover	DS page 27, datasheet-verified

Bond the breakout to the inside face so the standoff stays at or below 0.5 mm; counterbore from inside where the wall exceeds 1 mm, tapered wider than the cone; fit a 0.4 mm rib or black foam septum between Tx and Rx, the highest-value part of the assembly and free. Aimed at empty space, a good aperture returns "no target" and a bad one a stable 50-300 mm. Ignore any sub-40 mm return - minimum ranging distance is 4 cm (DS 3.3, datasheet-verified).

Radar through plastic

Radar suits a costume robot because it hides behind the shell where ToF cannot. Insertion loss in dB through PLA, PETG or ABS at 24 or 60 GHz is **not published** by any source here; what is published is permittivity, the thickness rule, and one measurement of what a wrong thickness does. Hi-Link and Waveshare publish the same 24.125 GHz table (datasheet-verified - [LD2410 manual.pdf](#)) Table 3, [Waveshare radome guide](#)):

Medium	ϵ_r	Half-wave at 24.125 GHz	1/8 wave
Air	1.00	6.20 mm	1.55 mm
ABS 1	1.50	5.06 mm	1.27 mm
ABS 2	2.50	3.92 mm	0.98 mm
PC	3.00	3.58 mm	0.89 mm
PMMA acrylic 1	2.00	4.38 mm	1.10 mm
PMMA acrylic 2	5.00	2.77 mm	0.69 mm

PLA is absent from it. Measured work gives ϵ_r about 2.75 from microwave through 60 GHz, so PLA half-wave is 3.75 mm at 24.125 GHz and 1.51 mm at 60 GHz, eighth-wave 0.94 mm and 0.38 mm (computed). The 60 GHz eighth-wave is thinner than one printed perimeter, so at 60 GHz half-wave is the only usable option. Printed ϵ_r falls with infill, so a 40 % infill wall is not 2.75. **PETG ϵ_r is not published**: bound it between the ABS 2 row (2.50, 3.92 mm) and the PC row (3.00, 3.58 mm) and

measure, or print the window in ABS or PLA.

Wall thickness $D = (m/2) \cdot c_0 / (f \cdot \sqrt{\epsilon_r})$ $m = 1, 2, 3 \dots \pm 20 \%$
Antenna standoff $H = (m/2) \cdot c_0 / f$ (half-wave in AIR, $\pm 1.2 \text{ mm}$)

Datasheet-verified from Hi-Link 7.2 and Waveshare, identical to [TI SWRA705](#) Equations 3-5. At half-wave the inner-face reflection cancels. If half-wave is impossible go to 1/8 wave or thinner; between the two is the worst of both. Air half-wave at 24.125 GHz is 6.20 mm, so preferred standoff is 12.4 or 18.6 mm.

TI measured the cost of getting it wrong: a 2 mm ABS box over an IWR6843ISK at 62 GHz turned a smooth 105-120 dB pattern into nulls swinging roughly 80-125 dB (SWRA705 6.1). Correct ABS half-wave at 62 GHz is 1.53 mm (computed), so 2 mm is 1.31x optimum - a 0.47 mm error wrecked it. Sources disagree on ABS: Hi-Link and Waveshare tabulate two point values, 1.50 and 2.50, while TI Table 3-1 gives 2.0-3.5 with $\tan \delta$ 0.0050-0.019. Prefer TI's range, a material spread rather than one vendor's point value; it moves the 24 GHz half-wave from 4.39 to 3.32 mm, a 24 % swing wider than the tolerance, so get ϵ_r from the filament supplier or measure it. Curvature helps too: a flat radome is tuned only at boresight, while with the antenna at the centre of a spherical radome the path is equal at every angle (SWRA705 4.2), radius $n \cdot \lambda_0/2$ and wall $n \cdot \lambda_{m/2}$.

Material	Status
Metal in the wall	Opaque (LD2410 5.5)
Metallic paint or any conductive coating	Opaque (SWRA705 3, 8; Waveshare)
CF-filled, conductive, anti-static or metal-powder filament	Opaque to attenuating (inferred from the no-conductor rule)
Frosted or textured surface	Degrading, raises reflection and loss (Waveshare)
Sparse infill, voids, multi-layer wall	Degrading, must be solid with no bubbles (SWRA705 3)

The last row bites printing hardest: a 3-perimeter 20 %-infill wall is a solid-sparse-solid sandwich of unknown effective ϵ_r . Reserve a 100 %-infill puck across the aperture plus margin and print it smooth-side-out. One deliberate use of metal: Hi-Link recommends a metal backplane behind each radar to kill the back lobe, which on a ring also cuts radar-to-radar coupling.

No vendor here publishes a minimum separation between two 24 GHz modules: the DFRobot SEN0395 and SEN0610 wikis and the Hi-Link LD2410 manual are silent on it, and Seeed says only that the MR60BHA2 "must avoid interference from nearby radars" without a distance (vendor-page-verified). All these modules sit in the same 24.00-24.25 GHz band with a 250 MHz sweep (LD2410 manual Table 2, datasheet-verified), so two overlapping beams dechirp each other and no frequency plan separates them. The LD2410 beam is ± 60 deg (datasheet-verified), so three radars at 120 deg spacing cover 360 deg with the -3 dB edges just meeting; three is the count that minimises overlap. Time-multiplexing radars is not practical - they take seconds to settle their background estimate after a power cycle and expose no gate input, so unlike ToF they cannot be round-robined.

UART fan-in for multiple radars

These modules stream unsolicited frames rather than answering polls. The LD2410 has one UART and one GPIO at 3.3 V (datasheet-verified); SEN0395 runs 115200 baud; C4001/SEN0610 runs 9600 baud and also offers I2C at 0x2A or 0x2B, covering exactly two radars (vendor-page-verified).

Ring size	Fan-in	Parts
2 radars	Two hardware UARTs	ESP32 or RP2040
3-4 radars	2 hardware + 2 PIO UARTs	One RP2040
5-8 radars	RP2040 concentrator, 8 PIO UART RX	One Pico-class board
8+, distributed over metres	Satellite MCU + RS-485/Modbus	MAX3485, SN65HVD75 class

The RP2040's two PIO blocks hold four state machines each, so one chip receives eight streams at 115200-256000 baud with DMA and no per-bit CPU cost. Put it in the base as a concentrator that also owns the ToF firing schedule and publishes one sector vector upstream (inferred). SoftwareSerial is reliable only to about 38400 baud and cannot receive on two instances at once. An SC16IS752 (2 UARTs, 64-byte FIFOs, 5 Mbit/s, datasheet-verified) works on SPI but not I2C: four bridges polled at 50 Hz cost roughly 15 kB/s against about 30 kB/s usable, on the same bus as the ToF ring (computed). An analog mux (74HC4051) suits command/response sensors only, and RS-485 needs a per-node MCU because no module

here has a node address.

Power budget

From DS Table 16 (16 mA typ, 40 mA peak with VCSEL, 20 uA between measurements) and the LD2410 manual Figure 11 oscilloscope capture (80.62 mA average, 152.17 mA maximum measured), plus about 3 mA per breakout for LED and LDO (inferred). All rows computed.

6-sensor ring	Average	Peak	At 5 V
6 ToF concurrent	114 mA	240 mA	0.57 W
6 ToF, 2 groups of 3	66 mA	120 mA	0.33 W
6 ToF sequential	34 mA	40 mA	0.17 W
6 x LD2410 24 GHz	484 mA	913 mA	2.42 W
4 ToF + 2 LD2410	243 mA	464 mA	1.22 W

12-sensor ring	Average	Peak	At 5 V
12 ToF concurrent	228 mA	480 mA	1.14 W
12 ToF, 2 groups of 6	132 mA	240 mA	0.66 W
12 x LD2410 24 GHz	967 mA	1.83 A	4.84 W avg, 9.13 W peak
8 ToF + 4 LD2410	486 mA	929 mA	2.43 W avg, 4.65 W peak

ToF power is irrelevant - time-multiplex for crosstalk, never for power. Radar power is not: twelve 24 GHz modules need their own buck converter, which argues for three or four radars in a hybrid ring. The DFRobot SEN0395 is worse again where a figure is published, 90 mA average each (vendor-page-verified), so six of them draw 540 mA and 2.70 W. Cut the LED jumper on every breakout, 36 mA at twelve nodes.

Radar is a pulsed load, and the LD2410 manual demands a supply capable of over 200 mA per module. Do not daisy-chain power through Qwiic cable: 28 AWG is 0.2326 ohm/m, so 0.93 A over a 1 m round trip drops 216 mV and 1.83 A drops 426 mV, against 49 mV and 97 mV on 22 AWG (computed). Run a 22 AWG power star from a 5 V/3 A buck, 1000 uF at the distribution board, 100 uF plus 100 nF at each radar, and Qwiic red and black for the last 100 mm only. Treat 200 mA as the working limit per JST SH contact (no vendor publishes a rating), and keep one common ground - a radar returning chirp current through an I2C cable injects it into SDA.

Latency budget

The robot tips before it skids - $a_{tip} = g \cdot (t/2)/h$ is 2.86 m/s² at a 600 mm centre of gravity on a 350 mm track - so design to 1.5 m/s², giving 30 mm braking at 0.3 m/s and 83 mm at 0.5 m/s (computed). The chain is $T_{total} = T_{revisit} + T_{integration} + T_{processing} + T_{actuation}$: integration 33 ms (the datasheet-referenced 10 Hz / 33 ms point, DS Table 16), processing 10 ms, actuation 50 ms (inferred - measure yours). Refresh follows from motion alone, since the robot must move no more than a quarter of its radius between looks at one bearing (inferred): 44 mm at 0.5 m/s gives **full perimeter refresh at or below 88 ms - 10 Hz minimum, 20 Hz target**, relaxing to 146 ms at 0.3 m/s. Reaction distance is $d_{react} = v_{close} \cdot T_{total} + v^2/(2a) + d_{margin}$, with 83 mm braking and 100 mm margin:

Ring configuration	T_revisit	T_total	Human 1.9 m/s	Cat 4.5 m/s
8 ToF simultaneous (zero overlap only)	33 ms	126 ms	0.423 m	0.750 m
8 ToF, 2 groups of 4	66 ms	159 ms	0.485 m	0.899 m
8 ToF, 4 groups of 2	132 ms	225 ms	0.611 m	1.196 m
8 ToF sequential	264 ms	357 ms	0.861 m	1.790 m
12 ToF sequential	396 ms	489 ms	1.112 m	2.384 m

All computed. Read the cat column against the ambient table above: sequential firing of eight sensors demands 1.79 m on a small dark animal, and the VL53L1X reaches 1.70 m on 17 % grey **in the dark**, 0.68 m at 200 kcps/SPAD. Strictly sequential firing is not viable for pet detection with this sensor class. Two groups of four is the first configuration with real margin, and one of only two that meet the 88 ms rule - the other is all eight fired at once, which is legal only with zero cone overlap. Four ToF sequential (132 ms) passes at 0.3 m/s and fails at 0.5 m/s. Radar frame rate is **not published** and community figures

cluster near 10 Hz, so treat radar as a presence layer at 100 ms-class latency and let ToF own collision avoidance (inferred).

The bus is not the bottleneck: a result read is about 17 bytes, 0.4 ms at 400 kHz, plus 45 us per mux select, so twelve sensors per sweep is 5.3 ms - 16 % of one integration window, or 21 ms at 100 kHz, still inside one window (computed). **Run the ring at 100 kHz:** 3.3x the capacitance headroom, possibly no LTC4311, and nothing the latency budget can measure.

Failure modes and safe behaviour

A single-modality perimeter system will eventually hit something. Every candidate sensor has at least one **fail-to-danger** mode, where it reports "clear" while a person, a pet or a stair edge is there, and each has one physics cause, so a second sensor of the same physics does not remove it. Nothing here is safety-rated: a VL53L5CX, an RPLIDAR C1, an LD2410 and an MLX90640 are sensors, not electro-sensitive protective equipment. Each has one output, an undeclared failure behaviour, and a response time that varies with the target.

The failure catalogue

Time-of-flight (VL53L0X / L1X / L5CX / L7CX)

Measured values are from Caroleo, Albin and Maiolino, Sensors 26(5):1639, 2026 (MDPI).

Failure	Cause and condition	Mitigation
Dark absorbing surfaces	Measured, black vinyl: 18.8 % valid returns at 25 cm, 20.3 % at 40 cm, 34.4 % at 60 cm ; deviation on a black board peaks near 50 mm past 400 mm.	Never read "no return" as "clear". Use the per-zone target_status; slow as valid zones fall; cover with radar.
Mirrors and glass	Vendor-stated: through a transparent barrier it ranges to the barrier ; a 45 deg mirror returns the folded path, so the corridor seems to continue (US10594920B2).	One forward ultrasonic, the one modality that sees glass reliably; lidar return intensity can also flag it. Map mirrors and glass doors as keep-out zones.
Direct sunlight	Measured: 270 to 0.5 lux negligible, but a 500 W halogen at about 3 klux drove rejection to almost 45 % . A VL53L1X at 14 to 93.6 klux gave 33 % useful data (ST Community). ST publish 0 and 5 klux only.	Indoor duty. Treat a sunlit patch or patio door as a blind patch, not free space.
Cover-glass crosstalk	Vendor-stated: immune beyond 60 cm, but below 60 cm crosstalk can exceed the true return (AN5856) -- the band the protective field occupies.	Calibrate with the final cover glass fitted, again after any shell change. Keep the AN5856 air gap.
Dusty or greasy window	Vendor-stated: grease on the cover glass raises crosstalk, so day-one calibration drifts.	Track baseline crosstalk on a clear bearing; angle the window down so dust sheds.
Retroreflectors	Datasheet-verified: certified practice adds a Z_ref1 term unless retroreflectors are excluded (Banner AG4). Hi-vis tape and reflective collars saturate a SPAD.	Add Z_ref1 = 200 mm in a home.
Multipath	Inferred: in a corner two surfaces return a path longer than either. Magnitude here is not published .	Reject zones beyond the geometric maximum for their bearing; one odd zone among consistent neighbours is suspect.
Thermal drift	Measured: the part self-heats to 52 deg C after about 900 s ; within-frame variability is 0.5 to 2 mm .	Calibrate warm, not cold. Allow 15 min before trusting sub-centimetre accuracy.

The safety lane calls this part "65 deg diagonal". Prefer the catalogue, **45 x 45 deg horizontal by vertical, 63 deg diagonal** (VL53L7CX **60 x 60 deg, 90 deg diagonal**), and quote the horizontal figure: the diagonal is separately measured and cannot be converted.

mmWave radar (24 GHz LD2410 class, 60 GHz IWR6843 class)

Failure	Cause and condition	Mitigation
Ego-motion clutter	Measured: a moving radar gives every static return a Doppler shift, so every wall reads as moving (EM-Fall).	Subtract ego-Doppler from odometry and IMU. Real signal work, not a config flag.
Metal ghosts	Measured: reflective static objects create multipath and shadow ghosts , placing false targets further away than the real one (PMC12158235).	Require track persistence; drop tracks outside known room bounds. Ghosts cost availability, not safety.
Vibration	Measured: a 60 GHz radar resolves micron-scale motion (arXiv 2107.10993), and the drivetrain is inside that sensitivity; TI warn that strong vibration or moving debris needs filtering (TI SWRA818).	Isolate the module; notch motor and gearbox frequencies; validate with the drive running.
Fans and curtains	Vendor-stated: fan blades give Doppler signatures like human motion, and fabric in an air current triggers alarms.	Range-Doppler filtering; reject fixed-position periodic returns.

Failure	Cause and condition	Mitigation
Wall penetration	Vendor-stated: the LD2410 detects a person through two layers of sheetrock , and through panelling up to 5 cm (ESPHome) . A moving robot cannot tune a fixed gate.	Range-gate hard, or use 60 GHz: TI state higher absorption and reduced wall penetration .
Blind to a still person	Vendor-stated: detection then rests on respiration and heartbeat signatures, with a 1 to 2 s dwell .	Never let radar be the last thing between the robot and a person.
Pets read as humans	Vendor-stated: TI list pets and small animals as motion signatures resembling human presence; removing them takes a trained non-human filter, not a threshold.	Treat a moving thing at pet height as stop-and-yield, not as a classification to settle before braking.

Thermal arrays (MLX90640 class)

Failure	Cause and condition	Mitigation
Ambient near body temperature	Vendor-stated: person logic keys on 35 to 40 deg C ; the part is +/-2 deg C over 0 to 100 deg C (Adafruit 4407) . At 35 deg C ambient a clothed person sits inside the error bar, and 40 deg C is ISO 3691-4's own declared short-term ambient maximum, so the failure sits inside the standard's operating envelope.	Corroborating channel only, allowed to fail silent. Threshold on the scene median.
Sun-warmed floor	Vendor-stated: heat sources degrade performance. A sunlit rug at 38 deg C reads as a person.	Require shape and motion consistency; 32x24 can reject a large uniform blob.
Glass opacity	Vendor-stated: optical glass is opaque at 8 to 14 microns ; LWIR needs germanium, zinc selenide or silicon.	No normal shell window. Budget a germanium aperture or an open hole, and its cleaning.
Slow frames	Datasheet-verified: 4 or 8 Hz on I2C for noise. Chain latency 255 ms at 8 Hz, 380 ms at 4 Hz ; at 4 Hz and 0.5 m/s separation is 1060 mm .	Warning field and classification only, never the stop path.
Pets too small at range	Inferred: a 32x24 array over 55 deg puts a 300 mm cat at 1 m across 2 to 3 pixels, and fur insulates, so the coat surface is much cooler than the core.	Expect no pet detection beyond about 1.5 m . Thermal sees pets at close range only.

PIR

Vendor-stated: two pyroelectric elements cancel in a stable scene, and the chain responds only to changes faster than about 1 to 2 s, so a **motionless person is invisible by design**. Warm air from a vent moves infrared energy across a zone boundary and triggers falsely ([Industrial Monitor Direct](#)). Inferred: on a moving robot the differential principle fires continuously. **PIR is disqualified as a safety sensor**; it is a wake-from-sleep trigger while parked.

2D lidar (RPLIDAR A1 / C1 class)

Failure	Cause and condition	Mitigation
Planar blind slice	Inferred from datasheet geometry. Below the plane: a cat lying flat, a toy, a step edge. Above it: a table top, an open drawer, a 900 mm counter overhang. Certified practice keeps the plane below 200 mm .	Insufficient alone on a 600 to 1200 mm robot. Add a downward-tilted ToF array below, and an upward element for overhangs.
Black absorbing surfaces	Datasheet-verified by omission: the A1M8-R4 range of 0.15 to 6 m (0.15 to 12 m on the R5) is qualified "White objects" ; low-reflectance range is not published .	As for ToF. IEC 61496 Type 3 closes this gap; a hobby lidar never proves it.
Dust and window film	Datasheet-verified: Type 3 devices must prove resistance to dirt, backgrounds and obstruction.	Scheduled cleaning, plus a self-test on a known static feature's return intensity.
Shadows and mutual interference	Datasheet-verified: certified practice makes the installer remove the shadow effect and needle- or cone-shaped field artefacts, and offset two scanners 100 mm or more vertically, or shield them (Banner AG4). A pillar or a table leg casts a radial shadow.	Use more than one sensor position; never claim 360 deg from a single vantage point.
Eye safety	Datasheet-verified: Class I , 785 nm typical, 3 mW typical and 5 mW peak maximum, per 21 CFR 1040.10 and 1040.11 with Laser Notice 50.	Safe as supplied. Replacing the housing or driving the emitter from your own firmware voids it.

Ultrasonic

Ultrasonic is here because it answers glass and mirrors, so its own failures must be stated. Vendor-stated: soft materials absorb sound, and most HC-SR04 units are unreliable on soft, uneven targets such as people beyond about **1 m**, where the successful-measurement rate falls below **50 %**. Angling the sensor more than about **15 deg** off perpendicular removes the echo, and an object closer than **2 cm** returns 0 or an erratic reading. Two units fired together saturate each other, so a **33 ms** interleave is needed and eight sensors round-robin take **264 ms**, about 3.8 Hz ([Last Minute Engineers](#)). That latency is fatal in the stop path. Use ultrasonic for hard specular targets only -- glass, mirrors, gloss doors -- in the warning field, and read a missing echo as "unknown", never "clear".

Failure direction

Fail-to-danger modes report clear when occupied: a black sock to a ToF, a still person to a radar, a cat under the lidar plane. Each needs a second modality with different physics. **Fail-to-safe** modes report occupied when clear: radar ghosts, PIR on a draught, a dark rug read as a cliff. Those cost availability only. So: **OR the stop signals, AND the clear signals**. Any sensor may stop the robot; all must agree before it resumes.

Safe behaviour

The standards worth knowing

ISO 13482:2014 classifies a 350 mm household machine as a *mobile servant robot*. It gives no number for how hard a robot may hit a person, only a process: risk estimation per ISO 12100, then inherently safe design before safeguarding, with software limits on space, speed and force and a mandatory protective stop. The often-cited line that the standard admits no injury limits exist **could not be confirmed at a primary source**; treat it as unverified.

ISO 3691-4 carries the usable numbers. Clause 4.8.2 requires that trucks *"shall stop before contact between the rigid parts of the truck or load and a stationary person"* (TUV Rheinland), replacing EN 1525's weaker wording, which required only a signal enabling a stop. Personnel detection and braking must both reach **PL d**, implying a Category 3 redundant architecture. Keep **0.3 m/s**, the speed accepted when personnel detection is muted, with floor marking required in the same row. Those Annex A values are verified against the **2020 first edition**, not the 2023 text, whose free preview stops at Clause 3.

ANSI/A3 R15.08 is the North American analogue. Part 1 (R15.08-1-2020) covers the vehicle, Part 2 (ANSI/A3 R15.08-2-2023) the system and the application, and Part 3, for users, is in preparation. It separates an AGV, which follows a fixed guide path, from an AMR, which generates its path from the environment; a household robot is an AMR. Its numeric zone and stopping values are paywalled and **not published** in any free source.

IEC 61496 governs the sensor as a safety device. A safety laser scanner is Type 3 and must detect **1.8 % reflectance minimum** in its protective zone; the certified Keyence SZ-V declares minimum detectable objects of 20, 30, 40, 50, 70 or 150 mm, depending on the setting, at *"Reflectance 1.8 % min."* (Keyence). That is why a certified scanner sees a black sock and a VL53L5CX does not. The same sheet requires **20 % or more in the warning zone**, so even certified hardware degrades on dark targets in the slow-down tier.

Protective versus warning field. The warning field is outer and triggers a non-safety reaction: slow, announce, illuminate. The protective field is inner and triggers the stop within the declared response time. **Speed and separation monitoring** ties them to motion: the protective field must be at least the stopping distance, and must grow with speed. The builder's version is three speed tiers, each with its own derived polygons.

Stopping distance

The computed appendix carries the tables; two facts govern them. The deceleration ceiling is **tip-over, not traction**: 175 mm from centre to leading edge caps braking at **2.86 m/s²** with the centre of gravity at 600 mm, and **5.72 m/s²** at 300 mm. And the closing speed is not the robot's: ISO 13855 uses **K = 1600 mm/s** for a walking person, so a robot at 0.5 m/s closes at **2.1 m/s**. Separation including that approach term, at 2.0 m/s² with 100 mm measurement and 100 mm degradation allowances, inferred from datasheet-verified frame rates:

Sensor chain	Reaction	S at 0.3 m/s	S at 0.5 m/s	S at 1.0 m/s
VL53L5CX, 60 Hz, 4x4	97 ms	406 mm	466 mm	701 mm
VL53L5CX, 15 Hz	147 ms	501 mm	571 mm	831 mm
RPLIDAR C1, 15 Hz	167 ms	539 mm	613 mm	883 mm
LD2410 or IWR6843	200 ms	602 mm	682 mm	970 mm
MLX90640, 4 Hz	380 ms	944 mm	1060 mm	1438 mm
Certified scanner	120 ms	450 mm	514 mm	762 mm

In protective-field form a VL53L5CX ring at 15 Hz and 0.5 m/s needs **386 mm**, or **586 mm** with the retroreflector adder; adding the 175 mm half-width, the sensor must see reliably to **561 mm**, and to **822 mm** at 1.0 m/s. Measure your own braking distance and substitute it.

Classification must never gate the stop. With the warning field at twice the protective field, the gap between edges is **167 to 323 ms** for a moving target. A 4 Hz thermal frame is 250 ms; a micro-Doppler classifier needs 1 to 2 s. Neither fits. Stop on geometry, using the fastest and dumbest sensor available, and let classification modulate behaviour after the stop. A design that identifies before it brakes is a design that hits people.

Compliance is cheap and not optional. A 15 kg robot at 1.0 m/s with a rigid shell delivers about **1500 N** mean into a shin, against the ISO/TS 15066 transient limit of **130 N** for a lower leg. At 0.5 m/s with a **50 mm** crushable bumper it delivers **38 N**. Those are collaborative-robot human limits, the nearest published tolerance data; no pet limit exists.

Cliff detection is separate and mandatory

A horizontal scanner at 150 to 200 mm sees nothing at a stair nosing: the floor simply stops. Travel after the edge is seen, at 3.0 m/s² (inferred):

Speed	50 ms loop	100 ms loop	150 ms loop	250 ms loop
0.3 m/s	30 mm	45 mm	60 mm	90 mm
0.5 m/s	67 mm	92 mm	117 mm	167 mm
1.0 m/s	217 mm	267 mm	317 mm	417 mm

At 1.0 m/s every row exceeds the 175 mm half-footprint: the robot goes over at every plausible latency. At 0.5 m/s and 100 ms it must look at least **92 mm** ahead of the front contact patch, and preferably 150 mm. Use **four to six downward ToF sensors at 50 Hz or better**, wired so a lost reading counts as a cliff. Downward ToF measures a distance, so a black floor still returns a range; reflective IR is intensity-only and is the classic failure. **Never disable cliff detection to cure a dark-rug false positive.** That turns a nuisance into a fall.

The minimum set, and what a thinner one hits

Below this the robot will eventually hit something: a 0.5 m/s speed cap, dropping to 0.3 m/s in unmapped or occupied space; a 20 to 50 mm compliant leading surface over a bumper switch; four to six downward cliff sensors; a 360 deg geometric field from a six to eight module ToF ring, or lidar plus a ToF array; a physical emergency stop; and two-tier speed-dependent field logic. Worth adding: 60 GHz radar for presence, never 24 GHz; one or two forward ultrasonic sensors for glass, warning field only, since an eight-sensor round-robin at a 33 ms interleave takes 264 ms; a thermal array for classification after the stop.

- **Lidar only.** A cat asleep on its side at 120 mm is below the plane, as are a black boot, a glass table leg and a stair edge.
- **ToF only.** A person in black jeans at 600 mm returns valid data in under a fifth of zones. The robot reads an empty corridor and walks into a shin at about 250 N.
- **mmWave only.** A person still in a doorway while the robot moves: ego-Doppler swamps the return, micro-motion needs 1 to 2 s, the budget is 184 ms. Meanwhile it brakes for a fan two rooms away.
- **Thermal only.** At 35 deg C indoors the contrast is inside the +/-2 deg C error bar.
- **Ultrasonic only.** A human in a wool coat at 1.2 m is detected under half the time.
- **PIR only.** Disqualified.

You will not reach PL d with uncertified parts. Use the industrial arithmetic anyway, adopt the Z-factor discipline at the larger hobby values, cap speed so residual risk stays small when detection fails completely, and test the adversarial set: a black sock, a mirror, a glass door, a sunlit patch, a 120 mm sleeping-cat mannequin, and a person standing still in black clothing.

Recommendation: three builds, one per budget

Each tier below is one recommendation, not a menu. Prices are USD as read on 2026-09-12 and exclude tax and shipping. Nothing here has been purchased or physically tested; every figure is from a datasheet or a vendor page, and the ones that are not are marked as estimates.

Tier 1 - collision-ring: do not hit anything

USD 455, 724 mA average at 5 V. This tier answers requirement (a) and nothing else. It has no human channel and no pet channel and it is not honest to claim otherwise.

Qty	Part	Vendor / SKU	Unit	Extended
12	VL53L8CX 8x8 multizone dToF carrier	Pololu 3419	24.95	299.40
6	VL53L4CD 1-1300 mm dToF carrier (cliff)	Pololu 3692	13.95	83.70
2	PCA9548 8-channel I2C mux, STEMMMA QT	Adafruit 5626	6.95	13.90
12	Flexible Qwiic cable, 500 mm	SparkFun PRT-17257	2.75	33.00
6	STEMMA QT cable, 100 mm	Adafruit 4210	0.95	5.70
1	RP2040 concentrator, Pico class	estimate	4.00	4.00
1	Compliant bumper, 20-50 mm closed-cell foam plus 4 microswitches	estimate	15.00	15.00
Total				about 455

Mounting. The twelve ring sensors go at **200 mm** height, boresight **horizontal with no cant**, at **30 deg azimuth spacing**. The six cliff sensors go at **150 mm** with a **23 deg down-cant**, four across the front arc at plus and minus 20 and 60 deg, and two at the rear at 150 and 210 deg.

Why twelve and not eight. The VL53L8CX is 45 deg horizontal. Eight of them total exactly 360 deg, which is the one count that must be avoided: at exactly 360 the edge rays of adjacent sensors are parallel and the wedge between them never closes at any range. Twelve gives 180 deg of designed overlap and closes the wedges 513 mm from the platform centre, 338 mm past the skin. The compliant bumper covers what is left, which is why the bumper is in the bill of materials rather than in a list of nice-to-haves.

Why 200 mm and level. Low enough that a 250 mm standing cat is inside the vertical beam from the skin outward; high enough that the floor does not enter the frame until 483 mm, so the ring is not reporting carpet in every zone.

Tier 2 - collision-ring + animate-layer: the recommendation

USD 909, 999 mA average at 5 V. This is the build to make. It adds requirements (b) and (c), and it closes the two failure modes that would otherwise put the robot on the floor or on the cat: dark targets in a lit room, and glass.

Qty	Part	Vendor / SKU	Unit	Extended
-	Everything in Tier 1			455.00
1	Slamtec RPLIDAR C1 360 deg dToF scanner	DFRobot	69.00	69.00
8	Panasonic AMG8833 8x8 thermal array, STEMMMA QT	Adafruit 3538	44.95	359.60
2	RCWL-1601 ultrasonic ranger	Adafruit 4007	3.95	7.90
1	PCA9548 8-channel I2C mux, third	Adafruit 5626	6.95	6.95
8	STEMMA QT cable, 300 mm	Adafruit 4210	1.25	10.00
Total				about 909

Why the scanner, at 100 mm, and why the C1 rather than the A1

The RPLIDAR C1 is the highest value per dollar in the whole study, because one USD 69 part covers both structural weaknesses of a ToF ring:

- **Dark targets.** The C1 is specified 0.05 to 12 m at 70 percent reflectivity and 0.05 to 6 m at 10 percent. A black cat on a dark carpet is a 10 percent class target: the ToF ring loses it past about 1.55 m in a lit room and the C1 holds it to 6 m.

- **Ambient light.** The C1 is rated to 40,000 lux. The VL53L8CX is characterised at 0 and 5 kLux only, and a sunlit patch of floor at roughly 100 kLux reads as *empty space* - a wall of zones reporting "signal rate too low", not noisy numbers. Empty space is the dangerous failure.
- **A lying cat.** A scan plane at 100 mm intersects a cat lying on its side at 120 mm. No fixed ring at 200 mm does.
- **A static background map.** 500 points per revolution is enough to scan-match. Once the robot has a map, anything that does not fit it is dynamic, and dynamic is the strongest animate cue available without a camera. A ToF ring has far too few points to ever do this.

The runner-up, **rejected**, is the RPLIDAR A1 - Adafruit 4010, the part this document also covers as a standalone guide. It is triangulation rather than direct time of flight; its own datasheet qualifies the range as "White objects" with no low-reflectance figure published and says "without direct sunlight exposure"; it is 170 g against 110 g, 5.5 Hz against 10 Hz, and 0.15 m minimum range against 0.05 m. On a 350 mm chassis the A1's blind radius nearly reaches the skin. It costs USD 31 more to be worse on every axis that matters here. Buy the A1 if you already own it or you want the better-documented ROS path; buy the C1 for this robot.

Why eight thermal arrays, at 250 mm, level

Thermal is the **only** modality that separates animate from inanimate, and the only one that sees a *motionless* target. A sleeping cat is invisible to motion-gated radar and invisible to PIR, and plainly visible to thermal.

At 250 mm with a 60 x 60 deg array, the floor first enters frame at 433 mm, the top of a 200 mm cat enters the beam 87 mm from the skin so the animal is in frame essentially always, and a standing adult's head stays in frame to 2.51 m with the torso held beyond that. Ring closure needs $\text{floor}(360/60) + 1 = 7$; eight gives 15 deg of overlap, closes at 670 mm, and leaves a largest hideable sphere of 117.5 mm, smaller than a cat in any dimension.

Rejected: the STHS34PF80 (Adafruit 6426, USD 14.95). A genuinely good part - 10 uA, no Fresnel lens, holds a static human - but it has exactly one sensing element. It reports that something warm is somewhere in an 80 deg cone: no bearing, no size, no contribution to human versus pet. Keep it as a parked-robot wake-up layer only.

Rejected: PIR of every kind, including pet-immune PIR. Pet immunity is a geometric trick - the lens's lowest beam is cut away and the unit is mounted at 2.2 to 2.75 m. At 1.0 m on a robot that elevation separation collapses, and the human's torso and head are now above a downward-fanning lens, so a pet-immune PIR on a low robot can see the pet and miss the person, the exact inversion of its purpose. Worse, PIR responds to the rate of change of flux, so on a driving robot every scene moves and the output is a random number generator.

Two ultrasonic sensors forward, USD 7.90. Glass and mirrors are the one target class the entire optical stack fails on: a ToF sensor ranges to the pane, a lidar returns a dropout or a phantom room, a thermal array sees room-temperature glass. Ultrasonic sees glass reliably. Constrain it hard - warning field only, never in the protective-stop path, and never as the human detector, because soft targets absorb 40 kHz.

The fusion logic

Two rings, both body-fixed, extrinsically calibrated once so a bearing in one is a bearing in the other. The thresholds are a starting point to be calibrated in place.

```

# Tier 0: collision. Geometry alone, at ring frame rate. No classifier in this path.
if min_range_in_swept_corridor < protective_field(v): STOP
elif min_range_in_swept_corridor < warning_field(v): SLOW to 0.3 m/s and announce

# Tier 1..4: classification. Runs in parallel. Changes behaviour only AFTER the stop.
match clusters across modalities by bearing (+/-10 deg) and range (+/-150 mm):

if dT < 1.5 K and jitter < 8 mm and planar and world_velocity ~ 0:
OBJECT # box, wall, furniture, door frame

elif dT >= 2.0 K and (h_top > 900 mm or cluster touches the top zone row) and width > 250 mm:
HUMAN # tall AND warm - the high-confidence rule

elif dT >= 1.5 K and h_top < 450 mm and width < 600 mm and (jitter > 8 mm or moving):
PET # low AND warm AND non-rigid. Cat versus dog is NOT attempted.

elif dT >= 1.5 K:
UNKNOWN_LIVING # crouching adult, toddler, large dog -> BEHAVE AS HUMAN

else:
UNKNOWN_OBSTACLE # stop anyway

```

Three rules are embedded in that, and each one is worth more than the code around it:

- **Height is the primary human-versus-pet discriminator, not temperature and not breathing rate.** A resting beagle breathes at 13 to 25 per minute and a resting human at 12 to 20. The distributions overlap almost completely, so a rule of the form `if rate > 20 then pet` will call a calm human a dog. Temperature says *alive*; height says *which*.
- **Never threshold on absolute temperature.** A long-coated dog's flank reads 28.1 degC in a 21 degC room, a 6 K contrast that falls to 2 K in a warm room, while an AMG8833's absolute accuracy is plus or minus 2.5 degC. Apparent temperature is mostly a range measurement in disguise: a cat at 3 m reads 2.5 K and a clothed human at 5 m reads 3.8 K, and moving either by a metre flips the ordering. Use a per-pixel background moving average at about 30 s, which cancels warm-up drift, chassis self-heating and room ambient in one step.
- **The ambiguous class defaults to human.** A crouching adult, a toddler and a Great Dane all land in UNKNOWN_LIVING. Over-classifying as human is the safe failure.

Tier 3 - full-classification-stack

USD 1,440, 3.8 A average at 5 V. Buy this only if labelled classification out to 10 ft is a requirement rather than a wish. The jump from Tier 2 is USD 531 and 2.8 A for one capability.

Qty	Part	Vendor / SKU	Unit	Extended
16	VL53L8CX carrier, 22.5 deg spacing	Pololu 3419	24.95	399.20
6	VL53L4CD carrier (cliff)	Pololu 3692	13.95	83.70
1	VL53L8CH CNH histogram head, 900 mm, own bus	price not published	~35	~35
6	MLX90640-ESF-BAA 32x24 thermal, 110 x 75 deg	Adafruit 4469, out of stock 2026-09-12	74.95	449.70
1	Slamtec RPLIDAR C1	DFRobot	69.00	69.00
2	RCWL-1601 ultrasonic	Adafruit 4007	3.95	7.90
4	PCA9548 8-channel mux	Adafruit 5626	6.95	27.80
1	Raspberry Pi 5	-	80.00	80.00
1	Hailo-8L AI kit	-	70.00	70.00
4	Wide-FoV camera, about 120 deg H, at 900-1000 mm	not priced in any source	~35	~140
-	Concentrator, cabling, bumper	estimate	~79	~79
Total			about 1,440	

Sixteen ToF instead of twelve puts the wedge tip at 343 mm from the centre - 168 mm past the skin, inside the robot's own footprint - and drops the largest sphere that can hide in a seam from 79.4 mm to 56.0 mm. This is the tightest closure worth buying, and it is what lets the compliant bumper stop being part of the safety argument.

What no tier can do, at any range

- **Direct sunlight above about 5 kLux on any optical sensor.** ST publishes 0 and 5 kLux only; the RPLIDAR C1 stops at 40 kLux; direct sun is 100 to 120 kLux. No number exists past that. Do not extrapolate one.
- **Mirrors.** A mirror at 45 deg returns a folded path length and the robot believes the corridor continues. Map known mirrors as permanent keep-out zones.
- **Cat versus dog without a camera.** No combination of ToF, 32x24 thermal and one hobby radar does it.
- **Radar vital signs on a moving robot.** Ego-motion writes a Doppler bias across the scene orders of magnitude larger than a 5 mm chest excursion.
- **Any safety rating.** None of this is IEC 61496 electro-sensitive protective equipment. A certified Type 3 scanner must prove detection at 1.8 percent minimum target reflectance in its protective field; that is the pass mark a VL53L8CX does not meet, and the reason a certified scanner sees a black sock and this stack does not. Use the industrial arithmetic and the Z-factor discipline anyway. Do not claim the rating.

Three decisions that override the parts list, and cost nothing

1. **Make the platform round.** A 350 mm square has a 494.975 mm diagonal and a 72.487 mm corner protrusion. Four 45 deg sensors on the face centres leave the corners permanently uncovered, and moving them to the corners is strictly worse. If it must be square, put the sensors on the face centres, chamfer the corners at 45 deg, and treat the corners as mechanically protected.
2. **Cap speed at 0.5 m/s, and 0.3 m/s in unmapped or occupied space.** A 15 kg robot at 1.0 m/s with a rigid shell delivers a mean 1500 N into a shin against a 130 N ISO/TS 15066 transient limit. The same robot at 0.5 m/s with a 50 mm compliant bumper delivers 38 N. Speed is the cheapest safety component in the build.
3. **Never let classification gate the stop.** Braking runs on raw geometry at frame rate. Classification runs in parallel and only changes what happens after the robot has stopped.

Standalone guide: ST multizone grid time-of-flight

This chapter stands alone. The platform it sizes for is a 350 mm-footprint indoor mobile robot, 600-1200 mm tall, that needs 360 deg obstacle detection and wants to tell a person from a pet. Everything needed to reach a buying decision is repeated here.

The latest part is the VL53L9CX, and it is not the one to buy

As of 2026-09-12 the newest ST grid time-of-flight sensor is the **VL53L9CX**, not the VL53L8CX. The evidence is four independent sources: ST's own press release ([newsroom.st.com p4783](https://newsroom.st.com/p4783)), the ST product page ([st.com vl53l9cx](https://st.com/vl53l9cx)), independent trade coverage dated 2026-06-22 ([CNX Software](#)), and a DigiKey listing for the STEVAL-VL53L9 evaluation board (P/N 29294599, status Active). Mass production started early July 2026. This is a product, not a rumour.

Two absence findings matter as much as the presence finding.

- **There is no VL53L5CH.** Searches of ST's line, DigiKey and the stm32duino GitHub organisation return only VL53L7CH and VL53L8CH. No VL53L5CH product page, datasheet or driver exists. Confidence: vendor-page-verified (absence).
- **There is no VL53L10 and no VL53L9CH.** Nothing later than the VL53L9CX was found in any source. Treat the VL53L9CX as the end of the line on 2026-09-12.

One sourcing caveat, stated plainly because it changes how much weight each number carries. During research st.com was unreachable from the research host; direct requests returned HTTP 000 and zero bytes. The VL53L5CX, VL53L7CX, VL53L8CX and VL53L8CH datasheets were therefore read in full from byte-identical distributor mirrors (Pololu, Farnell, MikroElektronika) and are **datasheet-verified**. The VL53L9CX datasheet could not be retrieved at all, so every VL53L9CX figure below is **vendor-page-verified** only.

The family

Part	Grid	FoV H x V	Headline range	Host interface	Status
VL53L5CX	4x4 / 8x8	45 x 45 deg	400 cm	I2C 1 MHz	Active, stocked
VL53L7CX	4x4 / 8x8	60 x 60 deg	350 cm	I2C 1 MHz	Active, stocked
VL53L7CH	4x4 / 8x8 + CNH	60 x 60 deg	350 cm	I2C / SPI	Active
VL53L8CX	4x4 / 8x8	45 x 45 deg	400 cm	I2C 1 MHz / SPI 20 MHz	Active, stocked
VL53L8CH	4x4 / 8x8 + CNH	45 x 45 deg	400 cm	I2C 1 MHz / SPI 3 MHz	Active
VL53L9CX	54 x 42 (2268)	55 x 42 deg	8.8 m	MIPI I3C / MIPI CSI-2	Newest, MP July 2026

Every figure in the "headline range" column is unconditioned. The CX and CH headlines are dark-room, white-88%-target, inner-zone, 4x4 numbers, and the datasheets publish the conditioned tables that replace them (below). The VL53L9CX's 8.8 m has no such table behind it at all: ST publishes no per-reflectance or per-ambient maximum-range data for that part.

Field of view, stated exactly. ST prints a *diagonal* FoV on the front page of every VL53 datasheet. That diagonal is a separately measured optical figure, not the geometric diagonal of the square detection volume, so it cannot be converted into anything and must never be used to size a ring. The horizontal and vertical figures are the only usable ones: VL53L5CX and VL53L8CX and VL53L8CH are **45 x 45 deg** horizontal by vertical (63 and 65 deg diagonal); VL53L7CX and VL53L7CH are **60 x 60 deg** (90 deg diagonal); VL53L9CX is **55 x 42 deg** (71 deg diagonal). For scale, a true 45 x 45 deg square has a geometric diagonal of 60.7 deg, not 65 deg. The numbers do not agree because they are not the same measurement.

The VL53L8CX detection volume is characterised, verbatim, against a *"white 88% reflectance perpendicular target, in full FoV, located at 1 m from the sensor, without ambient light (dark conditions), with an 8x8 resolution, with a 14% sharpener (default value), in continuous mode, at 15 Hz."* The collector exclusion zone -- the angle from which stray light can reach the array -- is wider at 57.9 x 57.9 deg.

Zones, rate and what a frame contains

Every CX and CH part is the same 8x8 SPAD array read out in one of two modes (UM2884 Table 2):

Resolution	Min ranging rate	Max ranging rate
4x4, 16 zones	1 Hz	60 Hz
8x8, 64 zones	1 Hz	15 Hz

Default is 4x4 at 1 Hz, and resolution must be set *before* frequency because the legal frequency range depends on it. **8x8 tops out at 15 Hz. There is no 8x8 fast mode.**

Each zone reports distance, target status, signal, ambient, reflectance, range sigma and a per-zone firmware motion indicator. Up to **four targets per zone** are available, set by the VL53L5CX_NB_TARGET_PER_ZONE macro in platform.h (not through the API); the default is 1. The default target order is **Strongest**, which is the wrong default for a robot: a dark obstacle at 400 mm in front of a white wall at 2 m reports the *wall* until you call vl53l5cx_set_target_order() with Closest or read multiple targets.

Range, by reflectance and by ambient light

These are the numbers that decide the design. All rows: target fills 100% of the FoV; Munsell N4.75 (17%), N8.25 (54%), N9.5 (88%); AVDD 3.3 V; 23 degC; **90% detection rate**; 5 kLux realised as 2 W/m2 target irradiance at 940 nm; no cover glass; crosstalk margin 0 kcps. Datasheet-verified.

VL53L8CX and VL53L8CH, continuous 8x8 at 15 Hz (DS14161 Table 17; DS14310 Table 20 is identical value for value, because the CH differs from the CX only in firmware):

Target	Zone	Dark typ	Dark min	5 kLux typ	5 kLux min
White 88%	Inner	4000 mm	4000 mm	1550 mm	1100 mm
White 88%	Corner	3950 mm	2900 mm	1400 mm	1100 mm
Grey 54%	Inner	3300 mm	2350 mm	1400 mm	1000 mm
Grey 54%	Corner	3100 mm	2100 mm	1250 mm	950 mm
Grey 17%	Inner	2450 mm	1500 mm	1150 mm	900 mm
Grey 17%	Corner	1950 mm	1300 mm	950 mm	700 mm

VL53L8CX and VL53L8CH, continuous 4x4 at 30 Hz (DS14161 Table 16; DS14310 Table 19):

Target	Zone	Dark typ	Dark min	5 kLux typ	5 kLux min
White 88%	Inner	4000 mm	4000 mm	2850 mm	2850 mm
White 88%	Corner	4000 mm	4000 mm	2850 mm	2700 mm
Grey 54%	Inner	4000 mm	4000 mm	2600 mm	2550 mm
Grey 17%	Inner	4000 mm	4000 mm	1650 mm	1600 mm
Grey 17%	Corner	3950 mm	3900 mm	1550 mm	1500 mm

The single most important line in this chapter: at 8x8, 15 Hz, against a 17% grey target under 5 kLux ambient, a VL53L8CX **corner** zone is specified at 950 mm typical and 700 mm minimum. A person in dark trousers, in an ordinary lit room, at the edge of the field, is a sub-metre sensor. Dropping to 4x4 buys back most of that range and throws away three quarters of the zones.

VL53L5CX, continuous 8x8 at 15 Hz (DS13754 Table 18). Only two reflectances are characterised, and ST footnotes its own 17% grey chart as *"measured 13% in IR at 940 nm"*:

Target	Zone	Dark typ	Dark min	5 kLux typ	5 kLux min
White 88%	Inner	3500 mm	2600 mm	1100 mm	950 mm
White 88%	Corner	3100 mm	1700 mm	1000 mm	800 mm
Grey 17%	Inner	1300 mm	900 mm	800 mm	600 mm
Grey 17%	Corner	1100 mm	600 mm	650 mm	400 mm

VL53L7CX and VL53L7CH, continuous 8x8 at 15 Hz (DS13865 Table 18). This is what the wide field costs:

Target	Zone	Dark typ	Dark min	5 kLux typ	5 kLux min
White 88%	Inner	2000 mm	1700 mm	500 mm	400 mm
White 88%	Corner	1900 mm	1100 mm	500 mm	400 mm
Grey 54%	Inner	1600 mm	1500 mm	400 mm	400 mm

Target	Zone	Dark typ	Dark min	5 kLux typ	5 kLux min
Grey 17%	Inner	800 mm	700 mm	350 mm	250 mm
Grey 17%	Corner	750 mm	450 mm	250 mm	200 mm

The wide FoV is paid for one-for-one in range. **In a lit room the VL53L7CX at 8x8 is a 0.2-0.5 m sensor.** At 4x4/30 Hz it reaches only 500-650 mm in 5 kLux. Six of them give 360 deg of geometry and almost no range.

Accuracy (VL53L8CX Table 18), 8x8 rows: from 20-200 mm, **+/-11 mm** against a white 88% target in the dark but **+/-14 mm** against a 17% target under 5 kLux; from 200-4000 mm, **+/-5%** in the dark at all three reflectances, degrading to **+/-8%** against a 17% target under 5 kLux. Every accuracy figure carries a reflectance and an ambient-light condition; none of them is a single number. ST adds, verbatim, that *"the accuracy of the corner zone data compared to the center 4 zones may degrade by up to 4%"* and that assemblies should allow *"typically an additional 1~2%"* for PCB tilt and housing.

Power (VL53L8CX): 215 mW in continuous mode at either resolution; **6.7 mW for 8x8 autonomous at 1 Hz**, and 1.6 mW for 4x4 at 1 Hz. That autonomous figure, with a host interrupt on a programmed distance or motion threshold, is the reason this family suits a battery robot at all.

The CH histogram variants

The VL53L8CH and VL53L7CH are **firmware variants, not different silicon**. The VL53L8CH is pin-to-pin compatible with the VL53L8CX and driver compatible with the VL53L7CH, and ST titles its datasheet (DS14310 Rev 3) *"Artificial intelligence enabler, high performance 8x8 multizone Time-of-Flight (ToF) sensor."* The CH firmware adds the **compact and normalized histogram (CNH)**: the raw per-zone return-signal-versus-range profile, on top of all the normal ranging outputs.

CNH parameter	Value
Histogram bin width	250 ps
Bin equivalent range	37.5 mm
Bins in the core histogram	128
CNH buffer maximum size	6160 bytes
Maximum zones per CNH aggregate	64

This is the only part in the family that carries material and shape information rather than range alone. A flat cardboard box face normal to the sensor puts essentially all its return into 1-2 bins (37.5-75 mm of depth spread). A cat is a curved furry body roughly 200-300 mm deep inside one zone at 2-3 m, so its return spreads over about seven bins with a long low fur tail. A human leg is a roughly 150 mm cylinder, about four bins. A pet standing in front of a wall shows as two separated peaks that the processed "distance" output collapses into one number.

The cost is bus time. ST's own Table 18 gives a 64-zone, 18-bin CNH frame as **6108 bytes and 56 ms of transfer at 1 MHz I2C**, for 15 fps. The VL53L8CH's SPI runs at only 3 MHz against the CX's 20 MHz. **Two CH sensors saturate a 1 MHz I2C bus at 15 Hz. Ten is impossible.** Put CNH on exactly one sensor, on its own bus.

The VL53L9CX

Vendor-page-verified throughout; the datasheet was not retrievable.

Parameter	Value
Zones	up to 54 x 42 = 2268
FoV	55 x 42 deg H x V, software-reducible
Range	under 5 cm to 8.8 m, no reflectance or ambient condition published
Frame rate	up to 100 Hz of processed data
Outputs	depth, 2D IR with and without active illumination, reflectance, confidence
Size	12.8 x 6.1 x 4.6 mm

Each zone subtends about 1.02 deg horizontally against 5.625 deg for an 8x8 VL53L8CX. That is a genuine generational jump and it fixes the CX line's worst weakness. It is still the wrong part for this robot, for four reasons that are each sufficient: it is **MIPI CSI-2 / I3C only**, so it needs a Linux-class SoC with a camera port per sensor and cannot hang off a microcontroller; the evaluation board is **\$80.39** and the X-NUCLEO-53L9A1 is **\$96.44**; both boards read **0 in stock** at DigiKey on 2026-09-12,

with one STEVAL unit due 2026-10-28 and no ETA at all for the X-NUCLEO; and there is **no microcontroller driver** -- no stm32duino repository, no Arduino library, no CircuitPython driver.

Integration facts you must budget for

Firmware upload. `v153l5cx_init()` "copies the firmware (~84 kbytes) to the module by loading the code over the I2C interface and performing a boot routine" (UM2884 Rev 2). The image lives in host flash as a const array and **one copy serves every sensor of the same part number**. At 1 MHz I2C the upload is roughly 0.8-1.0 s per sensor, so ten sensors is an 8-10 second boot unless you use SPI or parallel buses (calculated). Both Pololu and SparkFun state outright that 8-bit MCUs cannot run these parts.

RAM and bus bandwidth. With one target per zone and all outputs enabled, a VL53L8CX_ResultsData frame is **1357 bytes**. Disabling ambient-per-SPAD, SPADs-enabled, signal-per-SPAD, range-sigma and reflectance brings it to **397 bytes**. At 1 MHz I2C, nine bits per byte:

Configuration	Bytes/frame	Bus time	Sensors at 15 Hz, 80% bus
Full output, 1 target/zone	1357	12.2 ms	4
Minimal output	397	3.6 ms	14
CH, 64 zones x 18 bins CNH	6108	56 ms	1

I2C addressing. Every part in the family boots at address 0x52 (8-bit form; 0x29 in 7-bit form). To build a ring you pull down the LPn pin of every device except the one being reprogrammed, call `v153l5cx_set_i2c_address()`, then raise the pins, repeating per device. **The new address is volatile and must be re-applied after every power cycle**. That means one host GPIO per sensor, or an I/O expander at a fixed non-colliding address. This is the most common integration surprise with this family; put it in the schematic on day one.

Mutual interference and the real ring refresh rate. Every unit emits at 940 nm with no documented coded modulation, and **ST publishes no mutual-interference rejection figure for co-located units**. The VL53L8CX does carry a SYNC pin for frame synchronisation, listed in Table 15 alongside I2C, SPI and INT and documented in UM3109. The safe architecture is to time-slice the ring so that sensors with overlapping fields never integrate at the same moment; opposite pairs can fire together because their fields do not overlap. Twelve sensors in four time slices at 15 Hz give a full-ring refresh of about **0.27 s**. Size the stopping distance from that number, not from the 60 Hz headline. Confidence: inferred.

Cover glass. ST's histogram algorithms give crosstalk immunity "above 60 cm"; below 60 cm the crosstalk can exceed the correction. Run the Xtalk calibration plugin if the sensors sit behind a bezel.

Drivers.

Platform	Support	Last touched
ST ULD (bare C)	UM2884 L5CX, UM3109 L8CX, UM3183 L8CH	reference, BSD-3
STM32Cube	STMicroelectronics/x-cube-tof1	2025-04-03
Arduino	stm32duino/VL53L8CX, VL53L8CH, VL53L7CX, VL53L7CH, VL53L5CX	2026-07-01 down to 2023-08-23
CircuitPython	sensebox/CircuitPython_VL53LxCX, community, no Adafruit driver	2025-09-19
Linux	ST publishes a driver; no mainline IIO driver confirmed (unverified)	--

The maintenance signal is worth reading: the VL53L8CX Arduino library was updated 2026-07-01, the VL53L5CX library not since 2023-08-23.

Where to buy, prices read 2026-09-12

Product	Vendor	SKU	Price USD	Stock
VL53L5CX carrier	Pololu	3417	19.95	Active and Preferred
VL53L7CX carrier	Pololu	3418	19.95	Active and Preferred
VL53L8CX carrier	Pololu	3419	24.95	Active and Preferred
Qwiic ToF Imager L5CX	SparkFun	SEN-18642	32.50	Backorder
Qwiic Mini ToF Imager L5CX	SparkFun	SEN-19013	25.95	In stock

Product	Vendor	SKU	Price USD	Stock
SATEL-VL53L8	DigiKey	497-SATEL-VL53L8-ND	32.95	190, 13 wk lead
VL53L5CX-SATEL	DigiKey	497-VL53L5CX-SATEL-ND	22.70	437, 51 wk lead
X-NUCLEO-53L8A1	DigiKey	497-X-NUCLEO-53L8A1-ND	38.24	40, 4 wk lead
VL53L8CX bare chip	DigiKey	VL53L8CXV0GC/1	8.77	listed
STEVAL-VL53L9	DigiKey	29294599	80.39	0, 1 due 2026-10-28

Adafruit stocks no multizone grid part. A search on 2026-09-12 returns only the single-zone VL53L0X, VL53L1X, VL53L4CD, VL53L4CX and VL6180X. No VL53L8CH or VL53L7CH price is published by any source read. Nothing in the family is discontinued or NRND, but "Active" is not "buyable": note the 51-week lead behind the VL53L5CX-SATEL stock and the SparkFun backorder.

On a 350 mm robot

Sensor count. The naive arithmetic $360 / 45 = 8$ is wrong. The sensors sit on the body skin, not at the robot's centre, so with a 175 mm body radius and spacing exactly equal to the FoV the adjacent edge rays are *parallel* and the blind wedge between them never closes. Computed by ray intersection from the datasheet FoV figures:

Sensor	FoV H	N	Spacing	Seam closes at	Blind slot
VL53L8CX	45 deg	8	45.0 deg	never	134 mm, permanent
VL53L8CX	45 deg	12	30.0 deg	513 mm	338 mm deep
VL53L8CX	45 deg	16	22.5 deg	343 mm	168 mm deep
VL53L7CX	60 deg	6	60.0 deg	never	175 mm, permanent
VL53L7CX	60 deg	10	36.0 deg	421 mm	246 mm deep

Twelve VL53L8CX is the buildable ring. Ten VL53L7CX would also close, more tightly and for \$199.50 against \$299.40, but its 5 kLux 8x8 range against a 17% target is 250-350 mm versus the VL53L8CX's 950-1150 mm. A ring you cannot use in a lit room is not a ring. Toeing each sensor outward a few degrees shortens the closure distance substantially and is the cheapest fix the mechanical design can offer.

Note what the seams are made of. Two adjacent fields meet at their corner zones, and the corner zone is the weak zone: in the 8x8 table above, under 5 kLux against a 17% target, the VL53L8CX corner is specified at 950 mm typical against the inner zone's 1150 mm, and ST states corner accuracy may degrade by up to 4% against the centre four zones. The ring is worst exactly where two sensors hand over.

Height and tilt. Mount the ring at **300 mm with the axes horizontal**. At 45 deg vertical the 8x8 row centre elevations run -19.9, -14.5, -8.8, -3.0, +3.0, +8.8, +14.5, +19.9 deg. From 300 mm, rows 0-2 strike the floor at 0.88, 1.20 and 1.96 m slant range, which gives a stable floor baseline: a return shorter than baseline is an obstacle standing on the floor, and a return longer or invalid is a step down. That is free and robust. The price is that the top of the field at 1.5 m range is only **921 mm above the floor** -- a standing adult's head and shoulders are outside the frame at every useful range. Tilting the ring up 10-15 deg recovers height at the cost of the floor baseline. Do not try to do both with one ring; add a second sensor higher up instead.

What an adult and a cat look like. Zone boundaries are linear in tangent space, so at 8x8 a 45 x 45 deg part has a zone width of 0.10355 x range:

Range	Zone size	Adult torso 450 mm	Adult height 1700 mm	Cat height 250 mm	Cat length 500 mm
1 ft, 305 mm	31.6 mm	fills all 8	fills all 8	7.9 rows	fills all 8
3 ft, 914 mm	94.7 mm	4.8 col	fills all 8	2.6 rows	5.3 col
5 ft, 1524 mm	157.8 mm	2.9 col	fills all 8	1.6 rows	3.2 col
8 ft, 2438 mm	252.5 mm	1.8 col	6.7 rows	1.0 row	2.0 col
10 ft, 3048 mm	315.6 mm	1.4 col	5.4 rows	0.8 row	1.6 col

Read that table together with the range tables above. At 1 ft everything fills the frame and there is no silhouette at all, only a distance field: emergency-stop range. At 3-5 ft both angular resolution and range performance are simultaneously adequate, and this is the only working envelope. At 8-10 ft a cat is one zone or less -- and, more to the point, at 8x8 under 5 kLux against a 17% target the sensor is specified to 950-1150 mm, so **a cat at 8 ft in a daylight room does not appear at all**. The

honest statement is "no detection at 8 ft in daylight; one zone at 8 ft in a dark room".

Can the grid silhouette alone classify human, pet and box? No. Four independent reasons, each sufficient. Vertical truncation: from any height a 350 mm robot can carry, a standing human's head is outside the 45 deg vertical field inside 3 m, so height -- the most discriminative feature -- is not measurable, only "taller than the frame". Angular starvation: a pet is 1-2 zones beyond 2.4 m, and two zones cannot carry shape. Reflectance confound: whether the sensor reports at all depends strongly on reflectance and ambient light, so a black cat and no cat look identical in a lit room at 2 m; absence of return is not absence of target. Static ambiguity: a sleeping cat, a backpack and a cardboard box produce the same 2 x 3 blob.

What does work, in priority order: **height above a calibrated floor plane** (pet if the cluster maximum is under 450 mm, human if over 900 mm), which needs the target's top inside the FoV and so needs a higher or tilted sensor; the **per-zone motion indicator**, initialised with `vl53l5cx_motion_indicator_init()`, which is free, runs in sub-10 mW autonomous mode and is the correct animate-versus-inanimate primitive; **temporal cadence** (a walking human's leg zones oscillate at roughly 1.8-2.2 Hz, a trotting cat at 3-5 Hz -- inferred, ST publishes no gait data), as confirmation only; and the **CNH histogram**, which is the one feature that can say "box" rather than merely "not moving". Everything that falls outside the human and pet branches must still raise an obstacle, because avoidance can never depend on classification succeeding.

Verdict

Buy it for: a dense near-field obstacle and floor-plane sensor at 0.3-1.5 m, in a ring, on a 32-bit host. Twelve VL53L8CX at 300 mm, 8x8 at 15 Hz, minimal output mode, target order `Closest`, four interference time slices, about \$300 of sensors. Buy one VL53L8CH for a forward classification head at about 900 mm on its own I2C bus, where the CNH histogram supplies the box-versus-body discrimination the grid cannot. Use the ring's autonomous mode at 1 Hz with thresholds on the motion indicator as an always-on presence wake-up: about 80 mW for all twelve.

Do not buy it for: room-scale sensing, outdoor or conservatory use (ST characterises only 0 and 5 kLux; direct sunlight is 100-120 kLux and is **not published**), dark-target detection beyond about a metre in a lit room, glass or mirrors (a 940 nm dToF ranges straight through a window -- **not published**), or any ring built on the assumption that eight sensors cover 360 deg. Do not buy the VL53L9CX for this robot: it needs a MIPI CSI-2 host, both evaluation boards are at zero stock, and no microcontroller driver exists. Design it in for 2027.

The one thing that will disappoint you: it is not a depth camera. An 8x8 zone subtends 5.625 deg, so the "image" is 64 blurred distance averages, not a picture -- and the 400 cm on the box is a dark-room, white-target, inner-zone, 4x4 number that becomes 950 mm at the corner against dark clothing in an ordinary lit room. Anyone expecting a silhouette they can classify will get a handful of blobs whose most reliable feature is that they are moving.

Standalone guide: Adafruit 4010, the Slamtec RPLIDAR A1

What this part is, and what it is not

Adafruit product **4010** is the **Slamtec RPLIDAR A1**, manufacturer model **A1M8**: a 360 deg two-dimensional spinning laser triangulation scanner ([product page](#), read 2026-09-12). It is 96.8 x 70.3 x 55 mm on the [Slamtec parameter page](#) and 170 g in the datasheet; Adafruit's own page lists the outline differently, as 98.5 mm wide by 60 mm high. In the box: the scanner head, a USB-to-serial adapter board, and a cable.

Do not confuse product 4010 with the Vishay VCNL4010, a single-point infrared proximity and ambient-light sensor sold by Adafruit as product **466**. A parts search for "4010" returns both. If it is a small breakout board it is the wrong part; if it spins, it is the LiDAR.

Revision matters for the range figure. Adafruit states only that "As of June 2, 2021" Slamtec revised the unit, labelled A1M8. No suffix is printed and neither Slamtec page mentions "R6", which appears only on reseller pages, so **treat "R6" as inferred, not vendor-stated**. What matters is R5 or later, which the 2021 date and the 12 m figure both imply.

Specification, with every stated condition

From the Slamtec LD108 A1M8 datasheet rev 2.1 (2018-02-05, Adafruit-hosted) and v2.2 (2019-02-14). Every row is datasheet-verified except the accuracy row, which is vendor-page-verified.

Item	Typical	Min / Max	Stated condition
Range, R5 and later	0.15 - 12 m	-	Comments column reads only "White objects"
Range, R4 and earlier	0.15 - 6 m	-	Comments column reads only "White objects"
Angular range	0 - 360 deg	-	-
Scan field flatness	-	-1.5 / +1.5 deg	Plane wobble; v2.2 only
Distance resolution	< 0.5 mm	-	Below 1.5 m only
Distance resolution	< 1 % of distance	-	All distance range
Angular resolution	1 deg	-	At 5.5 Hz scan rate
Sample frequency	8000 Hz	8010 Hz	R1/R2 capped at 2000 Hz; R3/R4 need firmware 1.24+
Scan rate	5.5 Hz	1 - 10 Hz	"measured when RPLIDAR A1 takes 360 samples per scan"
Points per revolution	1450	-	Datasheet Introduction, at 5.5 Hz, needs 8 kHz sampling
Laser	785 nm, 3 mW peak	775 - 795 nm, 5 mW	Class I, 110 us pulse, 21 CFR 1040.10/.11, Notice 50
Accuracy	1 % of range	-	To 3 m; 2 % from 3 - 5 m (parameter page)

The field of view is 360 deg horizontal and effectively zero vertical. There is no vertical fan and no diagonal figure to misread; the only vertical extent is the +/-1.5 deg flatness wobble, a defect tolerance rather than coverage.

Reflectance is this datasheet's weak point. Slamtec states only "White objects" against the 12 m figure: no reflectivity percentage, no dark-target range, no ambient figure in lux. The phrase "white objects with 70 % reflectivity" circulates in third-party copy, but full-text search of rev 2.1 and v2.2 on 2026-09-12 returned zero hits for "70", "reflectivity" or "reflectance" in any specification row, and the parameter page does not carry it either. **Dark-target range is not published.** Derate substantially for a black cat, a dark jacket or a matte bumper, and verify empirically.

The datasheet claims only operation "in all kinds of indoor environment and outdoor environment **without sunlight**". Adafruit's page lists "Distance Resolution: 0.2cm" - where page and datasheet differ, **use the datasheet**. Class I with an invisible 785 nm beam is the right rating for a machine scanning at cat eye height, and the datasheet says the short pulse "can make sure its safety to human and pet." The unit shuts the laser down on excess transmit power, power-on failure, unstable rotation or abnormal sensor behaviour, and the host can query that health state.

Output and rate

Each sample carries Distance in mm from the rotating core, Heading in degrees, Quality as a signal-strength proxy, and a Start Flag marking a new revolution. That is the entire output. **There is no object, no track, no velocity, no classification and no presence flag.**

Two published sampling conditions coexist: **1 deg resolution at 360 samples per scan** in the specification table, and, in the Introduction, "5.5 hz when sampling 1450 points each round" - about 0.25 deg, but needing 8 kHz sampling and so R5 or later. **Design to 1 deg.** The Adafruit guide also warns that "a single revolution is not guaranteed to give a reading for each possible angle": scans are sparse and irregular, with float angles, not a clean 360-element array.

Power, wiring and the USB adapter

All rows datasheet-verified.

Rail	Typical	Max	Comment
Scanner voltage	5 V	4.9 - 5.5 V	"may damage the core" above max
Scanner ripple	20 mV	50 mV	"High ripple may cause the core working failure"
Scanner start current	500 mA	600 mA	"Underpower may cause the startup failure"
Scanner run current	300 mA	350 mA	Work mode, 5 V input
Motor voltage	5 V	5 - 10 V	Adjust voltage according to speed
Motor current	100 mA	-	5 V input

Budget 400 mA at 5 V, so 2.0 W continuous, with about 700 mA and 3.5 W of inrush at start - twenty to forty times a single-zone ToF part, and it never stops while scanning. Waveshare's A1 page lists "System Current: 100 mA, Power Consumption: 0.5 W", which cannot be reconciled with 300 mA scanner plus 100 mA motor and looks like a reseller transcription error. **Design to the datasheet: 2 W.** Slamtec requires the scanner and motor to be **powered separately** "in order to ensure data accuracy". Sharing a rail with drive motors is the commonest cause of flaky scans.

Data interface: **3.3 V TTL UART, 115200 bps, 8N1** on a PH2.54 7-pin connector, outputs 2.9 - 3.5 V high and 0.4 V or less low, inputs accepting 1.6 - 3.5 V high. Motor enable is MOTCTL, a 0 - 5 V PWM signal; through the bundled adapter the driver asserts **DTR** to spin the motor, unlike the A2 and A3 which use PWM.

To a Raspberry Pi: use the bundled adapter, confirm /dev/ttyUSB0, and feed it from a supply that holds up under the 600 mA start current - under-volting gives the classic fault of a motor that spins while health never reaches Good. Without the adapter the Pi GPIO UART is 3.3 V and directly compatible, but you supply both 5 V rails and drive MOTCTL yourself.

To an ESP32: UART pins are 3.3 V, so TX, RX and ground connect straight to the header with no level shifting; supply both 5 V rails from the robot, not the ESP32 board. Note that the Adafruit library's docstring says "The Current Version does NOT support CircuitPython", so on a microcontroller you port the packet parser yourself. Note also that the link is fixed at 115200 bps, and that the 1450-point condition needs 8000 samples per second, so the driver's express mode (scan_type=2) is the path to it. **Design to 1 deg and verify the denser mode on your own unit with iter_measurements before relying on it.**

Drivers, libraries and ROS

pip install adafruit-circuitpython-rplidar gives a thin MIT-licensed wrapper adopted from the Skoltech rplidar library: .info, .health returning ('Good'|'Warning'|'Error', code), .start_motor(), .set_pwm(), .start(scan_type), .iter_measurements() yielding (new_scan, quality, angle, distance), and .iter_scans() yielding one list per revolution. **Maturity: adequate, not strong** - still linked from a guide edited 2024-06-03, but the CircuitPython port promised in 2019 has not landed.

Past a demo, use the vendor stack: the C++ [rplidar_sdk](#) and [rplidar_ros](#) for ROS 1, or [sl_lidar_ros2](#) for ROS 2, both publishing sensor_msgs/LaserScan. **That output format is the A1's real advantage** - the hardware is not special, but [slam_toolbox](#), [nav2](#) costmaps and the leg-detection packages consume LaserScan already. Reported field problems: the **bundled USB adapter is the most common failure item** (motor spins, no data, adapter LED dark), missing cables, enumeration failures on SBCs lacking the USB-UART kernel driver, and under-volting at startup.

Price, stock and service life

Read 2026-09-12: Adafruit lists **USD \$99.95, 19 in stock**, actively stocked, not discontinued, not end-of-life. DFRobot sells the same scanner at **\$99.00**.

This is the only sensor in this document with a wear-out mechanism, and its service life is not published. The rotor turns continuously whenever the unit scans. Slamtec's product page states it uses **OPTMAG optical-magnetic coupling**

instead of a slip ring, removing the sliding contact that normally limits a spinning LiDAR's life; that is the only durability claim either document makes. **No MTBF and no service-life hours are published for the A1M8**, and neither document gives a replacement interval for the spinning assembly. Treat the rotor as a consumable of unknown life: log run hours and hold a spare. Operating temperature is 0 - 45 deg C in the datasheet and 0 - 40 deg C on the parameter page; design to 0 - 40 deg C.

On a 350 mm robot: mounting and the planar slice

Mount the A1 on the **rotational centre axis** with a clear 360 deg optical window. This is not a sensor you distribute; it is one sensor you centre, and any mast, handle or antenna in the plane is a permanent dead sector to mask in software. The 150 mm blind radius is harmless - the 175 mm body half-radius exceeds it, so the blind zone hides inside the footprint - but the corollary is unforgiving: anything that gets under or against the chassis was never seen, so bumpers and cliff sensors are not optional.

Mounting height decides what the robot can see, and no height works for everything. Target dimensions from geometry.json:

Target	Height above floor	Plane at 200 mm	Plane at 400 mm
Adult, standing	1700 mm	Seen - two ankle clusters	Seen - two shin/knee clusters
Toddler	850 mm	Seen - shins	Seen - thigh or hip
Large dog (labrador)	650 mm at shoulder	Seen - body	Seen - upper body
Small dog (terrier)	330 mm at shoulder	Seen - body	Missed entirely
Cat, standing	250 mm at shoulder	Seen - top 50 mm above plane	Missed entirely
Cat, lying	130 mm profile	Missed - plane passes 70 mm over	Missed entirely
Table overhang	40 mm slab, 700 mm gap	Legs seen, slab invisible	Legs seen, slab invisible

A sleeping cat is invisible at any plane height you can usefully pick. Clearing a 130 mm lying profile needs a plane below about 130 mm, and there the -1.5 deg flatness edge strikes the floor at $130 / \tan(1.5 \text{ deg}) =$ about 5.0 m, filling the far field with floor returns. At 200 mm the floor strike is about 7.6 m; at 400 mm about 15 m, beyond range. *Arithmetic from the published flatness spec, not a Slamtec table.* **The table overhang is never visible:** a plane below the slab sees the legs and reports clear air where the tabletop is, which is how a robot drives its mast into a table edge. The plane is a wedge, not a sheet: 26 mm thick at 0.5 m, 52 mm at 1 m, 128 mm at 2.4 m.

Scan points on target

Counts for the adult, the leg or cat and the chair leg are from the computed catalogue in capability.json; the toddler column is the same angular-subtense arithmetic applied to the 250 mm toddler width in geometry.json. All are at 1 deg resolution. Arc spacing per degree is 5.3 mm at 1 ft, 16.0 mm at 3 ft, 26.6 mm at 5 ft, 42.6 mm at 8 ft, 53.2 mm at 10 ft.

Distance	Adult torso 450 mm	Leg or cat 140 mm	Toddler 250 mm	Chair leg 30 mm
1 ft (0.30 m)	73 pts	26 pts	45 pts	5.6 pts
3 ft (0.91 m)	28 pts	9 pts	16 pts	1.9 pts
5 ft (1.52 m)	17 pts	5.3 pts	9 pts	1.1 pts
8 ft (2.44 m)	11 pts	3.3 pts	6 pts	0.7 pts, nothing
10 ft (3.05 m)	8 pts	2.6 pts	5 pts	0.6 pts, nothing

geometry.json gives a human leg and a standing cat the same 140 mm width, so **at every distance a human leg and a cat return the same number of points.** The research lane file assumed a 120 mm shin and reported about 23, 8, 5, 3 and 2 points - the same calculation on a narrower target. Prefer the table above: it uses the document's shared target dimensions and angular subtense rather than a flat arc division. Read the chair-leg column twice. **A 30 mm chair leg falls below one scan point beyond about 6 ft and can be missed entirely between samples.**

What 5.5 Hz means on a moving robot

One revolution takes **182 ms**. At 0.5 m/s the robot travels 91 mm while a revolution is collected, so its first and last points came from positions 91 mm apart - compensate with odometry or accept about 90 mm of angular smear. Targets move too: a person approaching at the ISO 13855 walking speed of 1.6 m/s covers 291 mm per revolution, more than half a 450 mm torso width.

Worst-case data age is a full revolution, because any bearing is sampled once per 182 ms. The computed stopping appendix sizes this chain at 5.5 Hz with a firm 2.0 m/s² brake: about **107 mm of stopping distance at 0.3 m/s, 203 mm at 0.5 m/s and 532 mm at 1.0 m/s**, against 136 mm at 0.5 m/s for a VL53L5CX ToF chain at 15 Hz on the same robot. *Computed, not vendor-stated*. At 1.0 m/s that is more than a robot length, so the A1 alone should not be the last line of defence at speed.

The consumer must keep up: the Adafruit guide warns that if the callback "takes too long to process a scan, data from the RPLIDAR will eventually be dropped." There is no flow control. **Finish the whole perimeter decision inside 180 ms.**

From a range ring to a perimeter decision

None of this ships with the sensor.

1. **Ingest and de-skew** - consume `iter_scans()` or the ROS LaserScan fast enough to avoid drops, with odometry compensation across the revolution.
2. **Mask dead sectors** from the robot's own structure; discard returns inside the chassis radius.
3. **Filter by quality** - the main defence against specular ghosts from glass and mirrors.
4. **Cluster** with adaptive-breakpoint or DBSCAN segmentation on a distance-dependent threshold, because arc spacing grows with range.
5. **Extract cluster features** - width, depth, curvature, point count, mean range, linearity: what the classical ROS `leg_detector` random forest consumes.
6. **Track** with a constant-velocity Kalman filter, so every cluster carries a velocity. **Velocity is the only strong life-versus-furniture cue available.**
7. **Classify** - leg-pair geometry for humans, small low moving clusters for pets, stationary non-leg shapes as inanimate. Treat every unclassified cluster as potentially a person.
8. **Decide** on the minimum range in the swept corridor ahead, not the global minimum.
9. **Fuse** with bumpers, cliff sensors and a presence channel.

Steps 4 to 7 are a multi-week software project, or a decision to adopt ROS 2 and take nav2 plus an off-the-shelf person detector. Budget for that, not for the \$99.95.

Direct answers

Any obstacle it could collide with? Yes - within the plane, and only within the plane. That is what the sensor is for, and it does it well: 360 deg, 5.5 - 10 Hz, millimetre-class range on every surface that returns 785 nm light. What it misses is categorical, not marginal:

- Anything **below** the plane - a doorstep, a shoe, a power strip, a sleeping cat.
- Anything **above** the plane - a tabletop, a counter overhang, an open drawer, an outstretched arm.
- **Glass, mirrors, polished metal and still water.** A glass door returns almost nothing at normal incidence and a phantom object at the mirrored distance off-axis. The A1 has no defence against this, and it is the classic domestic-robot failure.
- **Matte black and dark fabric at range**, where no derated range is published and dropouts, not wrong ranges, are the expected symptom.
- **Thin objects** - chair legs at range, cable runs, pedestal spokes - which fall between 1 deg samples from about 3 m out.
- **Descending edges.** A horizontal plane cannot see a hole.

One more coverage risk, **inferred from the multi-LiDAR crosstalk literature and the absence of any published mitigation, not vendor-stated**: Slamtec publishes no crosstalk specification and no multi-unit coordination mechanism for the A1, so two of these robots in one room should be expected to produce spurious returns.

It classifies nothing on its own. Any statement about what an object *is* is your software, and the 2D-LiDAR person-detection literature is unanimous that "the lack of identifying information in 2D range data" is the core drawback. DR-SPAAM, PeTra and the FROG dataset beat the geometric baseline, but all infer over time with known false-positive rates. A chair leg and a cat leg at 3 m are the same two points.

Distance	Moving human	Moving cat	Static human	Static pet
1 ft	Yes, 73 pts, high confidence	Yes, 26 pts, plane at 200 mm	Object only	Object only
3 ft	Yes, 28 pts, good leg-pair	Yes, 9 pts	Object only	Object only
5 ft	Yes, 17 pts	Yes, 5.3 pts, shape marginal	Object only	Object only
8 ft	Yes, 11 pts, pair separation poor	Yes, 3.3 pts, chair-leg confusable	Object only	Object only
10 ft	Yes, 8 pts	Yes, 2.6 pts, dark fur may drop out	Object only	Object only

A human? Yes, as a human-sized object, at all five distances, provided the plane crosses the body. As a *human* specifically, only by inference from leg-pair geometry and motion over time, and only reliably to roughly 4 - 5 m indoors. *Inferred from the sampling geometry and the people-tracking literature, not a vendor figure.*

A pet? Only with the plane low enough to cross a standing animal, so at or below 200 mm for a cat. A lying cat is missed at every usable height, and a black cat is the worst optical case for a 785 nm triangulation LiDAR with no derated range published.

A static human? No, not as a human, ever. The A1 measures geometry, not life: no micro-motion channel, no respiration signal, no thermal signature. A person standing still at 3 m is a pair of unmoving clusters, which is also the description of a table leg. **The A1 cannot meet a "motionless person in the room" requirement and no software will make it.**

Verdict

Buy it, and buy it as the obstacle layer, not the presence layer. On a 350 mm robot, an A1 at the rotational centre at knee height, **350 - 450 mm above the floor**, is the correct primary obstacle sensor: it captures the human leg-pair signature the tracking literature is built around, crosses a standing labrador, and sits below most tabletops so it sees table legs. It explicitly sacrifices the standing cat, the small dog, the sleeping cat and every overhang. If pets matter more than reach, drop the plane to 200 mm and accept floor strike at about 7.6 m. The pairing that fixes the weaknesses is three layers:

- 1. **RPLIDAR A1**, \$99.95 - 360 deg obstacle geometry, mapping, navigation, moving-target tracking.
- 2. **A presence channel** - a 60 GHz mmWave presence radar or a thermal IR array - for the static human and the sleeping pet, which one horizontal plane structurally cannot do.
- 3. **Contact and cliff** - bumpers for the sub-plane blind zone, downward ToF or IR for descending edges, which a horizontal plane cannot see because it cannot see a hole.

If only one sensor fits and the requirement is "do not collide with anything," the A1 is the right choice and nothing in this price class is close. If the requirement is "know whether a person or a cat is in the room," the A1 is the wrong anchor, and a presence radar or thermal array should be.

Options beyond the two catalogues: the rest of ST, and scanning lidar

Two things are worth buying outside the Adafruit and DFRobot catalogues this reference has covered so far, and only two. The first is the **ST VL53L3CX multi-target ranger** on a Pololu carrier, at \$19.95. The second is a **360 deg planar dTOF scanning lidar** - the Slamtec RPLIDAR C1 at \$69.00 or the LDROBOT LD19 at \$99.00. Everything else in this chapter is either a part with no hobby breakout, a part whose datasheet could not be read, or a scanner that costs more mass, more power or more money than the two named above for no gain on a 350 mm robot.

The second half of the chapter answers the question the operator actually asked: a spinning lidar is not a better ToF ring, it is a different sensor with a different blind spot, and the two blind spots are complementary rather than overlapping.

Part 1 - the rest of ST's FlightSense portfolio

VL53L3CX: the only ST part worth leaving the Adafruit catalogue for

The VL53L3CX reports **up to four ranges at once inside one cone** (datasheet-verified, ST DS13204 Rev 2, [mirror](#)). No other single-cone ST part does this, and Adafruit does not stock it.

Parameter	Value	Condition
Field of view	25 deg, the datasheet's single cone figure	full FoV assumed covered by all range rows
Range, white 88 %	310 cm typical at 94 % detection	30 ms budget, 23 C, indoor, no infrared
Range, grey 17 %	170 cm typical	same
Range, outdoor overcast	100 cm at 88 %, 70 cm at 17 %	ST equates this to about 5 kLux daylight
Active current	16 mA typical, 40 mA peak	23 C, 2v8
Host cost	not recommended for 8-bit MCUs	histogram driver needs RAM and code space

Vendor overclaim, flagged. Pololu's carrier [#3416](#) is titled "500cm Max" and its body text says 5 m. There is no 5 m row anywhere in DS13204 Rev 2. Use **310 cm against white 88 % and 170 cm against grey 17 %, both indoors with no infrared**, both datasheet-verified.

The multi-target output is a weak but free classification cue. A cone that reports "something at 0.6 m **and** something at 2.4 m" is looking at a thin object with the room still visible behind it; a cone that reports only 0.6 m is looking at something broad enough to occlude the background. That separates a chair leg from a human torso. It does not separate a sleeping cat from a handbag.

Verdict: worth it, for one or two positions. Not for a whole ring - at \$19.95 against the VL53L4CX's \$14.95 it costs more and reaches less far.

VL53L1CB and VL53L4ED: not verified, and that is a tooling limit

Both parts' datasheets are hosted only on `st.com`, which was **unreachable from the research machine in every pass** (`curl` returned `Recv failure: Connection was reset` and a 40 s timeout; `WebFetch` timed out at 60 s). Nothing about either is asserted here beyond what ST's own product page text carries.

- **VL53L1CB** - ST's cover-glass-optimised sibling of the VL53L1X ranging core. **No VL53L1CB-specific number is verified.** No Pololu or Adafruit breakout exists. If you want the VL53L1X ranging core, buy the VL53L1X: Pololu [#3415](#), \$22.95, read 2026-09-12.
- **VL53L4ED** - ST's low-power variant of the VL53L4CD. ST's product page states **18 deg FoV, 1 mm to 1300 mm in standard conditions, and accurate measurement to 800 mm under 5 klx ambient with special settings**. That is **vendor-page-verified, not datasheet-verified**. There is no hobby breakout; it is a bare module aimed at battery-powered consumer products.

Neither part changes a design decision on this robot. Skip both.

Two traps worth restating while ST parts are on the page. First, the **VL53L1X's receiver field is programmable from 15 deg to 27 deg, and the 27 deg figure is a diagonal** (DocID031281 Rev 3). The horizontal field is narrower than the diagonal, so

sizing a ring from 27 deg overstates coverage; coverage-geometry.md owns that arithmetic. Second, the VL53L1X's headline reach is a dark-room number: long mode gives **360 cm typical in the dark but collapses to 73 cm under strong ambient** (200 kcps/SPAD, 100 ms budget, white 88 % target), while short mode barely moves, **136 cm dark to 135 cm ambient**. Size any single-cone perimeter on the ambient column, not the dark column.

SATEL evaluation boards

ST sells postage-stamp "satellite" carriers - VL53L4CX-SATEL, VL53L3CX-SATEL, VL53L1CB-SATEL, VL53L5CX-SATEL - that plug into the matching X-NUCLEO-53L*A1 expansion shield for evaluation on a Nucleo board. **Their pack quantity, contents and prices are not verified**: every SATEL page lives on st.com. They are listed here so the option is not silently omitted. Their real use is ST's reference cover-glass geometry and calibration flow, which matters if you intend to put a window in front of a ring. Price them at Mouser or DigiKey before committing to a custom carrier.

ST's **automotive and industrial FlightSense** parts are in the same unverified position and are practically irrelevant: AEC-Q100 extended-temperature parts on broker MOQ, offering no more range and no wider cone than the VL53L4CX. Qualification does not solve an information problem.

The two short-range ST parts, for completeness

The brief names them, so state the role and move on.

- **VL6180X** (DocID026171 Rev 7) is specified to **0 to 100 mm**, with the datasheet noting that ranging beyond 100 mm depends on target reflectance and external conditions. Under **5 kLux diffuse halogen** it holds **> 70 mm at 88 % and > 60 mm at 17 %** reflectance. On a 350 mm robot that means the obstacle is already under the bumper, so its only defensible role is bump-imminent or cliff detection. Pololu's carrier is **End-of-Life Rationing**, so do not design it in. That listing also markets **20 cm** by default and **60 cm** "at the cost of reduced resolution" - both above the datasheet's 100 mm, and neither carries a reflectance condition.
- **VL53L4CD** (DS13812 Rev 8) reaches **typ 1200 mm against white 88 % at 90 % detection indoors and typ 450 mm against grey 17 %**, falling to **550 mm and 400 mm** outdoor overcast. 45 cm against a dark target is a bumper sensor too.

Where to buy, and at what price

Part	Vendor and SKU	Price (read 2026-09-12)	Status
VL53L3CX	Pololu 3416	\$19.95	active
VL53L1X	Pololu 3415	\$22.95	active
VL53L4CD	Pololu 3692	\$13.95	active
VL6180X	Pololu 2489	\$19.95	End-of-Life Rationing
VL53L4CX	Adafruit 5425	\$14.95	in stock
SATEL boards	ST / Mouser / DigiKey	not published in this pass	unverified

Gap, stated plainly. Mouser's datasheet mirrors returned anti-bot JavaScript and a segfault, and no DigiKey page was read in this pass, so **no Mouser or DigiKey price is quoted in this chapter**. SparkFun sells a Qwiic dToF Imager built on the ams-OSRAM TMF8820/8821, not on an ST part; **its price was not read**. Pololu is the only non-Adafruit vendor with verified prices here.

Part 2 - scanning lidar under 500 USD

A scanning lidar puts one ranging head on a motor and spins it. One part, one cable, a full planar point cloud with no tiling arithmetic. Three tables follow because a portrait page will not carry thirteen columns.

Scan geometry and rate

Model	Scan Hz	Points/s	Ang. res.	Points/rev	Principle
RPLIDAR A1M8 (R5/R6)	5.5 typ	8000	see conflict below	see conflict below	triangulation
RPLIDAR C1	10 typ (8-12)	5000	0.72 deg	500	dTOF
RPLIDAR A2M12	10 (5-15)	16 000	0.225 deg	1600	not stated
RPLIDAR S2 / S2E	10	32 000	0.1125 deg	3200	not stated
LIDARBOT LD06	10 typ (5-13)	4500	1 deg	450	dTOF

Model	Scan Hz	Points/s	Ang. res.	Points/rev	Principle
LDROBOT LD19	10	4500	<= 1 deg	450	dTOF
YDLIDAR X2	6 default	3000	0.72 deg at 6 Hz	500	triangulation
YDLIDAR X4 PRO	6-12	5000	0.43 deg at 6 Hz	833	triangulation
YDLIDAR G4	5-12 (max 16)	9000	0.2 deg at 5 Hz	1800	triangulation
YDLIDAR T-mini Plus	6 default (6-12)	4000	0.54 deg	666	ToF
Benewake TF-Luna	not scanning	1-250 Hz frame	single beam	1	ToF
Benewake TFmini-S	not scanning	not published	single beam	1	ToF

The two research lanes disagree about the RPLIDAR A1, and the disagreement matters. The other-st-tof-and-lidar lane says the current R5/R6 part is an 8 kHz unit, so at 5.5 Hz it produces about **1450 points per revolution, 0.25 deg**, and cites DFRobot's own A1M8-R6 listing wording to that effect. The adafruit-tof-nonst lane says the datasheet conditions its 1 deg figure on 5.5 Hz and its 5.5 Hz figure on **360 samples per scan**, that 360 x 5.5 is 1980 and not 8000, and that the 1450 figure is therefore an unverified product of two specs measured under different conditions. **Prefer the conservative reading: design to 1 deg / about 360 points per revolution.** Slamtec's current spec page prints "Angular Resolution <= 1 deg" beside "Sampling Frequency 8K", which cannot both be true of one part, and a perimeter design must not be sized on the optimistic half of a contradiction. The C1 at 0.72 deg and 500 points/rev has no such ambiguity and is a better buy anyway.

Range and accuracy, with the reflectance condition

Model	Min	Max range	Condition	Low-reflectance range	Accuracy
RPLIDAR A1M8	0.15 m	12 m (R5+); 6 m (R4)	"White objects"	not published	< 1 % of distance
RPLIDAR C1	0.05 m	12 m	70 % reflection	6 m at 10 %	+/-30 mm
RPLIDAR A2M12	0.2 m	12 m	no condition on page	not published	not captured
RPLIDAR S2 / S2E	0.05 m	30 m	90 %	10 m at 10 %	not captured
LDROBOT LD06	0.02 m	12 m	70 % reflectivity	not published	30 mm typ, 45 mm max
LDROBOT LD19	0.02 m	12 m	no condition stated	not published	10 mm, 300-12000 mm
YDLIDAR X2	0.12 m	8 m	80 %, indoor	not published	2 cm below 1 m
YDLIDAR X4 PRO	0.12 m	10 m	80 %, indoor	not published	2 cm below 1 m
YDLIDAR G4	0.12 m	16 m	80 %, at 4 kHz	not published	2 cm below 1 m
YDLIDAR T-mini Plus	0.05 m	12 m	80 %	4 m at 10 %	20 mm, 0.05-12 m
Benewake TF-Luna	0.2 m	8 m	not published	not published	+/-6 cm, 0.2-3 m
Benewake TFmini-S	not published	12 m	not published	not published	not published

The LD19's 0.02-12 m carries **no reflectivity condition at all** on the Waveshare wiki, while the LD06 datasheet ties its identical 12 m to a **70 % target**. Assume the LD19 number is the same 70 %-class figure until LDROBOT says otherwise - that is inferred, not verified.

Physical, power, interface, eye safety, price

Model	Size (mm)	Mass	Power	Ambient	Class / price
RPLIDAR A1M8	96.8 x 70.3 x 55 kit	170 g	~2 W derived; 0.5 W on spec page	"no direct sunlight"	Class I; \$99.00 DFRobot, \$99.95 Adafruit
RPLIDAR C1	not captured	110 g	230 mA at 5 V (~1.15 W), reseller	40 000 lux	Class 1; \$69.00 DFRobot
RPLIDAR S2 / S2E	18 mm optical band	not captured	not captured	80 klux, IP65	Class 1; no USD price published
LDROBOT LD06	38.6 x 38.6 x 33.3	42 g	180 mA at 5 V, 0.9 W	30 kLux	Class 1; no price confirmable
LDROBOT LD19	38.6 x 38.6 x 33.5	not stated	180 mA at 5 V, 0.9 W	30 kLux	FDA Class 1; \$99.00
Waveshare D500	38.6 x 38.6 x 33.5	not stated	290 mA, 1.45 W	not stated	Class 1; not read
YDLIDAR X2	60.5 dia x 50.3 x 96	126 g	300 mA typ, 1000 mA start	0-2000 lux	Class I; not published
YDLIDAR T-mini Plus	38.6 x 38.6 x 33.9	not captured	340 mA typ	60 kLux	Class 1; not published
Benewake TF-Luna	35 x 21.25 x 13.5	< 5 g	<= 0.35 W	not published	not published; ~\$29.90
Livox Mid-360	not captured	265 g	6.5 W avg, 14 W heating	not captured	Class 1; not published

Interface and driver support. A1M8, C1, LD06, LD19, D500 and every YDLIDAR unit here stream over UART - 115200 (A1), 230400 (LD-class) or 460800 baud (C1). The Livox Mid-360 needs **100BASE-TX Ethernet** and a host that can take it. **Driver maturity is not published in the sources read for any unit in this table** - the wire protocols are documented, and community drivers are likely but unverified. The one host-side feature that is datasheet-verified is the A1M8's self-protection: it shuts the laser down on over-power, on an unstable or slow scan speed, or on a sensor fault, and the host can query that

health state over the same UART. The A1 also needs **two independent 5 V rails**, one for the scanner and one for the motor, which is real BOM on a small chassis.

Two entries should change a decision on their own. **YDLIDAR X4 PRO's published service life is 1500 hours**, about 62 days of continuous running - a consumable, not a component. **YDLIDAR X2's own datasheet lists a lighting environment of 0 to 2000 lux** - a triangulation scanner is not a sunlight sensor. Avoid the whole triangulation group (A1M8, X2, X4 PRO, G4) for a robot that will ever see a sunlit room, and avoid the A2 family's **0.2 m minimum range** on a 350 mm chassis whose skin is only 0.175 m from the centre.

The **Benewake TF-Luna and TFmini-S do not belong in this table as perimeter sensors** and are listed only because the brief names them. Both are single-beam rangefinders with roughly a 2 deg cone. Their honest role here is one cheap, long, low-power guard beam - a forward step detector or a beam on a servo. Nothing about them helps with classification.

Considered and set aside: Hokuyo, Unitree, the S3, and the one 3D unit

The brief names these, so each gets a line rather than a silent omission.

- **Hokuyo URG entry parts - not verified in this pass.** hokuyo-aut.jp returned PHP fatal errors instead of a page and the hokuyo-usa.com URG-04LX-UG01 path returned 404, so **no Hokuyo number is asserted here**. Two things can be said without inventing anything: the URG family is a **sector** scanner, not a full 360 deg one, and it is positioned a long way above the Chinese units on price. Confirm both against Hokuyo's own PDF before designing around it.
- **Unitree 4D LiDAR L1 / L2 - not verified.** unitree.com/LIDAR returned 404. No specification and no price is asserted.
- **Slamtec S3 - specs not captured in this pass.** It exists in the product line; nothing more is claimed here.
- **Livox Mid-360** is the only unit in this chapter that is not planar: **horizontal 360 deg with a vertical field of -7 deg to +52 deg**, 200 000 points/s on the first return, **40 m at 10 % reflectivity and 70 m at 80 %**, 0.1 m blind zone, angular precision below 0.15 deg. It genuinely sees the space above the robot. It costs 265 g, 6.5 W and an Ethernet host to do it, and its vertical field looks **up, not down**: mounted at 1000 mm, its lowest ray reaches the floor only at $1000/\tan(7 \text{ deg})$, about **8.1 m** out, so everything on the floor inside an 8 m radius is below the beam. That is the wrong blind spot for a cat.

Part 3 - the honest comparison

What a planar 360 deg lidar gives you that a ToF ring does not

1. **Angular density, by two orders of magnitude.** A C1 delivers 500 measurements per revolution with an angle attached to each. A 20-sensor VL53L4CX ring delivers 20 numbers. coverage-geometry.md owns the tiling arithmetic; the result it hands this chapter is that an 18 deg cone needs about 20 units to close 360 deg edge to edge, and more with overlap margin.
2. **Cluster width, therefore the only non-camera classification feature that works.** Because returns carry angles, a scan segments into clusters with measurable chord width. Two clusters 80-160 mm wide and 150-400 mm apart moving together are a person's shins. A cat is one 60-120 mm cluster low down. A ToF ring produces one distance per cone and no width at all.
3. **Scan matching, hence a background map.** 450-500 points per revolution is enough for SLAM. Once a map exists, anything that does not fit it is dynamic, and dynamic is the strongest animate-versus-inanimate cue available without a camera or a thermal sensor. A ToF ring has far too few points to scan-match, so it can never build that model.
4. **Range against dark targets.** C1: **6 m at 10 % reflection**. T-mini Plus: **4 m at 10 %**. The best single-cone ST part, the VL53L4CX, gives **2.1 m against a grey 17 % target indoors** and 1.1 m outdoors. A dTOF scanner sees a dark-trousered leg across a room; the ring sees it at arm's length.
5. **Ambient light.** Scanners quote 30 kLux (LD06/LD19), 40 kLux (C1), 60 kLux (T-mini Plus), 80 klux (S2). ST's "outdoor overcast" condition is only about **5 kLux**, and the VL53L4CX drops from 5.0 m to 1.6 m on white and 2.1 m to 1.1 m on grey-17 % there.
6. **One wire.** One UART, against 20 I2C devices that all boot at the same address and need 20 XSHUT lines or an expander plus a power-sequencing state machine.

What it fails at

1. **It sees exactly one plane, and the blindness does not shrink with distance.** A forward ToF cone narrows as you approach it. A planar lidar's blind region is a **half-space above and a half-space below, at every range**. A 300 mm standing cat is equally invisible at 0.3 m and at 8 m if the plane sits at 400 mm. The A1's scan field flatness is **+/-1.5 deg**, about +/-52 mm at 2 m - that does not help.
2. **No single plane works on this robot.** A 350 mm robot 600-1200 mm tall has a tall body to protect and a cat to avoid. A plane at 60-120 mm sees cat, floor clutter and ankles and misses every table top, counter overhang and leaning torso. A plane at 700-1000 mm sees torsos and worktops and misses the entire floor and every pet. Two planes means two lidars, two motors, and still no vertical continuity between them.
3. **It cannot separate a static pet from a static object at all.** Cluster width overlaps completely between a curled cat and a shoe. No temperature, no vertical extent, no micro-Doppler.
4. **Glass and mirrors.** Glass returns nothing or a ghost from the far side; a mirror produces a phantom room. The A1 datasheet publishes nothing on either - **not published**.
5. **It is a motor.** LD06 rated **10 000 h**, about 14 months continuous. YDLIDAR X4 PRO **1500 h**. A solid-state ToF ring has no wear-out mechanism.
6. **It needs an unobstructed optical window** at exactly its scan height, with no bracket or cable crossing it - a real industrial-design constraint on a 350 mm shell.

The penalty, in numbers

Quantity	Planar lidar	20 x VL53L4CX ring	Penalty
Hardware cost	\$69 (C1) to \$99 (LD19)	20 x \$14.95 = \$299	lidar is \$230 cheaper
Mass	42 g (LD19-class), 110 g (C1), 170 g (A1)	~20 g of breakouts	+22 g to +150 g
Power	0.9 W (LD19), ~1.15 W (C1), ~2 W (A1)	1.06 W continuous; ~0.2 W at 5 Hz duty	+0 to +0.95 W
Height intrusion	33 mm body plus a 360 deg window	none	a slot in the shell
Moving parts	1 motor, 1500-10 000 h rated	none	a wear item
Host compute	SLAM and clustering	trivial	a real CPU budget

Power is close to a wash - that is the surprise. An LD19 at 0.9 W costs about the same as a 20-sensor ring run flat out. The real penalties are **mass, the motor, the optical window and the CPU budget**. Cost runs the other way: the lidar is **three to four times cheaper** than a ring dense enough to close the perimeter.

Recommendation

- **Choose a planar dTOF lidar** when the robot needs to know where it is and what is around it - requirement (a), plus the static map that makes everything else tractable. Buy the **RPLIDAR C1** (\$69, dTOF, **0.05-12 m at 70 %**, **6 m at 10 %**, **0.72 deg**, **40 klux**, **IP54**, **110 g**) or the **LD19** (\$99, **0.02-12 m with no reflectivity condition published**, **450 points/rev**, **30 kLux**, **0.9 W**, **42 g class**). Buy the LD19 rather than the LD06: no LD06 price was confirmable anywhere, and Inno-Maker's LD06 URL now sells an LD19P.
- **Choose the ToF ring** when the robot must see vertical extent - overhangs, table tops, a hand reaching in, a lying cat - or when a spinning part is unacceptable. A ring has no wear-out mechanism and no optical window requirement.
- **Use both, and this is the right answer for this robot.** Mount the lidar plane **low, 60-120 mm**, where it catches the cat, the floor clutter and human ankles. Cover what it misses with a second sensor that has vertical extent - a multizone ToF, a thermal array or mmWave - not with a second lidar. Within ST's single-cone family the **VL53L4CX** (\$14.95, **18 deg**, **5.0 m white / 2.1 m grey-17 % indoors**, **19 mA**) is the only part worth a perimeter position, with one or two **VL53L3CX** where depth-behind-the-obstacle earns its \$19.95.
- **Neither option meets the classification requirement.** A planar lidar gives moving-versus-static and cluster width, which separates a walking human from a walking cat. It gives nothing for a sleeping cat against a backpack. That gap closes only with a sensor that measures something other than range: temperature, micro-motion or appearance.

Options for the three robot projects in this repository

The three robots in C:/dev/robots need three different perimeter builds, and the reason is electrical, not mechanical. dalek has the best shell for sensing and the worst electronics to hang off it. r2d2 has the thinnest shell of the three and 47 g of mass margin. fable-r2d2 has an entirely unused I2C bus, eighteen free GPIO, about 4.1 A of spare 5 V, and two body rings that have not been drawn yet. One answer repeated three times would be wrong on two of them.

Envelope figures are repo-verified from research/robot-project-envelopes.md, which cites the file each came from. Part figures are datasheet-verified or vendor-page-verified from data/sensors.csv. Coverage arithmetic is not repeated here; coverage-geometry.md and the computed appendix own it.

The three envelopes side by side

Property	dalek	r2d2 (revision C)	fable-r2d2
Shape at floor level	Circular fender, 300 mm diameter	Three separate feet, body suspended	Three separate feet, body suspended
Overall height	563.8 mm nominal	609.6 mm nominal	715.9 mm (driving stance)
Widest plan dimension	300 mm	~438 mm side-foot span	~486.7 mm (repo-derived)
Estimated assembled mass	~3.5 kg (repo-derived)	8.953 kg of a 9 kg ceiling	13.273 kg nominal
Drive	4 x TT 3777, skid steer	6 x TT 3777, 12 wheels, steering servo	6 x TT 3777, 12 wheels, casting rear
Controller	LilyGO TTGO T-Display (classic ESP32)	Adafruit HUZZAH32 3405	Raspberry Pi 4 4 GB + Adafruit KB2040
Battery	Bioenno BLF-1206A 12 V 6 Ah LiFePO4	Bioenno BLF-1206A 12 V 6 Ah LiFePO4	Power-Sonic PS-1270 F2 12 V 7 Ah SLA
Spare 5 V logic current	~0.3 A	not specified in the repo	~4.1 A
Free GPIO on the main MCU	essentially none	one pin (GPIO5)	Pi: 18. KB2040: one (D0/GP0)
Free I2C bus	shared 0x40 bus only	shared bus, 0x40 + 0x48 used	Pi I2C-1 + an unused KB2040 STEMMA QT bus
Head/dome wiring path	none (no slip ring)	none (no slip ring)	12-wire slip ring, all 12 assigned
Shell wall at a candidate aperture	1.8 mm PLA skirt / 4 mm PETG base ring	1.2 mm PETG body skin	3.6 mm PETG body / 3.0-3.2 mm dome
Existing ranging or presence sensor	none	none	none

dalek - the robot that must not be cut

Mounting surfaces and apertures

The base fender band, world Z13.8 to Z67.8, is the right height for a pet and a forbidden place to work. docs/mechanical.md states the rule that "the outside bumper remains continuous around the full circle", and cad/mechanical-checks.json carries a passing 2,880-sample check named continuous-bumper-missing-material. A window there breaks a released validation check.

That leaves the lower skirt, world Z67.8 to Z177.8, 110 mm tall, tapering 300 mm to 250 mm in 1.8 mm PLA. It is the thinnest wall on any of the three robots. It already carries twelve vertical seams at 30 deg pitch and twelve hemispheres per row at world Z93.8 and Z143.8, so a panel has to sit in the roughly 65 mm of clear arc between two seams. The head is unusable: no slip ring, and no head-angle encoder, so a head sensor would have no known bearing.

Because the bumper cannot be cut and the skirt is crowded, this is the one project where seeing *through* the shell is worth more than seeing through a hole. The 24 GHz parts in data/sensors.csv are recorded as sees_through_plastic = yes and need no aperture at all.

Bus, pins and current

docs/firmware.md allocates every pin the T-Display breaks out. No free interrupt line, no free UART, no free XSHUT line. The I2C bus on GPIO21/22 carries one device, a PCA9685 at 0x40; everything else is free, including 0x29, 0x2A and 0x2B.

That decides the part. A UART radar is unusable. A ring of ST multizone parts is unusable, because every VL53 unit boots at 0x29 with a volatile address and so needs one XSHUT GPIO per unit or an I2C multiplexer. The DFRobot SEN0610 is the one presence radar in the catalogue with a native I2C interface, strapped to 0x2A or 0x2B. Two addresses exist, so two units

is a hard ceiling, not a budget choice.

Spare current is 0.30 A on the 5V_LOGIC rail (Adafruit MPM3610 4739, 1.2 A unconditional over 6-21 V in, against a 0.90 A design load). That is the binding constraint on the whole robot. The two 5 V actuator rails are cut by the S2 stop switch, so a sensor must never sit on them: it would go blind at the moment the operator hits stop.

Recommendation

- 2 x DFRobot SEN0610 Gravity mmWave C4001, 100 x 80 deg (horizontal x vertical), I2C at 0x2A and 0x2B. USD 12.90 each.
- 1 x Adafruit 5425 VL53L4CX single-zone ToF, 18 x 18 deg, I2C 0x29, 19 mA. USD 14.95.
- Total USD 40.75.

Mount both radars flat inside the lower skirt, boresight horizontal, centred at world Z120, each between two seams, one on the front centreline and one at 180 deg. No aperture, no cut, no change to any released STL. Two 100 deg beams leave two side gaps; that is the honest cost of the two-address ceiling.

The VL53L4CX closes the radar's own blind zone, and that is why it is in the list. The C4001 datasheet gives no output below 1.2 m, which is exactly the collision zone on a 300 mm robot. Fit it on the front centreline at world Z120, tilted 5 deg down, through a single window in the same 1.8 mm skirt, sized from the ST package drawing. Put nothing over that window: the ams-OSRAM TMF8806 datasheet shows one layer of tape costing 340 mm of range - 1840 mm down to 1500 mm - on an 18 percent grey target under 170 lux halogen, the 1 klux sunlight-equivalent condition, and the same physics applies to any dToF cover.

Wiring is four conductors from the GPIO21/22 header to each module, powered from 5V_LOGIC only, with no new pull-ups.

Where this does not fit. DFRobot does not publish a supply current for the C4001, so the build cannot be closed against 0.30 A on paper. Measure one unit before buying the second. If the pair exceeds about 280 mA, the option that fits is a second Adafruit MPM3610 4739 off the 12 V bus upstream of the stop switch, giving the sensors their own 1.2 A rail. The MPM3610 rating carries no input-voltage condition, so that rail is honest at 12 V, unlike the D36V50F5 figures.

r2d2 - 47 g of margin and one free pin

Mounting surfaces and apertures

There is no skirt. The lower body mesh starts at world Z130 and the cylindrical wall at Z165; below Z130 is open air between three feet. The rear foot translates 119 mm fore-aft with posture (centre from Y = -128.926 to Y = -247.785), so anything mounted in it moves relative to the body. A floor-level ring is mechanically awkward here in a way it is not on dalek.

The body skin is a clean 1.2 mm PETG cylinder of 259.25 mm diameter, the thinnest shell of the three. Every "radar eye", logic display and utility arm on it is painted relief, and cad/exterior.scad body_panel_channels() engraves only to a remaining wall of 0.75 mm or more, so there is no existing through-cut to reuse. Cutting a fresh one disturbs no validated mechanism, unlike the Dalek's bumper. The dome is unusable: no wire crosses the rotating joint.

Mass, bus, pins and current

docs/verification.json gives 8.953 kg against a 9.000 kg design ceiling, and docs/mechanical.md states there is "no payload allowance". **47 g is the entire sensor budget, brackets and cable included.** That rules out an eight-unit ring before any electrical argument is made; the right answer is the fewest parts that reach capability level 2.

The HUZZAH32 has exactly one free pin, GPIO5. There is no free UART: GPIO16/17 are the post limits and the head bridge. The I2C bus uses 0x40 (INA219) and 0x48 (ADS1115), so a part defaulting to either collides; 0x29 and 0x2A are free. One free pin buys an interrupt line or one XSHUT line, not both, and that is the design decision on this robot.

Spare current is not specified in the repo. docs/electrical.md allots the whole 1.2 A of the 5V_LOGIC rail (Adafruit 4739, U6) to logic in its 60.9 W sizing, but never states what the Feather plus ADS1115 plus INA219 plus MAX98357A actually draw.

Recommendation

- 1 x Pololu 3419 VL53L8CX 8x8 multizone ToF, 45 x 45 deg, I2C 0x29, 93 mA. USD 24.95.

- 1 x DFRobot SEN0610 Gravity mmWave C4001, 100 x 80 deg, I2C 0x2A. USD 12.90.

- Total USD 37.85.

Cut one window in the lower body band at world Z200 on the front centreline and set the VL53L8CX behind it, tilted 8 deg down. One unit keeps its factory 0x29 and needs no XSHUT, so **spend GPIO5 on its interrupt line**, which is what makes a 15 Hz 64-zone part usable on a busy ESP32. Choose it over the cheaper VL53L5CX for ambient-light performance: 1650 mm inner and 1550 mm corner against a 17 percent grey target under 5 klux at 4x4 and 30 Hz.

Mount the radar behind the intact 1.2 mm skin at world Z220 on the rear centreline, with no aperture. Rearward is deliberate: the rear foot moves 119 mm during posture changes and nothing else on this robot can see behind it. Power both from 5V_LOGIC (U6). docs/electrical.md forbids powering anything through the Feather BAT/JST socket, and warns against adding 5 V pull-ups on top of the breakouts' own.

Where this does not fit. Two unknowns multiply: the rail's real headroom and the SEN0610's unpublished current. Measure the idle 5V_LOGIC draw before ordering. If the spare is below about 100 mA, drop the VL53L8CX and fit 2 x Adafruit 5425 VL53L4CX instead (USD 29.90, 38 mA, 18 x 18 deg each), using GPIO5 to hold the second in reset at boot so it can be readdressed. One XSHUT pin brings up exactly two units, which is why two is the count. That trades a 64-zone grid for two narrow beams and stays inside a 100 mA rail.

One further caution: cad/printed-frame.scad and docs/printed-frame-research.md are uncommitted and describe a revision D that would replace the cut-metal frame. Any bracket keyed to the revision C chassis may not survive it. Both mounts above are keyed to the skin instead.

fable-r2d2 - the only project that can take the full build

Mounting surfaces, apertures, and a door still open

cad/body.scad is a three-line stub, and cad/validation.json and bom/printed-parts.csv both record body_upper and body_lower as "pass_check": false, "error": "missing STL". **The two body rings have not been designed.** Apertures, flat mounting bosses and locally thinned panels go into them now at zero rework cost. Nowhere else in the repository is that true.

The lower body ring runs from about 205.8 mm to 341 mm above the floor in the driving stance, 3.6 mm PETG on a 317 mm diameter; the rear band carries the J1 charge jack and the SW1 rocker, and the front is clear. The skirt below, 164.5 to 205.8 mm, has flats only at Y = +/-94.85. The dome has real through-cuts and every one of them is occupied.

The stance is the trap. The body is tilted back 18 deg in the three-leg driving stance, so a sensor normal to the skin points 18 deg **up** at the front, level at the sides, and 18 deg down at the rear. A ring of identical bosses would aim the forward units at the ceiling. Each boss angle must carry its own azimuth correction on top of the common downtilt, and that correction is the single most important thing to get into body.scad before it is released.

Bus, pins and current

Pi I2C-1 uses 0x70-0x73 (four HT16K33) and 0x48 (ADS1115). Confirmed free: 0x29, 0x30, 0x40, 0x52, 0x68, 0x74, 0x75, 0x76, 0x77 - nine addresses for a nine-unit ring, an exact fit. Eighteen Pi GPIOs are free, so each unit gets its own XSHUT line and no multiplexer is needed. Separately, the KB2040's STEMMA QT connector is unpopulated and its whole I2C bus is unused, because board.SDA/board.SCL are GP12/GP13 while the motors use D2 to D10. Every address on it is free.

About 4.1 A of 5 V is spare against a 1.40 A design load. The Pololu measurement-condition caveat applies here as on the Dalek - 5.5 A is quoted "at 36 V in" against a family range of 3.5 A to 8 A - but the conclusion survives, because 1.40 A is a quarter of even the pessimistic 3.5 A end.

Recommendation

- 9 x Pololu 3419 VL53L8CX, 45 x 45 deg, 93 mA each, USD 24.95 each = USD 224.55.

- 1 x Adafruit 4407 MLX90640 32x24 thermal array, 55 x 35 deg, I2C 0x33, 23 mA. USD 74.95.

- 1 x Adafruit 6426 STHS34PF80 IR presence sensor, 80 deg, I2C 0x5A, 10 uA. USD 14.95.

- Total USD 314.45, about 860 mA.

Nine VL53L8CX close the circle at 40 deg boresight pitch. Nine is the count, not eight: `coverage-geometry.md` shows that eight 45 deg sensors total exactly 360 deg, which leaves each pair of adjacent edge rays parallel and eight wedges open to infinite range. Nine gives 5 deg of overlap and a ring that closes at a finite distance. Set them in body-frame bosses 90 mm above the skirt bottom edge, in the lower body ring, with a common 5 deg downtilt plus the per-azimuth stance correction. Height above the floor then varies around the circumference by up to about +/-49 mm (repo-derived: the 158.5 mm body radius times the sine of the 18 deg stance tilt); take the per-unit figure from the stance transform in CAD. Each boss is a flat pad with a through-window, not a thinned panel - for dToF the window is a hole.

Put the ring on Pi I2C-1 at 1 MHz, with nine Pi GPIOs as XSHUT lines, bringing each unit up in turn and readdressing it to one of 0x29, 0x30, 0x40, 0x52, 0x68, 0x74, 0x75, 0x76, 0x77. The addresses are volatile, so that sequence runs on every boot. Nine XSHUT lines still leave nine of the eighteen free Pi GPIOs unspent.

Put the MLX90640 and the STHS34PF80 on the **KB2040 STEMMA QT bus**, which is the point of that free resource. The MLX90640 is a 400 kHz part whose single frame costs about 42 ms of bus time; on the Pi bus it would drag the ring down from 1 MHz and cost it frame rate. On its own bus it costs nothing. Mount it on the front centreline tilted 20 deg down, which the catalogue records as the best single thermal configuration available, and it is what lifts this robot from capability level 2 to level 3-4 across its 55 deg forward field. The STHS34PF80 is the 10 uA always-on wake-up layer for a parked robot, and it holds a stationary human, which no PIR does.

Power everything from the 5 V rail. Three rules in `docs/electrical.md` must not be broken: never feed a DRV8833 from the 12 V bus, never connect the KB2040 RAW pin to the 5 V rail (it is powered only by the Pi's USB port), and power the Pi only through its USB-C input. Nothing crosses the slip ring; all twelve wires are assigned.

Two flags. Adafruit lists 4 in stock of the 55 deg MLX90640 (4407) and the 110 deg version (4469) is out of stock, so source the thermal part first. And the centre-of-gravity, foot-share and tip-back figures are marked invalid in `docs/mechanical.md`, because the built geometry has the centre foot trailing by 39.2 mm where `research/loads.md` assumed it led by 290 mm. This ring is low and light, but that re-run is owed before any mass goes high up.

Cliff and stair sensing - the layer none of the three builds carries

The brief asks the robot never to fall down a stair, and none of the three rings above answers that. `docs/the-problem.md` calls drop detection "a separate mandatory function" and asks for time-of-flight at 50 Hz or better, wired so that a lost reading counts as a cliff. `docs/coverage-geometry.md` adds that down-canted cliff sensors "are not part of the perimeter ring and must not be traded against it". So each build needs a second, cheap layer.

The catalogue part is the Adafruit 5396 VL53L4CD: USD 14.95, 22 mA, 18 x 18 deg, I2C 0x29, 100 Hz, 1.3 m of range and 0.45 m against a dark target. Its 100 Hz rate clears the 50 Hz rule. `docs/coverage-geometry.md` gives the geometry: at 200 mm mounting height and a 30 deg down-cant the look-ahead is 346 mm, the flat-floor return 400 mm, and a 180 mm stair riser moves that return to 760 mm, so a "beyond 500 mm means cliff" threshold has margin both ways. Below about 20 deg of cant the return off tile or polished wood goes specular and reads as a phantom cliff.

Every VL53 part boots at 0x29, so a cliff group needs one XSHUT line per unit or a multiplexer. That is where each robot stands.

Project	Where a cliff group can go	What it costs
da1ek	Nowhere on the present pin budget. 0x29 is taken by the VL53L4CX and no GPIO is free for an XSHUT line, so the 8-channel I2C multiplexer priced at USD 6.95 in <code>integration.md</code> is the only route. Its 0x70 default is free.	22 mA per unit against 0.30 A spare, which makes the second MPM3610 rail near-certain.
r2d2	The same multiplexer answer: GPIO5 is already spent on the VL53L8CX interrupt line, and 0x70-0x77 are free.	22 mA per unit on a rail whose real headroom is unmeasured, and each unit's mass comes out of the 47 g.
fab1e-r2d2	The KB2040 STEMMA QT bus, where every address is free and the thermal pair uses only 0x33 and 0x5A. Nine free Pi GPIOs remain after the ring's XSHUT lines.	22 mA per unit against about 4.1 A spare. Nothing binds.

How many units each robot needs is a coverage question, not an envelope question, and `coverage-geometry.md` owns it. What this chapter can say is that `fab1e-r2d2` is the only one of the three that absorbs the group without a new rail, a new multiplexer or a mass re-check.

Summary

Project	Tier	Parts	Cost	Current	Deciding constraint
dalek	Level 2 forward, 1-2 around	2 x SEN0610, 1 x VL53L4CX	USD 40.75	19 mA plus radar (not published)	0.30 A spare and zero free GPIO
r2d2	Level 2 forward, 1 rear	1 x VL53L8CX, 1 x SEN0610	USD 37.85	93 mA plus radar (not published)	47 g of mass margin
fable-r2d2	Level 2 all round, 3-4 forward	9 x VL53L8CX, 1 x MLX90640, 1 x STHS34PF80	USD 314.45	860 mA	Nothing binds; body.scad is unwritten

Capability levels are the `data/capability.json` scale: 1 presence only, 2 obstacle plus range, 3 living versus not, 4 human versus pet. The table covers the perimeter build only. Cliff and stair sensing is additional on all three, at USD 14.95 and 22 mA per VL53L4CD.

Two of the three totals cannot be closed on paper, both for the same reason: DFRobot does not publish a supply current for the C4001 modules. That is a one-meter measurement, not a research problem. Do it before ordering either radar.

What shipping robots actually use

Every shipping robot here splits perimeter sensing across separate sensors: non-semantic ranging to avoid collisions, a camera classifier to name floor objects, and - only on the expensive machines - a second camera at human height for identity. None does all three with one sensor. The two that tried, the Roomba j-series and Ohmni, carry the documented failure modes. Sources were re-verified against primary pages on 2026-09-12.

The shipping platforms at a glance

Platform	Perimeter stack (count)	Price, 2026-09-12	Source
iRobot Create 3	7 IR proximity pairs, 4 cliff IR, IMU, optical odometry	~\$299	vendor page; price unverified
TurtleBot 4 Standard	Create 3 set + RPLIDAR A1M8 + OAK-D-Pro	~\$2,000-2,400	datasheet; price unverified
Roomba j7 / j9	Bumper RGB camera + 2 IR windows + bumper + cliff IR	discontinued	layout inferred from photos
Roomba s9	Top vSLAM camera + 3D edge sensor, no classifier	discontinued	unverified, pages dead
Roomba Max 715 / 875	ClearView Pro LiDAR + PrecisionVision camera	\$599.99 / \$1,199.99	vendor page
Roborock Saros 10R	Solid-state LiDAR + 3D ToF + RGB + structured light	\$1,299.99	vendor page
Ecovacs T90 Pro Omni	RGB camera + 3D structured light + dToF	\$598.99-799.99	vendor page
Amazon Astro	Bezel cam, periscope cam, IR + ToF suite, custom cliff	\$1,599, invite-only	vendor page + teardown
Double 3	2 x RealSense D430 + 5 ultrasonic + IMU	\$3,999 launch	datasheet
temi V3	360 deg LiDAR, 3 depth/ToF cams, 6 linear ToF, cliff array	EUR 9,810 incl. VAT	vendor page
Starship	12 cameras incl. ToF and stereo, radar, ultrasonics	not sold to public	vendor page
Kiwibot 4	3 front cameras, 1 rear 180 deg camera, spotlights	not sold to public	unverified
Navimow X3	3 cameras / 300 deg, ToF 0.09-2.0 m (condition not published), solid-state LiDAR	~\$2,499-2,999	vendor page
Pudu BellaBot	3 RGBD cameras + LiDAR, 0.5 s stop response	~\$15,000-20,000 class	vendor page
Person Sensor	ESP32-S3 + 0.3 MP camera, 110 deg FOV	\$9.95, discontinued	datasheet

iRobot: three generations, three answers

The Create 3 is the best-documented cheap perimeter system in consumer robotics. **Seven pairs of IR proximity sensors** sit on a multizone bumper on a 339 mm chassis, spanning **+/-75 deg at 25 deg spacing**, with four cliff sensors underneath (vendor-page-verified). Ring height is set in the simulation URDF as `ir_intensity_z_pos = "-0.7*cm2m"`, that is **7 mm below** `base_link` (datasheet-verified). iRobot publishes no wavelength, no coupling method and no range; the widely repeated "940 nm, AC coupled, range depends on surface quality of the target" line is in no iRobot source and is **unverified**.

The failure modes are severe. Anything below bumper height - cables, socks, small toys - is found by collision, not detection. Dark IR-absorbing surfaces return nothing: a miss on the obstacle ring, a **false cliff** on the downward sensors, which is why vacuums of every brand refuse black rugs (inferred from reflective-IR physics, not vendor-stated). And a proximity sensor returns no class at all - table leg, ankle and cat read the same.

The j7 and j9 added PrecisionVision, a visible-light RGB camera on the bumper at about 60 mm (height inferred from the vacuum class, not vendor-published), recognising cords, socks, shoes and solid pet waste. iRobot backed that last class with the **P.O.O.P. promise**, a free replacement robot if a covered Roomba fails to avoid solid pet waste within one year - the only warranty in consumer robotics paying out on a perception failure. The dominant reported failure is light: reviewers say the camera cannot identify floor objects in a dark room, and whether the j9 has an illumination LED is **not published by iRobot**. Reports of dog toys missed and avoidance regressing after firmware updates are **unverified**.

The s9 is the counter-example: a 3D ranging sensor for edge work and **no object classifier**. Its headline figures are unrecoverable and its pages return HTTP 404, so both are **unverified**. The current Max line reverses the camera-only bet and pairs LiDAR with the camera - the company that argued hardest for one camera now ships ranging and classification on separate parts.

Roborock and Ecovacs: the pet-waste benchmark

Avoiding dog waste on a dark floor is the same perception problem as avoiding a sleeping cat, which makes this the most relevant prior art for pets. Roborock's StarSight 2.0 is a dual-transmitter solid-state LiDAR plus a **QVGA 3D ToF giving 21,600 points**, plus RGB, plus **vertical structured light** added to catch tall and narrow obstacles a floor-plane sensor misses - all in a 79.8 mm-tall robot. It sees objects "down to as small as 2cm wide and 2cm tall", conditioned only on "internal testing carried out by the manufacturer", which is a disclaimer, not a measurement condition. **Roborock publishes no detection range at all**; any "1-5 m" figure in circulation has no primary source. Class counts disagree: 301 on the StarSight page, 108 for the Saros 10R per Vacuum Wars.

Ecovacs pairs RGB with 3D structured light, names the method - "a mix of RGB cameras and infrared sensors" trained on "thousands of image datasets" - and states the ceiling: **"there is no 100% guarantee they will avoid pet waste in all situations"**. In the only consistent public obstacle-avoidance benchmark, **every top scorer pairs an active depth sensor with a camera classifier**; no camera-only or depth-only robot is in the top group. Quote it carefully: it is a **24-point** score over five scenarios, not 24 objects, and **no fleet average on that scale is published**.

Delivery, telepresence and the research bases

Starship's 697 x 569 mm sidewalk robot carries **12 cameras including ToF and stereo, plus radar, plus ultrasonics**, and grades behaviour: an adjacent object slows it, an object in front stops it. A classifying 360 deg perimeter fit for public sidewalks needed twelve sensors and three modalities. Kiwibot 4 runs three front cameras and one rear 180 deg camera and historically leaned on **remote human supervisors**. A thin stack is viable only when a teleoperator is the fallback classifier. Indoors, Pudu's BellaBot carries **three RGBD cameras plus LiDAR** and a stop response "as short as 0.5 seconds". Three depth cameras exist for one stated reason: to catch **low-lying and overhanging obstacles**, the two cases a single floor-plane scan misses. A tall robot on a narrow base has to answer that case deliberately.

temi V3 matches the target form factor exactly at **350 x 450 mm and 1000 mm tall**: a 360 deg LiDAR at the base front, two depth cameras, one ToF depth camera (5 m, 90 deg field of view, target reflectivity and ambient light not published), six linear ToF sensors, a 13 MP RGB camera on the motorised head at the top of the 1000 mm column, and a downward cliff array - for **EUR 9,810**. Double 3 backfills two active-IR stereo modules (0.2-10 m, **no reflectivity or ambient-light condition published**) with **five ultrasonic rangefinders**, because active IR stereo fails on glass, mirrors and dark matte surfaces; it classifies nothing and leaves judgment to the pilot. Ohmni is the other end: **\$1,995** and **no depth sensor at all**.

Astro does all four jobs with four mechanisms - Visual ID face recognition at screen height, one combined cat-and-dog model with no per-animal identity, an IR and ToF suite for geometry, and **custom** cliff sensors, which implies stock reflective parts were not good enough for a 9 kg robot. Its periscope camera reaches **1067 mm** at 132 deg **diagonal** field of view. **No teardown source identifies mmWave radar in Astro**.

TurtleBot 4 adds to the Create 3 set an RPLIDAR A1M8 (0.15-12 m, 360 deg, 5.5 Hz, **no reflectivity or ambient-light condition published**) at tower height plus an OAK-D-Pro with IR dot projector and flood LED. Classification there is purely a software problem.

Person detectors and mowers

The Useful Sensors Person Sensor was the reference drop-in: **\$9.95**, 7 Hz detection, up to four faces per frame over I2C, no raw image access. Its **range is never published** - the docs offer bounding-box size as a distance proxy. It is **discontinued**; the vendor left hardware entirely.

Mowers hold the only peer-reviewed pet-safety data. An Oxford study tested **19 models** against hedgehog cadavers: collision sensors on 8, wheel-current detection on 11, ultrasonic on 5, headlights on 4, **camera vision on zero**. The result: "Apart from one single incidence, all robotic lawn mowers had to physically touch the hedgehog carcasses to detect them", and that one non-contact detection could not be reproduced. Husqvarna's radar-equipped 435X AWD costs **\$2,999.99**; its support page says near-ground objects may be missed, vegetation reads as objects, avoidance is **disabled within 1.2 m of the boundary**, and then: **"It is not a safety feature. There is no guarantee that objects will not be run over."** Navimow X3 is the state of the art, with 24+ animal classes including hedgehogs and people detected to 3 m - and it publishes the honest cost of a camera-primary stack: **300 deg and 3 m by day collapse to 150 deg and 1.8 m at night**.

Transferable lessons, stated as rules

1. Split ranging from classification. Every successful platform uses different sensors for "do not collide", "is that a person" and "is that a pet".
2. Cover the forward arc at roughly 25 deg spacing, or buy one sensor that covers the arc. Seven IR pairs across +/-75 deg on a 339 mm bumper is the mass-produced reference layout.
3. Expect no non-contact pet detection without an active depth channel. Nineteen mowers with bumpers, wheel-current sensing and ultrasonics achieved effectively zero.
4. Budget a 2-4x haircut from datasheet range to useful range. Vector's SDK documents 1200 mm datasheet against about 300 mm useful on the robot.
5. Check that the robot's own mechanism does not occlude its sensors. Vector's lift arm blocks its ToF sensor, which then reports "clear" - readings the SDK itself calls not useful for object detection. This failure is free to avoid at design time.
6. Treat a dark target as "no return", and decide deliberately which way the robot fails on it.
7. Assume a camera-primary stack loses about half its range and half its field of view in the dark, or add an illuminator, or move ranging onto a sensor that ignores ambient light.
8. Add a non-optical near-field channel for glass, mirrors and dark matte surfaces.
9. Put identity at face height and protection at leg height, and copy the shipping height bands: cliff sensors 30-90 mm up and pointed down at the leading edge, near-field ranging at bumper height 40-90 mm, floor-object classification low and forward, identity at 900-1400 mm. A face detector is not a person detector.
10. Cover the low and the overhanging case explicitly. BellaBot ships three RGBD cameras because one floor-plane scan sees neither.
11. Do not build a perimeter on a single-source novelty module.
12. Treat speed as a perception budget. Less speed buys less range, and range is the expensive axis.
13. State the limits the way the vendors do, and put a mechanical layer - low mass, speed limit, compliant bumper, stop-on-contact - under the classifier.

What none of them attempts: **mmWave radar as an indoor perimeter sensor** (in no shipping consumer floor robot; only Husqvarna ships radar, band not published), **individual pet identity** (Astro identifies people, not animals), **certified human detection at hobby cost** (the cheapest safety-rated scanner found is \$3,111 for 275 deg), and **a detection range published with its measurement condition** - not one consumer vendor here does that.

Shopping lists

Three buildable lists follow: collision-ring at about **\$455**, collision-ring + animate-layer at about **\$909**, and full-classification-stack at about **\$1,440**. Every **[V]** price below was read from a vendor page by a source lane on **2026-09-12** and is in USD; the **[U]** prices were read from no vendor page at all. Prices **exclude tax and shipping**. **Nothing in these lists has been purchased, and no sensor named here has been physically tested** - every performance figure supporting these choices is datasheet-verified or vendor-stated, not measured.

Two markers are used in every table. **[V]** means the unit price was read on the vendor's own page in a source lane. **[U]** means no source lane priced the part: the figure is budgetary and must be confirmed before ordering. All three tier totals contain some [U] lines, so all three totals are approximate at the stated precision.

Tier 1 - collision-ring, about \$455

This tier answers "will I hit it?" and nothing else. It has no human channel and no pet channel.

Qty	Part	Vendor	SKU	Unit	Ext	What the line is for
12	VL53L8CX 8x8 multizone dToF carrier	Pololu	#3419	\$24.95 [V]	\$299.40	360 deg perimeter ring, h = 200 mm, level, 30.0 deg spacing
6	VL53L4CD 1-1300 mm dToF carrier	Pololu	#3692	\$13.95 [V]	\$83.70	Cliff and stair-edge, h = 150 mm, 23 deg down-cant
2	PCA9548 8-channel I2C mux	Adafruit	#5626	\$6.95 [V]	\$13.90	Addressing: every VL53 boots at 0x29
12	Flexible Qwiic cable, 500 mm	SparkFun	PRT-17257	\$2.75 [V]	\$33.00	Mux to ring
6	STEMMA QT cable, 100 mm	Adafruit	#4210	\$0.95 [V]	\$5.70	Mux to cliff sensors
1	RP2040 concentrator, Pico class	not priced	-	~\$4 [U]	~\$4	Ring timing and I2C concentration
1 set	Closed-cell foam bumper, 20-50 mm, plus 4 microswitches	not priced	-	~\$15 [U]	~\$15	Catches the 35 mm chair leg the ring can hide
Total	18 ToF modules, 0 people sensors			verified \$435.70	approx. \$455	724 mA average / 1.61 A peak at 5 V

Tier 2 - collision-ring + animate-layer, about \$909

Tier 1 in full, plus the three sensor additions below and their support parts. This is the recommended build.

Qty	Part	Vendor	SKU	Unit	Ext	What the line is for
1 set	Everything in Tier 1	-	-	-	~\$455	Collision layer, unchanged
1	Slamtec RPLIDAR C1 360 deg dTOF scanner	DFRobot	RPLIDAR C1	\$69.00 [V]	\$69.00	Dark targets (6 m at 10 pct reflectivity), 40,000 lux, cat lying at 120 mm, static map
8	AMG8833 Grid-EYE 8x8 thermal array	Adafruit	#3538	\$44.95 [V]	\$359.60	Animate vs inanimate ring, h = 250 mm, level, 45.0 deg spacing
2	RCWL-1601 ultrasonic ranger, 3 V/5 V	Adafruit	#4007	\$3.95 [V]	\$7.90	Glass and mirrors, forward arc, warning field only
1	PCA9548 8-channel I2C mux, third unit	Adafruit	#5626	\$6.95 [V]	\$6.95	Thermal ring branch
8	STEMMA QT cable, 300 mm	Adafruit	#4210	\$1.25 [V]	\$10.00	Mux to thermal ring
Total	19 ToF modules, 8 people sensors			addition \$453.45	approx. \$909	999 mA average / 1.95 A peak at 5 V

Tier 3 - full-classification-stack, about \$1,440

Labelled human / pet / object to 10 ft in light, and the same classes by thermal in darkness.

Qty	Part	Vendor	SKU	Unit	Ext	What the line is for
16	VL53L8CX carrier	Pololu	#3419	\$24.95 [V]	\$399.20	Tighter ring, h = 200 mm, 22.5 deg spacing
6	VL53L4CD carrier	Pololu	#3692	\$13.95 [V]	\$83.70	Cliff, h = 150 mm, 23 deg down
1	VL53L8CH CNH histogram head	bare part, carrier needed	VL53L8C H	~\$35 [U]	~\$35	Material and shape, h = 900 mm, own I2C bus
6	MLX90640-ESF-BAA 32x24 thermal, 110 x 75 deg	Adafruit (out of stock)	#4469	\$74.95 [V]	\$449.70	Thermal classification ring, h = 250 mm, 60 deg spacing
1	Slamtec RPLIDAR C1	DFRobot	RPLIDAR C1	\$69.00 [V]	\$69.00	Dark targets and ambient light
2	RCWL-1601 ultrasonic	Adafruit	#4007	\$3.95 [V]	\$7.90	Glass, forward arc
4	PCA9548 8-channel I2C mux	Adafruit	#5626	\$6.95 [V]	\$27.80	Bus splitting
1	Raspberry Pi 5	not stated	-	\$80.00 [V]	\$80.00	Camera host and inference
1	Hailo-8L AI Kit	not stated	-	\$70.00 [V]	\$70.00	YOLOv8n at 136.7 FPS, batch 8, on COCO classes
4	Wide-FoV camera, about 120 deg horizontal	not priced	-	~\$35 [U]	~\$140	Labelled human / cat / dog, h = 900-1000 mm, 90 deg spacing
1 set	RP2040 concentrator, cabling, bumper	not priced	-	~\$79 [U]	~\$79	Concentration, harness, contact backstop
Total	24 ToF modules, 10 people sensors			verified \$1,187.30	approx. \$1,440	3.8 A average / 5.2 A peak at 5 V

Shared support parts

These lines already appear inside the tier tables above - itemised in Tiers 1 and 2, and folded into Tier 3's single ~\$79 [U] concentrator-cabling-bumper line. The table is a cross-check, not an addition. Quantities scale with the sensor count, not with the tier name.

Support item	SKU	Tier 1	Tier 2	Tier 3	Note
PCA9548 8-channel I2C mux	Adafruit #5626, \$6.95 [V]	2	3	4	Mandatory. Every VL53 part boots at 0x29 and the address change is volatile
Qwiic cable, 500 mm	SparkFun PRT-17257, \$2.75 [V]	12	12	in the ~\$79 line	Mux to ring. Harness capacitance is not published by anyone; measure it
STEMMA QT cable, 100 mm	Adafruit #4210, \$0.95 [V]	6	6	in the ~\$79 line	Mux to cliff sensors
STEMMA QT cable, 300 mm	Adafruit #4210, \$1.25 [V]	-	8	in the ~\$79 line	Mux to thermal ring. See the SKU caution below
RP2040 concentrator	not priced, ~\$4 [U]	1	1	in the ~\$79 line	Ring timing, XSHUT GPIO per sensor
Level shifters	not priced in any source lane	-	-	-	Neither priced nor specified in any source lane
Mounting hardware, apertures, septa	not priced in any source lane	-	-	-	Printed into the shell. 1.50 mm dia. emitter / 1.80 mm dia. receiver at 1.0 mm standoff, with a septum
Compliant bumper and microswitches	not priced, ~\$15 [U]	1 set	1 set	in the ~\$79 line	Covers the object small enough to hide in a ring seam

Cautions before any order is placed

- **The 300 mm cable SKU is doubtful.** The source lists Adafruit **#4210** against both the 100 mm line at \$0.95 and the 300 mm line at \$1.25. One part number cannot be two lengths. Confirm the 300 mm part number on the vendor page before ordering; the \$1.25 unit price is [V] but the SKU it is attached to is not consistent.
- **Adafruit #4469 (MLX90640, 110 deg) was out of stock on 2026-09-12.** Tier 3's thermal ring must be sourced from Melexis, Waveshare or a distributor. The \$74.95 is the Adafruit list price.
- **The VL53L8CH price is not published in any source lane.** Confirm it, or substitute SATEL-VL53L8 at \$32.95 and lose the histogram output. SATEL-VL53L8 carries a 13-week lead time; VL53L5CX-SATEL carries **51 weeks**.
- **Every thermal aperture must be an open hole or thin LDPE.** Ordinary glass, polycarbonate, acrylic and PET are all opaque at 8 to 14 micrometres, so a visually clear cosmetic window silently zeroes the sensor. Budget the aperture, not a window, for every AMG8833 and MLX90640.
- **Two of Tier 3's four cameras must be USB.** The Pi 5 has two CSI ports. Confirm the camera choice against that before ordering four.
- **No tier fits dalek's present 5 V budget.** The repo records about 0.30 A spare against the 724 mA of Tier 1. A second 5 V buck, or a separate regulator off the 12 V pack, is a precondition, not an accessory. fable-r2d2 has about 4.1 A spare and is the right host, but its centre-of-gravity and tip-back figures are marked *unknown* in its own docs/mechanical.md and must be re-run before any mass goes high.

Sources

The 209 URLs below are the primary sources behind this reference, grouped by source type. They are a selection, not the whole corpus: the seventeen research files in `research/` cite 403 distinct URLs between them, and each of those files ends with its own complete source list. Where a vendor supplied many near-identical pages that only confirmed a price, a stock state or a part number, those are collapsed into one entry with a count.

Manufacturer datasheets and application notes

- ST single-zone ranging datasheets, the authority on range and ambient conditions: [VL53L0X](#), [VL53L1X](#), [VL53L3CX](#), [VL53L4CD](#), [VL53L4CX](#), [VL6180X](#).
- ST multizone datasheets, source of the horizontal and vertical fields of view used here: [VL53L5CX](#), [VL53L7CX](#), [VL53L8CX](#), [VL53L8CH](#), [VL53L9CX](#).
- ST user manuals and cover-glass notes, for zone geometry, driver limits and window design: [UM2884](#) ([VL53L5CX](#)), [UM3038](#) ([VL53L7CX](#)), [AN5231](#), [AN5856](#).
- Spinning-LiDAR datasheets: Slamtec [LD108 A1M8 rev 2.1](#), [LD108 A1M8 v2.2](#), [RPLIDAR A1](#), [RPLIDAR C1](#), [LDROBOT LD06](#), [Hokuyo URG-04LX](#).
- Single-point ranging datasheets: [Benewake TFmini](#), [Garmin LIDAR-Lite v3](#), [Sharp GP2Y0A21YK](#), [Sharp GP2Y0A02YK](#), [MaxBotix LV-MaxSonar-EZ](#), [Vishay VCNL4200](#).
- ams-OSRAM multizone dToF and eye safety: [TMF8801](#), [TMF8806](#), [TMF8828](#), [TMF8821 product page](#), [TMF8828 product page](#), [VCSEL eye-safety note](#).
- Thermal and infrared presence datasheets: [Panasonic Grid-EYE](#), [Grid-EYE reference specification](#), [Panasonic AMG8833 product page](#), [Melexis MLX90640](#), [ST STHS34PF80 datasheet DS13916](#).
- Radar datasheets and notes, for presence, micro-motion and radome design: [DFRobot C4001](#) ([SEN0609](#)), [Seeed MR60BHA2](#), [Seeed MR60FDA2](#), [Hi-Link HLK-LD2410 user manual V1.03](#), [Hi-Link LD2410 vendor page](#), [Hi-Link LD2450 vendor page](#), [Waveshare HMMD radome guide](#), [Infineon AN003623](#), [TI SWRA818](#), [60 GHz radar and false detections](#), [TI SWRA705](#), [mmWave radar radome design guide](#), [TI TIDUCV3B](#), [PIR signal-chain user guide](#), [ADI 24 GHz FMCW primer](#), [RFbeam](#), [understanding radar detection range](#).
- Bus and power parts for the ring wiring: [NXP P82B715](#), [NXP AN10710](#), [NXP SC16IS752](#), [ADI LTC4311](#).

Vendor product and wiki pages

- Adafruit: forty-one product pages read for price, stock and breakout wiring, anchored on [4010 RPLIDAR A1](#), [4407 VL53L4CD](#), [3538 AMG8833](#) and [6426 STHS34PF80](#), plus the [time-of-flight](#) and [thermal camera](#) listings, read to confirm the catalogue was complete.
- Pololu and SparkFun: [ST ToF carriers](#), [Qwiic VL53L5CX imager](#), [Qwiic TCA9548A mux](#), [Pimoroni VL53L5CX](#).
- DFRobot wikis, the primary documentation for the mmWave parts: [SEN0395](#), [SEN0609](#), [SEN0610](#), [SEN0623](#), [SEN0691 C4002](#), [SEN0557](#), with twenty-nine DFRobot store, blog and search pages for the 24, 60 and 77 GHz line-up and pricing.
- Scanner vendors: [Slamtec A1 parameters](#), [Slamtec C1](#), [Slamtec C1 datasheet mirror](#), [YDLIDAR triangulation range](#), [Waveshare D500](#), [Waveshare LD19](#), [Livox Mid-360](#), [Benewake TF-Luna](#).
- Parts with no retail breakout: [VL53L9CX](#), [VL53L4ED](#), [STEVAL-VL53L9](#) at [DigiKey](#), [X-NUCLEO-53L8A1](#).
- Other sensing vendors read for the discrimination chapter: [Seeed MR60BHA2](#), [Person Sensor at The Pi Hut](#) (discontinued), [Seeed YOLOv8 benchmarks on Pi 5](#) and the [AI kit](#), [TDK ultrasonic time-of-flight line-up](#), [DSC LC-100-PI pet-immune PIR](#), [Resideo IS335 pet-immune PIR](#), [Bosch Blue Line Gen2 installation guide](#).
- Adafruit learn guides, for wiring, addressing and driver behaviour: [RPLIDAR on Pi](#), [AMG8833](#), [MLX90640](#), [STHS34PF80](#), [TCA9548A](#), [I2C address list](#).

Academic papers and standards

- Machine-safety standards, the basis of the stopping-distance and protective-field argument: [ISO 3691-4:2023 preview](#), [ISO 13482](#), [ANSI/A3 R15.08-2-2023](#), [ANSI/RIA R15.08-1-2020](#), [The Robot Report on R15.08](#), [IEC 61496-1 explainer](#), [safety laser scanner and IEC 61496 guide](#), [IEC 60825-1 laser-class summary](#), [EN 50636-2-107 \(mowers\)](#), [SICK ESPE white paper](#), [nanoScan3 product information](#), [Keyence SZ-V specifications](#), [Banner AG4 application guide](#), [TUV Rheinland AGV white paper](#).
- Detection and classification literature, for what a 2D scan plane and a thermal array can separate: [leg detection from 2D range data](#), [person detection in LiDAR \(SPIE\)](#), [arXiv 1804.02463](#), [arXiv 2106.11239](#), [arXiv 2306.08531](#), [arXiv 2408.01951](#), [ACM IMWUT 3610902](#), [multi-person mmWave tracking](#), [EURASIP radar human sensing](#), [Caroleo et al., VL53L5CX characterisation](#), [Sensors 26\(5\):1639 \(full text, PMC12986679\)](#), [Expert Systems with Applications 2018](#), DOI [10.1016/j.eswa.2018.02.019](#), [RayPet](#), [Springer 2024](#), [MIRaSeg](#), [Springer](#), [UCT MSc](#), [FMCW micro-Doppler](#), [JRC78619](#), [pedestrian radar cross section at 24 and 77 GHz](#), [UWB MIMO detection of stationary humans](#), [PMC5134581](#), [Johnson criteria history](#), [OSTI](#).
- Physiology and animal-safety sources behind the pet and micro-motion claims: [PMC6888617](#), [PMC7070589](#), [PMC8944468](#), [PMC10497125](#), [PMC10777904](#), [PMC12158235](#), [Border collie and labrador gait](#), [PMC4687030](#), [canine gait parameters](#), [PMC5015230](#), [walk-to-run transition stride frequency](#), [PMC5435734](#), [forehead versus core temperature](#), [PMC9740153](#), [torso skin temperature under clothing](#), [PubMed 3349991](#), [Sensors 2024 24\(6\):1901](#), [PMC10975529](#), [PDSA resting respiratory rate](#), [Oxford hedgehog mower safety test](#).
- Materials, optics and building geometry: [PLA permittivity at mm-wave](#), [germanium LWIR windows](#), [germanium lenses and LWIR optics](#), [what long-wave infrared imaging is](#), [2018 IRC stair geometry](#), [Boise tread and riser sheet](#), [US10594920B2 \(cliff detection patent\)](#).

Community and teardown sources

- Teardowns and platform documentation, for prior-art sensor placement and ring height: [iFixit Amazon Astro teardown](#), [iRobot Create 3 hazards API](#), [Create 3 URDF](#), [TurtleBot 4 sensors](#), [TurtleBot 4 overview brochure](#), [Anki Vector technical reference](#), [Anki Vector SDK proximity sensor](#), [Starship robot specifications](#).
- Shipping-platform vendor pages behind the prior-art chapter: [Starship our-robots](#), [Kiwibot dimensions](#), [Pudu BellaBot](#), [temi 3 service robot](#), [iPresence on the mechanics of temi](#), [Ohmni telepresence robot](#), [IEEE Spectrum on Double 3](#), [Amazon Astro \(Wikipedia\)](#), [Roomba Max 705 retail listing](#), [Roborock StarSight \(US\)](#), [Husqvarna Automower 435X AWD](#), [Husqvarna support KA-70110](#), [Segway Navimow X3](#), [Navimow VisionFence](#), [rbtx SICK nanoScan3 pricing](#).
- Driver source read directly when the datasheet was silent: [rplidar_sdk](#), [rplidar_ros](#), [Adafruit CircuitPython RPLIDAR](#), [stm32duino VL53L1X class header](#), [stm32duino VL53L8CX API](#), [DFRobot_C4001](#), [DFRobot_C4002](#), [DFRobot_HumanDetection](#), [ESPHome dfrobot_sen0395](#), [LD2410 field notes](#), [ESP Easy P159 LD2410 plugin](#), [CircuitPython RPLIDAR docs](#), [rplidar_ros on ROS Index](#), [HC-SR04 beam tests](#).
- Forum threads and applications notes on multi-sensor interference, cover windows and sunlight limits: [ST: multiple sensors interfering](#), [ST: VL53L5CX sunlight range](#), [ST: VL53L7CH vs CX](#), [ST: cover-glass air gap](#), [Infineon BGT60TR13C FAQ](#), [Adafruit forum 190482](#), [Create 3 discussion 342](#), [PIR false-trigger troubleshooting](#), [multiple HC-SR04 crosstalk and interleave](#), [HC-SR04 tutorial](#), [pyroelectric and passive-infrared primer](#), [thermal detection](#), [recognition and identification explainer](#).
- Consumer-robot field reports and vendor claims, the evidence that dark, low-reflectance targets are the real field problem: [Vacuum Wars obstacle avoidance](#), [Samsung community: dark carpet cliff sensor](#), [black rug avoidance](#), [dark-rug cliff-sensor insight](#), [Ecovacs pet-waste avoidance](#), [iRobot Pet Owner Official Promise](#), [Roborock StarSight](#), [TechCrunch on Astro pet detection](#), [GeekWire on Astro for Business](#).
- Trade press and vendor announcements for parts with no datasheet-grade page yet: [CNX Software on VL53L9CX](#), [ST newsroom p4783](#), [Adafruit blog on the ams-OSRAM TMF8828](#).

Method and limits of this source set

All research was carried out on 2026-09-12, and every file in research/ carries that date. Sixteen lanes ran as parallel web-research agents, one per subject area, and a seventeenth file, research/recommendation.md, synthesised them. Numeric claims were then re-checked by an independent adversarial pass that re-read the primary document and was told to look for errors, not to confirm the finding. That pass is why several figures in the research files carry a correction note, including the diagonal-field-of-view error described in research/coverage-geometry.md, which the coverage chapter of this reference then uses.

Three limits apply. First, that adversarial pass is declared in sixteen of the seventeen files. research/dfrobot-mmwave-full-grid.md is the exception: it marks each row with its own confidence level but records no separate second pass, so its catalogue rows rest on a single reading of the vendor page or wiki.

Second, every price and stock state is a single read on 2026-09-12 and nothing was re-read later, so those figures age faster than the specifications beside them.

Third, **no part in this reference was purchased, bench-tested or physically measured by this project.** Every figure is marked in its source lane as datasheet-verified, vendor-page-verified, standard-verified, peer-reviewed-measured, inferred or unverified, and any measured number is somebody else's measurement, not ours. The dark-target and sunlight figures are the ones most worth confirming on the bench before a ring is committed to hardware, because those are exactly the conditions the datasheets leave open: Slamtec states only "White objects" against the A1M8 range and publishes no reflectivity percentage, and ST characterises the grid time-of-flight family at 0 and 5 kLux only, while direct sunlight is 100 to 120 kLux and ST publishes no figure for it. That reading is an inference from those two gaps, not a measurement.

Appendix: computed coverage tables

Every figure below is produced by radar-compare/scripts/geometry.py from first principles, not copied from a vendor page. Re-run that script to reproduce them.

Beam footprint width

Full lateral width of the cone, $w = 2 d \tan(\text{FoV}/2)$.

Field of view	1 ft	3 ft	5 ft	8 ft	10 ft
15 deg	80 mm / 3.2 in	241 mm / 9.5 in	401 mm / 15.8 in	642 mm / 25.3 in	803 mm / 31.6 in
20 deg	108 mm / 4.2 in	322 mm / 12.7 in	537 mm / 21.2 in	860 mm / 33.9 in	1075 mm / 42.3 in
25 deg	135 mm / 5.3 in	405 mm / 16.0 in	676 mm / 26.6 in	1081 mm / 42.6 in	1352 mm / 53.2 in
27 deg	146 mm / 5.8 in	439 mm / 17.3 in	732 mm / 28.8 in	1171 mm / 46.1 in	1464 mm / 57.6 in
30 deg	163 mm / 6.4 in	490 mm / 19.3 in	817 mm / 32.2 in	1307 mm / 51.4 in	1633 mm / 64.3 in
45 deg	252 mm / 9.9 in	758 mm / 29.8 in	1262 mm / 49.7 in	2020 mm / 79.5 in	2525 mm / 99.4 in
55 deg	317 mm / 12.5 in	952 mm / 37.5 in	1587 mm / 62.5 in	2539 mm / 99.9 in	3173 mm / 124.9 in
60 deg	352 mm / 13.9 in	1056 mm / 41.6 in	1760 mm / 69.3 in	2816 mm / 110.9 in	3520 mm / 138.6 in
63 deg	374 mm / 14.7 in	1121 mm / 44.1 in	1868 mm / 73.5 in	2988 mm / 117.7 in	3736 mm / 147.1 in
65 deg	388 mm / 15.3 in	1165 mm / 45.9 in	1942 mm / 76.4 in	3107 mm / 122.3 in	3884 mm / 152.9 in
80 deg	512 mm / 20.1 in	1534 mm / 60.4 in	2558 mm / 100.7 in	4092 mm / 161.1 in	5115 mm / 201.4 in
90 deg	610 mm / 24.0 in	1829 mm / 72.0 in	3048 mm / 120.0 in	4877 mm / 192.0 in	6096 mm / 240.0 in
100 deg	726 mm / 28.6 in	2180 mm / 85.8 in	3632 mm / 143.0 in	5812 mm / 228.8 in	7265 mm / 286.0 in
110 deg	871 mm / 34.3 in	2612 mm / 102.8 in	4353 mm / 171.4 in	6965 mm / 274.2 in	8706 mm / 342.8 in
120 deg	1056 mm / 41.6 in	3168 mm / 124.7 in	5279 mm / 207.8 in	8447 mm / 332.6 in	10559 mm / 415.7 in
150 deg	2275 mm / 89.6 in	6825 mm / 268.7 in	11375 mm / 447.8 in	18200 mm / 716.6 in	22751 mm / 895.7 in
180 deg	unbounded	unbounded	unbounded	unbounded	unbounded

Sensors needed to close a 360 degree ring

Bare closure needs $n \times \text{FoV} \geq 360$. The overlap columns add the margin that tolerates mounting error and one dead sensor.

Field of view	Bare 360 deg	With 5 deg overlap	With 10 deg overlap	With 25 percent overlap
15 deg	24	36	72	32
20 deg	18	24	36	24
25 deg	15	18	24	20
27 deg	14	17	22	18
30 deg	12	15	18	16
45 deg	8	9	11	11
55 deg	7	8	8	9
60 deg	6	7	8	8
63 deg	6	7	7	8
65 deg	6	6	7	8

Field of view	Bare 360 deg	With 5 deg overlap	With 10 deg overlap	With 25 percent overlap
80 deg	5	5	6	6
90 deg	4	5	5	6
100 deg	4	4	4	5
110 deg	4	4	4	5
120 deg	3	4	4	4
150 deg	3	3	3	4
180 deg	2	3	3	3

Where the blind wedge between adjacent sensors closes

Measured outward from the platform edge, sensors on a 175 mm radius aimed radially. "never" means $n \times \text{FoV} < 360$, where no range ever closes the gap.

Sensors in the ring	15 deg	25 deg	27 deg	45 deg	63 deg	65 deg	90 deg	100 deg	120 deg
3	never	never	never	never	never	never	never	never	never
4	never	never	never	never	never	never	never	1363 mm	411 mm
5	never	never	never	never	never	never	616 mm	379 mm	198 mm
6	never	never	never	never	3318 mm	1981 mm	303 mm	217 mm	128 mm
8	never	never	never	never	410 mm	366 mm	148 mm	115 mm	74 mm
10	never	never	never	679 mm	217 mm	200 mm	98 mm	78 mm	52 mm
12	never	never	never	338 mm	147 mm	138 mm	72 mm	59 mm	39 mm
16	never	1561 mm	866 mm	168 mm	89 mm	84 mm	48 mm	39 mm	27 mm

Centre-zone footprint of the grid imagers

One zone of the array, with how many zones the subject spans across.

Grid imager	1 ft	3 ft	5 ft	8 ft	10 ft
VL53L5CX / VL53L8CX 4x4	187 mm (adult 2.41, cat 0.75)	560 mm (adult 0.8, cat 0.25)	934 mm (adult 0.48, cat 0.15)	1494 mm (adult 0.3, cat 0.09)	1868 mm (adult 0.24, cat 0.07)
VL53L5CX / VL53L8CX 8x8	93 mm (adult 4.82, cat 1.5)	280 mm (adult 1.61, cat 0.5)	467 mm (adult 0.96, cat 0.3)	747 mm (adult 0.6, cat 0.19)	934 mm (adult 0.48, cat 0.15)
VL53L7CX 4x4	305 mm (adult 1.48, cat 0.46)	914 mm (adult 0.49, cat 0.15)	1524 mm (adult 0.3, cat 0.09)	2438 mm (adult 0.18, cat 0.06)	3048 mm (adult 0.15, cat 0.05)
VL53L7CX 8x8	152 mm (adult 2.95, cat 0.92)	457 mm (adult 0.98, cat 0.31)	762 mm (adult 0.59, cat 0.18)	1219 mm (adult 0.37, cat 0.11)	1524 mm (adult 0.3, cat 0.09)
AMG8833 Grid-EYE 8x8	88 mm (adult 5.11, cat 1.59)	264 mm (adult 1.7, cat 0.53)	440 mm (adult 1.02, cat 0.32)	704 mm (adult 0.64, cat 0.2)	880 mm (adult 0.51, cat 0.16)
MLX90640 32x24 (55 deg)	20 mm (adult 22.69, cat 7.06)	60 mm (adult 7.56, cat 2.35)	99 mm (adult 4.54, cat 1.41)	159 mm (adult 2.84, cat 0.88)	198 mm (adult 2.27, cat 0.71)
MLX90640 32x24 (110 deg)	54 mm (adult 8.27, cat 2.57)	163 mm (adult 2.76, cat 0.86)	272 mm (adult 1.65, cat 0.51)	435 mm (adult 1.03, cat 0.32)	544 mm (adult 0.83, cat 0.26)
MLX90641 16x12 (55 deg)	40 mm (adult 11.34, cat 3.53)	119 mm (adult 3.78, cat 1.18)	198 mm (adult 2.27, cat 0.71)	317 mm (adult 1.42, cat 0.44)	397 mm (adult 1.13, cat 0.35)
TMF8821 3x3	120 mm (adult 3.74, cat 1.16)	361 mm (adult 1.25, cat 0.39)	602 mm (adult 0.75, cat 0.23)	963 mm (adult 0.47, cat 0.15)	1204 mm (adult 0.37, cat 0.12)
TMF8821 4x4	123 mm (adult 3.65, cat 1.14)	369 mm (adult 1.22, cat 0.38)	616 mm (adult 0.73, cat 0.23)	985 mm (adult 0.46, cat 0.14)	1232 mm (adult 0.37, cat 0.11)

Vertical coverage, no down-tilt

Y means part of the subject is inside the beam at 1 / 3 / 5 / 8 / 10 ft; a dot means it is missed entirely. Floor strike is where the beam first reaches the floor.

Mount height	Vertical FoV	Floor strike	adult human standing	toddler	large dog	small dog	cat	cat lying	power cable	table top
100 mm	25 deg	451 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	Y
100 mm	45 deg	241 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	..YYY
100 mm	63 deg	163 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	..YYY
100 mm	90 deg	100 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY
200 mm	25 deg	902 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	...YY
200 mm	45 deg	483 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	..YYY
200 mm	63 deg	326 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY
200 mm	90 deg	200 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY
400 mm	25 deg	1804 mm	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	..YYY	...YY	..YYY
400 mm	45 deg	966 mm	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	.YYYY	..YYY	.YYYY
400 mm	63 deg	653 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	.YYYY
400 mm	90 deg	400 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	YYYYY
800 mm	25 deg	3609 mm	YYYYY	YYYYY	.YYYY	...YY	Y	Y		YYYYY
800 mm	45 deg	1931 mm	YYYYY	YYYYY	.YYYY	..YYY	..YYY	..YY	...YY	YYYYY
800 mm	63 deg	1306 mm	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	..YYY	..YYY	YYYYY
800 mm	90 deg	800 mm	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	.YYYY	.YYYY	YYYYY

Vertical coverage, 15 degree down-tilt

Same, sensor tilted down.

Mount height	Vertical FoV	Floor strike	adult human standing	toddler	large dog	small dog	cat	cat lying	power cable	table top
100 mm	25 deg	192 mm	YYY..	YYY..	YYY..	YYY..	YYY..	YYY..	YYY..	
100 mm	45 deg	130 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	
100 mm	63 deg	95 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	...YY
100 mm	90 deg	58 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	..YYY
200 mm	25 deg	384 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	
200 mm	45 deg	261 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	
200 mm	63 deg	190 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	...YY
200 mm	90 deg	116 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY
400 mm	25 deg	768 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	
400 mm	45 deg	521 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	...YY
400 mm	63 deg	380 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	..YYY
400 mm	90 deg	231 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY
800 mm	25 deg	1537 mm	YYYYY	YYYYY	YYYYY	.YYYY	..YYY	..YYY	...YY	YYY..
800 mm	45 deg	1043 mm	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	.YYYY	..YYY	YYYYY
800 mm	63 deg	759 mm	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	.YYYY	.YYYY	YYYYY
800 mm	90 deg	462 mm	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	.YYYY	YYYYY

Stopping distance against sensor update rate

Reaction distance is speed x (sensor period + 30 ms of compute). An obstacle closer than the total must never be the first time the robot sees it.

Speed	Sensor and update rate	Sensor lag	Braking	Reaction distance	Braking distance	Total stop
0.3 m/s	VL53L1X single zone @ 50 Hz	20.0 ms	firm 0.5 g	15 mm	9 mm	24 mm / 1.0 in
0.3 m/s	VL53L5CX 4x4 @ 60 Hz	16.7 ms	firm 0.5 g	14 mm	9 mm	23 mm / 0.9 in
0.3 m/s	VL53L5CX 8x8 @ 15 Hz	66.7 ms	firm 0.5 g	29 mm	9 mm	38 mm / 1.5 in
0.3 m/s	VL53L8CX 8x8 @ 15 Hz	66.7 ms	firm 0.5 g	29 mm	9 mm	38 mm / 1.5 in
0.3 m/s	typical 24 GHz mmWave @ 10 Hz	100.0 ms	firm 0.5 g	39 mm	9 mm	48 mm / 1.9 in
0.3 m/s	AMG8833 thermal @ 10 Hz	100.0 ms	firm 0.5 g	39 mm	9 mm	48 mm / 1.9 in
0.3 m/s	AMG8833 thermal @ 1 Hz	1000.0 ms	firm 0.5 g	309 mm	9 mm	318 mm / 12.5 in
0.3 m/s	MLX90640 32x24 @ 8 Hz	125.0 ms	firm 0.5 g	46 mm	9 mm	56 mm / 2.2 in
0.3 m/s	RPLIDAR A1 full scan @ 5.5 Hz	181.8 ms	firm 0.5 g	64 mm	9 mm	73 mm / 2.9 in
0.5 m/s	VL53L1X single zone @ 50 Hz	20.0 ms	firm 0.5 g	25 mm	26 mm	50 mm / 2.0 in
0.5 m/s	VL53L5CX 4x4 @ 60 Hz	16.7 ms	firm 0.5 g	23 mm	26 mm	49 mm / 1.9 in
0.5 m/s	VL53L5CX 8x8 @ 15 Hz	66.7 ms	firm 0.5 g	48 mm	26 mm	74 mm / 2.9 in
0.5 m/s	VL53L8CX 8x8 @ 15 Hz	66.7 ms	firm 0.5 g	48 mm	26 mm	74 mm / 2.9 in
0.5 m/s	typical 24 GHz mmWave @ 10 Hz	100.0 ms	firm 0.5 g	65 mm	26 mm	90 mm / 3.6 in
0.5 m/s	AMG8833 thermal @ 10 Hz	100.0 ms	firm 0.5 g	65 mm	26 mm	90 mm / 3.6 in
0.5 m/s	AMG8833 thermal @ 1 Hz	1000.0 ms	firm 0.5 g	515 mm	26 mm	540 mm / 21.3 in
0.5 m/s	MLX90640 32x24 @ 8 Hz	125.0 ms	firm 0.5 g	78 mm	26 mm	103 mm / 4.1 in
0.5 m/s	RPLIDAR A1 full scan @ 5.5 Hz	181.8 ms	firm 0.5 g	106 mm	26 mm	131 mm / 5.2 in
1.0 m/s	VL53L1X single zone @ 50 Hz	20.0 ms	firm 0.5 g	50 mm	102 mm	152 mm / 6.0 in
1.0 m/s	VL53L5CX 4x4 @ 60 Hz	16.7 ms	firm 0.5 g	47 mm	102 mm	149 mm / 5.9 in
1.0 m/s	VL53L5CX 8x8 @ 15 Hz	66.7 ms	firm 0.5 g	97 mm	102 mm	199 mm / 7.8 in
1.0 m/s	VL53L8CX 8x8 @ 15 Hz	66.7 ms	firm 0.5 g	97 mm	102 mm	199 mm / 7.8 in
1.0 m/s	typical 24 GHz mmWave @ 10 Hz	100.0 ms	firm 0.5 g	130 mm	102 mm	232 mm / 9.1 in
1.0 m/s	AMG8833 thermal @ 10 Hz	100.0 ms	firm 0.5 g	130 mm	102 mm	232 mm / 9.1 in
1.0 m/s	AMG8833 thermal @ 1 Hz	1000.0 ms	firm 0.5 g	1030 mm	102 mm	1132 mm / 44.6 in
1.0 m/s	MLX90640 32x24 @ 8 Hz	125.0 ms	firm 0.5 g	155 mm	102 mm	257 mm / 10.1 in
1.0 m/s	RPLIDAR A1 full scan @ 5.5 Hz	181.8 ms	firm 0.5 g	212 mm	102 mm	314 mm / 12.4 in